

Physical Science (Specialization) Reviewer
Compilation of Multiple-Choice Items with Answers/Rationalization and Calculations
(494 LET-Like Items)

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Introduction

Questions	Answers with Rationalization
Question #1: The height of an average person is probably (A). 1.00 m. (B). 1.60 m. (C). 2.00 m. (D). 2.60 m. Question #2: What is the mass of one-half liter of water? (A). 0.5 g. (B). 5 g. (C). 50 g. (D). 500 g. Question #3: One cc of water has a mass of about one ____ (what unit)? (A). mL (B). kg (C). g (D). cm Question #4: If you consider a very small portion of a material that is the same throughout, the density of the small sample will be (A). less than the larger sample. (B). the same as the larger sample. (C). more than the larger sample. (D). dependent on the shape of the larger sample. Question #5: Quantities, or measured properties that are capable of changing values are called (A). data. (B). variables. (C). constants. (D). impossible. Question #6: A sample of 1 mL of water has a mass of about (A). 1 cc (B). 2.2 kg (C). 1 g (D). 1.5 cm	Question #1: The height of an average person is probably (B). 1.60 m. From the Conversion Factors table in your text, we see that 1 m = 3.281 ft, so 2 m = 6.562 ft. Most people are between 3.2 and 6.5 feet tall, and sure enough, 1.60 m turns out to be about 5.2 ft. Question #2: What is the mass of one-half liter of water? (D). 500 g. By definition, a liter of water at a certain temperature has a mass of one kg. Thus one-half liter would have a mass of one-half kilogram, which is 500 gram (recall the prefix "kilo" means 1,000). Question #3: One cc of water has a mass of about one ____ (what unit)? (C). gram By definition, one liter of water has a volume of one cubic decimeter and a mass of one kilogram. A decimeter is 10 centimeters, so a cubic decimeter (1 decimeter by 1 decimeter by 1 decimeter) must have a volume of 1,000 cubic centimeters (10 centimeters by 10 centimeters by 10 centimeters). Since a kilogram of water has a mass of 1,000 grams, it must be that 1,000 cubic centimeters has a mass of 1,000 grams. Thus one cubic centimeter has a mass of one gram. Question #4: If you consider a very small portion of a material that is the same throughout, the density of the small sample will be (B). the same as the larger sample. Density is defined as the ratio of mass to volume. So a large volume would have a large mass and a small volume would have a small mass and the ratio remains the same, no matter what the volume. Question #5: Quantities, or measured properties that are capable of changing values are called (B). variables. This is the definition of a variable. Question #6: A sample of 1 mL of water has a mass of about (C). 1 g This sounds familiar. See the answer to question 3, and add the understanding that one liter has 1,000 milliliters (mL). A liter of water has a volume of one kilogram, or 1,000 grams. Thus 1,000 mL of water has a mass of 1,000 grams, and 1 mL has a mass of 1 g.

Question #7: What is the mass density of mercury if 20.0 cubic centimeters has a mass of 272 g?

- (A). 292 g/cc
- (B). 13.6 g/cc
- (C). 5732 g/cc
- (D). none of the above

Question #8: What is the mass of a 10.0 cubic centimeters of lead?

- (A). 10.0 grams
- (B). 13.6 grams
- (C). 114 grams
- (D). 2,200 grams

Question #9: What is the volume of a rock with a mass density of 3.00 grams per cubic centimeter and a mass of 600 grams?

- (A). 1800 cubic centimeters
- (B). 5 cubic centimeters
- (C). 200 cubic centimeters
- (D). none of the above

Question #7: What is the mass density of mercury if 20.0 cubic centimeters has a mass of 272 g?

(B). 13.6 g/cc

Since mass density is given
by the relationship $\rho = m/V$, then

$$\begin{aligned}\rho &= \frac{m}{V} = \frac{272 \text{ g}}{20.0 \text{ cm}^3} \\ &= \frac{272}{20.0} \frac{\text{g}}{\text{cm}^3} \\ &= \boxed{13.6 \frac{\text{g}}{\text{cm}^3}}\end{aligned}$$

Question #8: What is the mass of a 10.0 cubic centimeters of lead?
(C). 114 grams

The volume of a sample of lead is given and the problem asks for the mass. From the relationship of $\rho = m/V$, solving for the mass (m) gives mass density (ρ) times the volume (V), or $m = \rho V$.

The mass density of lead, 11.4 g/cm³, can be obtained from a table of densities (see your text), and

$$\begin{aligned}\rho &= \frac{m}{V} \\ V\rho &= \frac{m}{\cancel{V}} \\ m &= \rho V \\ m &= \left(11.4 \frac{\text{g}}{\text{cm}^3}\right)(10.0 \text{ cm}^3) \\ &= 11.4 \times 10.0 \frac{\text{g}}{\text{cm}^3} \times \text{cm}^3 \\ &= 114 \frac{\text{g} \cdot \cancel{\text{cm}^3}}{\cancel{\text{cm}^3}} \\ &= \boxed{114 \text{ g}}\end{aligned}$$

Question #9: What is the volume of a rock with a mass density of 3.00 grams per cubic centimeter and a mass of 600 grams?

(C). 200 cubic centimeters

Solving the relationship $\rho = m/V$

for volume gives $V = m/\rho$, and

$$\begin{aligned}\rho &= \frac{m}{V} \\ V\rho &= \frac{m}{\cancel{V}} \\ \frac{V\rho}{\rho} &= \frac{m}{\rho} \\ V &= \frac{m}{\rho} \\ V &= \frac{600 \text{ g}}{3.00 \frac{\text{g}}{\text{cm}^3}} \\ &= \frac{600}{3.00} \frac{\text{g}}{1} \times \frac{\text{cm}^3}{\text{g}} \\ &= 200 \frac{\cancel{\text{g}} \text{ cm}^3}{\cancel{\text{g}}} \\ &= \boxed{200 \text{ cm}^3}\end{aligned}$$

Question #10: A loaf of bread with a volume of 3,000 cubic centimeters and a density of 0.2 grams per cubic centimeter is crushed in the bottom of a grocery bag into a volume of 1,500 cubic centimeters. What is the density of the mashed bread?

- (A). 0.4 grams per cubic centimeter
(B). 600 g
(C). 0.2 grams per cubic centimeter
(D). 0.1 grams per cubic centimeter

Question #10: A loaf of bread with a volume of 3,000 cubic centimeters and a density of 0.2 grams per cubic centimeter is crushed in the bottom of a grocery bag into a volume of 1,500 cubic centimeters. What is the density of the mashed bread?

(A). 0.4 grams per cubic centimeter

The relationship between mass, volume, and density is $\rho = m/V$. The problem gives a volume, but not a mass. The mass, however, can be assumed to remain constant during the compression of the bread so the mass can be obtained from the original volume and density, or

$$\begin{aligned}\rho &= \frac{m}{V} \quad \therefore m = \rho V \\ &= \left(0.2 \frac{\text{g}}{\text{cm}^3}\right) (3,000 \text{ cm}^3) \\ &= 0.2 \times 3,000 \frac{\text{g}}{\text{cm}^3} \times \text{cm}^3 \\ &= 600 \text{ g}\end{aligned}$$

The density of the crushed bread is

$$\begin{aligned}\rho &= \frac{m}{V} \\ &= \frac{600 \text{ g}}{1,500 \text{ cm}^3} \\ &= \frac{600}{1,500} \frac{\text{g}}{\text{cm}^3} \\ &= 0.4 \frac{\text{g}}{\text{cm}^3}\end{aligned}$$

Motion

Questions	Answers with Rationalization
Question #1: A quantity of 60 km/h describes a property of motion known as (A). a vector. (B). acceleration. (C). velocity. (D). magnitude of velocity.	Question #1: A quantity of 60 km/h describes a property of motion known as (D). magnitude of velocity. The velocity of a moving object tells you the <i>speed and direction</i> of a moving object, sometimes referred to as the magnitude and direction of motion. (Magnitude means a property that can be quantitatively described, such as the volume of an eraser or the length of a pencil.) Since a direction was not specified, the quantity given is the magnitude of velocity alone.
Question #2: A ratio of $\Delta v / \Delta t$ is a measure of motion that is known as (A). speed. (B). velocity. (C). acceleration. (D). mass density.	Question #2: A ratio of $\Delta v / \Delta t$ is a measure of motion that is known as (C). acceleration. The delta symbol (Δ) means "change in," so Δv means "a change in velocity" and Δt means "a change in time." By definition, acceleration is a change of velocity per unit time, so Δv over Δt describes an acceleration.
Question #3: A ratio of $\Delta d / \Delta t$ is a measure of motion that is known as (A). speed. (B). velocity. (C). acceleration. (D). mass density.	Question #3: A ratio of $\Delta d / \Delta t$ is a measure of motion that is known as (A). speed. By definition, speed is a measurement of how fast something is moving (magnitude of velocity). It is defined as a change of distance per elapsed time -- or Δd over Δt.
Question #4: Ignoring air resistance, an object falling toward the surface of the earth has a velocity that is (A). constant. (B). increasing. (C). decreasing. (D). acquired instantaneously.	Question #4: Ignoring air resistance, an object falling toward the surface of the earth has a velocity that is (B). increasing. Falling objects are uniformly accelerated by the force of gravity and all objects in free fall experience a constant acceleration. The acceleration is 9.8 m/s^2, so during each second of fall an object gains a velocity of 9.8 m/s. Thus an object in free fall (meaning ignoring air resistance) has a velocity that is increasing during the fall.
Question #5: Ignoring air resistance, an object falling toward the surface of the earth has an acceleration that is (A). constant. (B). increasing. (C). decreasing. (D). dependent on the weight of the object.	Question #5: Ignoring air resistance, an object falling toward the surface of the earth has an acceleration that is (A). constant. Objects in free fall are uniformly accelerated by the force of gravity and experience a constant acceleration of 9.8 m/s^2. Thus during each second of fall an object experiences an increase of speed of 9.8 m/s.
Question #6: A ball rolling across the floor slows to a stop because (A). there are unbalanced forces acting on it. (B). the force that started it moving wears out. (C). all the forces are balanced. (D). the net force equals zero.	Question #6: A ball rolling across the floor slows to a stop because (A). there are unbalanced forces acting on it. Galileo understood that matter persists in its state of motion, and called the property <i>inertia</i>. Today, we understand that <i>inertia</i> describes the tendency of an object to remain in unchanging motion in the absence of an unbalanced force.

Question #7: After leaving the rifle, a bullet fired horizontally has how many forces acting on it (ignoring air resistance)?

- (A). One, from the gunpowder explosion.
- (B). One, from the pull of gravity.
- (C). Two, one from the gunpowder explosion and one from gravity.
- (D). Three, one from the gunpowder explosion, one from gravity, and one from the motion of the bullet.

Question #8: What is a sprinter's speed if a distance of 200.0 m is covered in 21.4 s?

- (A). 9.3454944 m/s
- (B). 179 m/s
- (C). 0.107 m/s
- (D). 9.35 m/s

Question #9: A pitcher throws a ball at 40.0 m/s and the ball is electronically timed to arrive at home plate 0.4625 s later. What is the distance from the pitcher to the home plate?

- (A). 0.0116 m
- (B). 18.5 m
- (C). 86.485 m
- (D). 86.5 m

Question #10: What is the velocity of a car that accelerates from rest at 9.0 feet per second squared for 8.0 seconds?

- (A). 1.125 ft/s
- (B). 0.89 ft/s
- (C). 72 ft/s
- (D). 17 ft/s

Question #7: After leaving the rifle, a bullet fired horizontally has how many forces acting on it (ignoring air resistance)?

- (B). One, from the pull of gravity.

The force from the gunpowder explosion can act only while the expanding gases can push on the bullet, that is, when it is still inside the rifle. Once it leaves the rifle the only force acting on the bullet is gravity.

Question #8: What is a sprinter's speed if a distance of 200.0 m is covered in 21.4 s?

- (D). 9.35 m/s

The sprinter's average speed is

$$\begin{aligned}\bar{v} &= \frac{d}{t} \\ &= \frac{200.0 \text{ m}}{21.4 \text{ s}} \\ &= \frac{200.0}{21.4} \frac{\text{m}}{\text{s}} \\ &= \boxed{9.35 \text{ m/s}}\end{aligned}$$

The least number of significant figures in the problem is three, so your answer should have three significant figures.

Question #9: A pitcher throws a ball at 40.0 m/s and the ball is electronically timed to arrive at home plate 0.4625 s later. What is the distance from the pitcher to the home plate?

- (B). 18.5 m

The relationship between average velocity ($\bar{v}$), distance (d), and time (t) can be solved for distance:

$$\begin{aligned}\bar{v} &= \frac{d}{t} \quad \therefore \quad d = \bar{v}t \\ d &= \left(40.0 \frac{\text{m}}{\text{s}}\right)(0.4625 \text{ s}) \\ &= 40.0 \times 0.4625 \frac{\text{m} \cdot \cancel{\text{s}}}{\cancel{\text{s}}} \\ &= \boxed{18.5 \text{ m}}\end{aligned}$$

Question #10: What is the velocity of a car that accelerates from rest at 9.0 feet per second squared for 8.0 seconds?

- (C). 72 ft/s

The question is asking for a final velocity (v_f) of an accelerating car. The acceleration ($a = 9.0 \text{ ft/s}^2$) and the time ($t = 8.0 \text{ s}$) are given. The initial velocity (v_i) is zero since the car accelerates from "rest." Therefore the question involves a relationship between acceleration (a), final velocity (v_f), and time (t), and

$$\begin{aligned} a &= \frac{v_f - v_i}{t} \\ at &= v_f - v_i \\ v_f &= at + v_i \\ v_f &= \left(9.0 \frac{\text{ft}}{\text{s}^2}\right)(8.0 \text{ s}) + 0 \frac{\text{ft}}{\text{s}} \\ &= 9.0 \times 8.0 \frac{\text{ft} \cancel{\text{s}}}{\cancel{\text{s}} \cdot \text{s}} \\ &= 72 \text{ ft/s} \end{aligned}$$

Question #11: A ball thrown straight up climbs for 3.0 seconds before falling. Neglecting air resistance, with what velocity was the ball thrown?

- (A). 3.0 m/s
- (B). 9.8 m/s
- (C). 14.5 m/s
- (D). 29 m/s

Question #11: A ball thrown straight up climbs for 3.0 seconds before falling. Neglecting air resistance, with what velocity was the ball thrown?

(D). 29 m/s

A ball thrown straight up decelerates to a velocity of zero, then accelerates back to the surface just as a dropped ball would do from the height reached. Thus the time required to decelerate upwards is the same as the time required to accelerate downwards. The ball returns to the surface with the same velocity with which it was thrown (neglecting friction).

Therefore:

$$\begin{aligned} a &= \frac{v_f - v_i}{t} \\ at &= v_f - v_i \\ v_f &= at + v_i \\ &= \left(9.8 \frac{\text{m}}{\text{s}^2}\right)(3.0 \text{ s}) \\ &= (9.8)(3.0) \frac{\text{m}}{\text{s}^2} \times \text{s} \\ &= 29 \frac{\text{m} \times \cancel{\text{s}}}{\cancel{\text{s}} \times \text{s}} \\ &= 29 \text{ m/s} \end{aligned}$$

Question #12: A car with an initial velocity of 88.0 ft/s is able to come to a stop over a distance of 100.0 ft when the brakes are applied. How much time was required for the stopping process?

- (A). 1.14 s
- (B). 2.27 s
- (C). 4.54 s
- (D). Need more information to answer.

Question #12: A car with an initial velocity of 88.0 ft/s is able to come to a stop over a distance of 100.0 ft when the brakes are applied. How much time was required for the stopping process?

(B). 2.27 s

For this type of problem, note that the velocity, distance, and time relationship calls for $\bar{v}$, or *average* velocity. The 88.0 ft/s is the *initial* velocity (v_i) that the car had before stopping ($v_f = 0$). The *average* velocity must be obtained before solving for the time required:

$$\begin{aligned}\bar{v} &= \frac{v_f + v_i}{2} \\ &= \frac{0 + 88.0 \text{ ft/s}}{2} = 44.0 \text{ ft/s}\end{aligned}$$

$$\begin{aligned}\bar{v} &= \frac{d}{t} \\ t &= \frac{d}{\bar{v}} \\ t &= \frac{100.0 \text{ ft}}{44.0 \text{ ft/s}} \\ &= \frac{100.0}{44.0} \text{ ft} \times \frac{\text{s}}{\text{ft}} \\ &= 227 \frac{\cancel{\text{ft}} \text{ s}}{\cancel{\text{ft}}} \\ &= \boxed{227 \text{ s}}\end{aligned}$$

Motion Patterns

Questions	Answers with Rationalization
Question #1: A change in the state of motion is evidence of (A). a force. (B). an applied force that is unbalanced. (C). a force that has been worn out after an earlier application. (D). any of these.	Question #1: A change in the state of motion is evidence of (B). an applied force that is unbalanced. First, recall that any change in a state of motion is known as acceleration. From Newton's second law of motion, $F = ma$, you can see that an unbalanced force and acceleration are directly related. Then if the acceleration is zero then the net force must also be zero. Thus any change that occurs in the state of motion must mean that an unbalanced force was applied.
Question #2: Considering the forces on the system of you and a bicycle as you pedal the bike at a constant velocity in a straight line (A). the applied force you are exerting on the pedal is greater than other forces. (B). all forces are in balance. (C). the resisting forces of friction are greater than the applied forces. (D). air and tire friction forces are less than the applied force.	Question #2: Considering the forces on the system of you and a bicycle as you pedal the bike at a constant velocity in a straight line (B). all forces are in balance. From Newton's first law of motion we understand that an object moves at a constant velocity only when all the forces are balanced. The second law of motion, $F = ma$, also tells us that anytime the acceleration is zero the force is also zero, so all the forces must be in balance.
Question #3: The extent of resistance to a change of motion is determined by an object's (A). weight. (B). density. (C). mass. (D). any of these.	Question #3: The extent of resistance to a change of motion is determined by an objects (C). mass. By definition, inertia is a resistance to a change in motion. Also by definition, mass is a measure of inertia. Thus the extend of resistance to a change of motion is determined by the mass of an object, which is measured in kilograms.
Question #4: With all other factors equal, if you double the unbalanced force on an object of a given mass the acceleration will be (A). doubled. (B). increased fourfold. (C). increased by one-half. (D). increased by one-fourth.	Question #4: With all other factors equal, if you double the unbalanced force on an object of a given mass the acceleration will be (A). doubled. Recall Newton's second law of motion is $F = ma$. Thus $a = F/m$, and we can see that the acceleration is directly proportional to the force. If you double the unbalanced force on an object of a given mass the acceleration will also be doubled. After all, is this not why a car with the most powerful engine will always have a greater acceleration if the cars are otherwise alike?
Question #5: With all other factors equal, If you double the mass of an object while a constant unbalanced force is applied, the acceleration will be (A). doubled. (B). increased fourfold. (C). one-half as much. (D). one-fourth as much.	Question #5: With all other factors equal, If you double the mass of an object while a constant unbalanced force is applied, the acceleration will be (C). half as much. Again, recall that Newton's second law of motion is $F = ma$. Thus $a = F/m$, and we can see that the acceleration is inversely proportional to the mass when a constant unbalanced force is applied. Thus doubling the mass will result in $1/2$ the acceleration.
Question #6: Which of the following is a unit that can be used for a measure of weight?	Question #6: Which of the following is an unit that can be used for a measure of weight? (B). newton

- (A). kilogram
- (B). newton
- (C). kg x m/s
- (D). none of these

Question #7: Which of the following is a unit for a measure of resistance to a change of motion?

- (A). lb
- (B). kg
- (C). N
- (D). none of the above

Question #8: The immediate source of the force that accelerates a car over a road comes from

- (A). the engine.
- (B). the tires.
- (C). the road.
- (D). all of these.

Question #9: Doubling the distance between an orbiting satellite and the earth will result in what change in the gravitational attraction between the two?

- (A). one-half as much
- (B). one-fourth as much
- (C). twice as much.
- (D). four times as much.

Question #10: If a ball swinging in a circle on a string is moved twice as fast the force on the string will be

- (A). twice as great.
- (B). four times as great.
- (C). one-half as much.
- (D). one-fourth as much.

Question #11: What net force is needed to accelerate a 1.25 kg book 5.00 m/s²?

- (A). 0.250 N
- (B). 4.00 N
- (C). 6.00 N
- (D). 6.25 N

Weight is the force of gravity acting on a mass. From Newton's second law of motion ($F = ma$) we understand that the acceleration from the pull of gravity is g , and the weight (w) therefore is $w = mg$. The N is therefore the unit of weight (recall that the kg is an unit of mass, not weight).

Question #7: Which of the following is an unit for a measure of resistance to a change of motion?

- (B). kg

As discussed in question three, inertia is a resistance to a change in motion. Also by definition, mass is a measure of inertia. Thus the extend of resistance to a change of motion is determined by the mass of an object, which is measured in kilograms.

Question #8: The immediate source of the force that accelerates a car over a road comes from

- (C). the road.

From Newton's third law of motion we understand that every force is immediately matched by a second force that is equal in magnitude and opposite in direction. In addition, an acceleration always occurs in the same direction as an applied unbalance force (this is correct, isn't it?). So, the car pushes on the road and the road pushes back on the car, which accelerates it over the road. How else could it accelerate in a direction opposite to the direction it is pushing on the road?

Question #9: Doubling the distance between an orbiting satellite and the earth will result in what change in the gravitational attraction between the two?

- (B). one-fourth as much

Newton's universal law of gravitation tells us the relationship between the force, or gravitational attraction, is inversely proportional to the square of the distance between two object. Thus doubling the distance between two object will result in 1/2 squared, or 1/4 the attraction.

Question #10: If a ball swinging in a circle on a string is moved twice as fast the force on the string will be

- (B). four times as great.

The force varies with the square of the velocity of a ball moving in a circle. Thus if you move a ball twice as fast, two squared is four and the force will be four times as great.

Question #11: What net force is needed to accelerate a 1.25 kg book 5.00 m/s²?

- (D). 6.25 N

$$\begin{aligned}
 F &= ma = (1.25 \text{ kg}) (5.00 \frac{\text{m}}{\text{s}^2}) \\
 &= (1.25)(5.00) \text{ kg} \times \frac{\text{m}}{\text{s}^2} \\
 &= 6.25 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \\
 &= \boxed{6.25 \text{ N}}
 \end{aligned}$$

Question #12: What net force does the road exert on a 70.0 kg bicycle and rider in order to give them an acceleration of 2.0 m/s^2 ?

- (A). none.
- (B). 70.0 kg
- (C). 140.0 kg
- (D). 140.0 N

Question #13: A 1500 kg car accelerates uniformly from 44.0 km/hr to 80.0 km/hr in 10.0 s. What was the net force exerted on the car?

- (A). 113 N
- (B). 150 N
- (C). 1500 N
- (D). 113,400 N

Question #14: What is the weight of a 70.0 kg person?

- (A). 70.0 kg
- (B). 686 N
- (C). 79.8 N
- (D). 7.14 N

Question #12: What net force does the road exert on a 70.0 kg bicycle and rider in order to give them an acceleration of 2.0 m/s^2 ?

(D). 140.0 N

The bicycle tire exerts a backwards force on the road and the equal and opposite reaction force of the road on the bicycle produces the forward motion. (The motion is always in the direction of the applied force.) Therefore,

$$\begin{aligned} F = ma &= (70.0 \text{ kg})(2.0 \frac{\text{m}}{\text{s}^2}) \\ &= (70.0)(2.0) \text{ kg} \times \frac{\text{m}}{\text{s}^2} \\ &= 140 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \\ &= \boxed{140 \text{ N}} \end{aligned}$$

Question #13: A 1500 kg car accelerates uniformly from 44.0 km/hr to 80.0 km/hr in 10.0 s. What was the net force exerted on the car?

(C). 1500 N

This problem involves two separate equations:

Newton's second law of motion and equation for acceleration.

These may be combined as follows:

$$\begin{aligned} F = ma \quad \text{and} \quad a &= \frac{v_f - v_i}{t} \quad \therefore \quad F = m \left(\frac{v_f - v_i}{t} \right) \\ \text{and,} \quad F &= (1500 \text{ kg}) \left(\frac{22.2 \text{ m/s} - 12.2 \text{ m/s}}{10.0 \text{ s}} \right) \\ &= (1500 \text{ kg}) \left(\frac{10.0 \text{ m/s}}{10.0 \text{ s}} \right) \\ &= 1500 \times 1.00 \frac{\text{kg} \cdot \frac{\text{m}}{\text{s}}}{\text{s}} \\ &= 1500 \frac{\text{kg} \cdot \text{m}}{\text{s}} \times \frac{1}{\text{s}} \\ &= 1500 \frac{\text{kg} \cdot \text{m}}{\text{s} \cdot \text{s}} \\ &= 1500 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \\ &= 1500 \text{ N} \end{aligned}$$

Question #14: What is the weight of a 70.0 kg person?

(B). 686 N

$$\begin{aligned} W &= mg \\ &= (70.0 \text{ kg})(9.80 \frac{\text{m}}{\text{s}^2}) \\ &= 70.0 \times 9.80 \text{ kg} \times \frac{\text{m}}{\text{s}^2} \\ &= 686 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \\ &= \boxed{686 \text{ N}} \end{aligned}$$

Question #15: A 1,000.0 kg car at rest experiences a net force of 1,000.0 N for 10.0 s. What is the final speed of the car?

- (A). 10.0 m/s
- (B). 100.0 m/s
- (C). 1.0 m/s
- (D). 100,000 m/s

Question #16: On Earth, an astronaut and equipment weigh 1960.0 N. While weightless in space, the astronaut fires a 100 N rocket backpack for 2.0 s. What is the resulting velocity of the astronaut and equipment?

- (A). 0.5 m/s
- (B). 1 m/s
- (C). 2 m/s
- (D). Weightless means this will not work.

Question #15: A 1,000.0 kg car at rest experiences a net force of 1,000.0 N for 10.0 s. What is the final speed of the car?

- (A). 10.0 m/s

You were given a mass (m), a force (f), and a time (t) and asked to find a final speed (v_f). These relationships are found in Newton's second law of motion and in the definition of acceleration. The two equations can be combined and solved for v_f :

$$F = ma \quad \text{and} \quad a = \frac{v_f - v_i}{t} \quad \therefore \quad F = m \left(\frac{v_f - v_i}{t} \right)$$

$$Ft = m(v_f - v_i)$$

$$\frac{Ft}{m} = v_f - v_i$$

$$v_f = \frac{Ft}{m} + v_i$$

Now the solution becomes a matter of substituting values and carrying out the mathematical operations:

$$v_f = \frac{\left(1,000.0 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \right) (10.0 \text{ s})}{1,000.0 \text{ kg}} + 0 \frac{\text{m}}{\text{s}}$$

$$= \frac{(1,000.0)(10.0)}{1,000.0} \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \frac{\text{s}}{1} \times \frac{1}{\text{kg}}$$

$$= 10.0 \frac{\cancel{\text{kg}} \cdot \cancel{\text{m}} \cdot \cancel{\text{s}}}{\cancel{\text{kg}} \cdot \cancel{\text{s}} \cdot \cancel{\text{s}}} = 10.0 \frac{\text{m}}{\text{s}}$$

Question #16: On Earth, an astronaut and equipment weigh 1960.0 N. While weightless in space, the astronaut fires a 100 N rocket backpack for 2.0 s. What is the resulting velocity of the astronaut and equipment?

- (B). 1 m/s

(a) Newton's laws of motion consider the resistance to a change of motion, or mass, and not weight. The astronaut's mass is

$$W = mg \quad \therefore \quad m = \frac{W}{g} = \frac{1960.0 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}}{9.80 \frac{\text{m}}{\text{s}^2}} = \frac{1960.0}{9.80} \frac{\text{kg} \cdot \cancel{\text{m}}}{\cancel{\text{s}^2}} \times \frac{\text{s}^2}{\cancel{\text{m}}} = 200 \text{ kg}$$

(b) From Newton's second law of motion, you can see that the 100 N rocket gives the 200 kg astronaut an acceleration of:

$$F = ma \quad \therefore \quad a = \frac{F}{m} = \frac{100 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}}{200 \text{ kg}} = \frac{100 \text{ kg} \cdot \cancel{\text{m}}}{200 \cancel{\text{s}^2}} \times \frac{1}{\text{kg}} = 0.5 \text{ m/s}^2$$

(c) An acceleration of 0.5 m/s^2 for 2.0 s will result in a final velocity of

$$a = \frac{v_f - v_i}{t} \quad \therefore \quad v_f = at + v_i$$

$$= (0.5 \text{ m/s}^2)(2.0 \text{ s}) + 0 \text{ m/s}$$

$$= 1 \text{ m/s}$$

Energy

Questions	Answers with Rationalization
Question #1: According to the scientific definition of work, pushing on a rock accomplishes no work unless there is (A). an applied force greater than its weight. (B). a net force greater than zero. (C). an opposing force. (D). movement in same direction as the force.	Question #1: According to the scientific definition of work, pushing on a rock accomplishes no work unless there is (D). movement in same direction as the force. Work is understood to be the product of the magnitude of the applied, unbalanced force and the parallel distance through which the force acts, or $W = Fd$. Thus there must be an unbalanced force and parallel movement, or the work is equal to zero.
Question #2: The metric unit of a joule (J) is a unit of (A). potential energy (B). work (C). kinetic energy (D). all of the above	Question #2: The metric unit of a joule (J) is a unit of (D). all of the above By definition, a joule is a newton of force acting through a distance of one meter and this is the unit obtained by measuring work ($W = Fd$), potential energy ($PE = mgh$), and kinetic energy ($KE = 1/2mv^2$). All have the same units and this has meaning about the close relationship between energy and work.
Question #3: Power is (A). the rate at which energy is expended. (B). work per unit of time. (C). the rate at which work is done. (D). any of the above.	Question #3: Power is (D). any of the above. (a) When you do work on something you give it energy of position (potential energy) or you give it energy of motion (kinetic energy). In turn, objects that have kinetic or potential energy can now do work on something else. (b) By definition, power is work per unit of time, or $P = W/t$. Considering these two statements, you can see that all the choices are correct.
Question #4: A Nm/s is a unit of (A). energy. (B). power. (C). force. (D). work.	Question #4: A Nm/s is a unit of (B). power. Work is found from $W = Fd$ and a newton of force times a meter of distance (Nm) is a joule (J). Power is $P = W/t$, or work per unit of time, so Nm/s is a joule/second (J/s), a unit of power know as a watt.
Question #5: The kilowatt-hour is a unit of (A). power. (B). work. (C). time. (D). force.	Question #5: The kilowatt-hour is a unit of (B). work. Since power is $P = W/t$ it follows that $W = Pt$. Thus power (W/t) times the time (t) tells you how much work (W) was done. A kilowatt-hour is power times time, so it is a unit of work.
Question #6: The potential energy of box on a shelf, relative to the floor, is a measure of (A). the work done putting the box on the shelf from the floor. (B). the weight of the box times the distance above the floor. (C). the energy the box has because of its position above the floor. (D). any of these.	Question #6: The potential energy of box on a shelf, relative to the floor, is a measure of (D). any of these. When you do work on the box you give it energy of position (potential energy). In turn, the box now has potential energy can now do work on something else. Considering these two statements, you can see that all the choices are correct.

Question #7: Which quantity has the greatest influence on the amount of kinetic energy that a large truck has while moving down the highway?

- (A). mass.
- (B). weight.
- (C). velocity.
- (D). size.

Question #8: Most all energy comes to and leaves the earth in the form of

- (A). nuclear energy.
- (B). chemical energy
- (C). radiant energy.
- (D). kinetic energy.

Question #9: How many U.S. gallons of petroleum does the accounting device of a "barrel of oil" hold?

- (A). 24
- (B). 42
- (C). 55
- (D). 100

Question #10: The law of conservation of energy is a statement that

- (A). energy must be conserved and you are breaking a law if you waste energy..
- (B). the supply of energy is limited so we must conserve.
- (C). the total amount of energy is constant.
- (D). energy cannot be used faster than it is created.

Question #11: Energy is

- (A). the ability to do work.
- (B). the work needed to create potential or kinetic energy.
- (C). the work that can be done by an object with potential or kinetic energy.
- (D). all of the above.

Question #12: When you do work on something, you

- (A). leave it with less energy than it had before.
- (B). move it in a direction 90° to the direction of the applied force.
- (C). give it energy.

Question #7: Which quantity has the greatest influence on the amount of kinetic energy that a large truck has while moving down the highway?

(C). velocity.

Kinetic Energy is $KE = \frac{1}{2}mv^2$ so you can see that it is proportional to the mass of a moving object but velocity has a greater influence. The velocity is squared, so a truck with twice the velocity has 4 times as much energy; a truck with 3 times the velocity has 9 times as much energy.

Question #8: Most all energy comes to and leaves the earth in the form of

(C). radiant energy.

By definition, radiant energy is the only form of energy that can travel through space.

Question #9: How many U.S. gallons of petroleum does the accounting device of a "barrel of oil" hold?

(B). 42

A barrel of oil is considered to be 42 United States gallons. This size of barrel does not exist in the real world as oil drums hold fifty-five United States gallons.

Question #10: The law of conservation of energy is a statement that

(C). the total amount of energy is constant.
The law of conservation of energy states that energy is never created or destroyed. Energy can be converted from one form to another but the total energy remains constant.

Question #11: Energy is

(D). all of the above.

When you do work on something you give it energy of position (potential energy) or you give it energy of motion (kinetic energy). In turn, objects that have kinetic or potential energy can now do work on something else. Thus all the choices are correct.

Question #12: When you do work on something, you

(C). give it energy.

When you do work on something you give it energy of position (potential energy) or you give it energy of motion (kinetic energy).

(D). none of these.

Question #13: An automobile engine with more power

- (A). can do the same work at a lesser rate.
- (B). can do less work over a greater time period.
- (C). requires less time to do the same amount of work.
- (D). none of these.

Question #14: A light bulb rated at 100 W does not mention a time factor because the time is

- (A). not used in power ratings of light bulbs.
- (B). already included in the definition of a watt.
- (C). considered in energy consideration other than light bulbs.
- (D). not important because bulbs always use the same amount of energy.

Question #15: Energy is conserved

- (A). in any and all energy transformations.
- (B). only when energy changes from one form to another.
- (C). very rarely, since some is needed to run machines.
- (D). most of the time, but not always.

Question #16: How much work is done in raising a 10.0 kg backpack from the floor to a shelf 1.50 m above the floor?

- (A). 150 J
- (B). 155 J
- (C). 15.0 J
- (D). 15.1 J

Question #17: If 5,000 J of work is used to raise a 100 kg crate to a

Question #13: An automobile engine with more power

- (C). requires less time to do the same amount of work.
- Power is defined as work per unit of time, $P = W/t$ so more power means that the same work can be done in less time.

Question #14: A light bulb rated at 100 W does not mention a time factor because the time is

- (B). already included in the definition of a watt.
- Power is defined as work per unit of time, $P = W/t$ so 100 watts means 100 joules per second. Thus time is already in the definition of a watt.

Question #15: Energy is conserved

- (A). in any and all energy transformations.
- The law of conservation of energy states that energy is never created or destroyed. Energy can be converted from one form to another but the total energy remains constant.

Question #16: How much work is done in raising a 10.0 kg backpack from the floor to a shelf 1.50 m above the floor?

- (A). 150 J

Note that 10.0 kg is a mass quantity, and not a weight quantity. Weight is found from $W = mg$, a form of Newton's second law of motion. Thus the force that must be exerted to lift the backpack is its weight, or $(10.0 \text{ kg}) \times (9.8 \text{ m/s}^2)$ which is 98 N. Therefore, a force of 98 N was exerted on the backpack through a distance of 1.5 m, and

$$\begin{aligned} W &= Fd \\ &= \left(98 \frac{\text{kg} \cdot \text{m}}{\text{s}^2}\right)(1.5 \text{ m}) \\ &= 147 \text{ N} \times \text{m} \\ &= 150 \text{ J} \end{aligned}$$

Question #17: If 5,000 J of work is used to raise a 100 kg crate to a high shelf in a warehouse, how high was the crate raised?

- (A). 5 m

high shelf in a warehouse, how high was the crate raised?

- (A). 5 m
- (B). 50 m
- (C). 980 m
- (D). 4020 m

Question #18: A 60.0 kg student runs up a 5.00 meter high stairway in a time of 3.94 seconds. How many watts of power did she develop?

- (A). 76.1 W
- (B). 15.2 W
- (C). 300 W
- (D). 750 W

Question #19: How many horsepower is a 1400 W blowdryer?

- (A). 1.9 hp
- (B). 2.5 hp
- (C). 2.0 hp
- (D). The answer to this question is not possible.

Question #20: What is the kinetic energy of a 2,000 kg car that is moving at 72 km/h?

- (A). 5184 kJ
- (B). 400 kJ
- (C). 20 kJ
- (D). 40 kJ

$$m = 100 \text{ kg}$$

$$g = 9.8 \text{ m/s}^2$$

$$W = 5,000 \text{ J}$$

$$h = ?$$

$$W = mgh \quad \therefore \quad h = \frac{W}{mg}$$

$$h = \frac{5,000 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \text{m}}{(100 \text{ kg}) \left(9.8 \frac{\text{m}}{\text{s}^2} \right)}$$

$$= \frac{5,000}{100 \times 9.8} \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \frac{\text{m}}{1} \times \frac{1}{\text{kg}} \times \frac{\text{s}^2}{\text{m}}$$

$$= \frac{5,000}{980} \text{ m}$$

$$= 5 \text{ m}$$

Question #18: A 60.0 kg student runs up a 5.00 meter high stairway in a time of 3.94 seconds. How many watts of power did she develop?

(D). 750 W

$$m = 60.0 \text{ kg}$$

$$g = 9.8 \text{ m/s}^2$$

$$h = 5.00 \text{ m}$$

$$t = 3.94 \text{ s}$$

$$P = ?$$

$P = \frac{mgh}{t}$ is already solved for power, and

$$P = \frac{mgh}{t}$$

$$= \frac{(60.0 \text{ kg}) \left(9.8 \frac{\text{m}}{\text{s}^2} \right) (5.00 \text{ m})}{3.94 \text{ s}}$$

$$= \frac{(60.0)(9.8)(5.00)}{(3.94)} \frac{\left(\frac{\text{kg} \cdot \text{m}}{\text{s}^2} \right) \times \text{m}}{\text{s}}$$

$$= \frac{2940}{3.94} \frac{\text{N} \cdot \text{m}}{\text{s}}$$

$$= 746.19 \frac{\text{J}}{\text{s}}$$

$$= 750 \text{ W}$$

Question #19: How many horsepower is a 1400 W blowdryer?

(A). 1.9 hp

$$\frac{1.00 \text{ hp}}{746 \text{ W}} \times 1400 \text{ W}$$

$$\frac{1400}{746} \frac{\text{hp} \cdot \text{W}}{\text{W}}$$

$$1.9 \text{ hp}$$

Question #20: What is the kinetic energy of a 2,000 kg car that is moving at 72 km/h?

(B). 400 kJ

$$m = 2,000 \text{ kg}$$

$$v = 72 \text{ km/h}$$

$$KE = ?$$

$$\frac{0.2778 \frac{\text{m}}{\text{s}}}{1.0 \frac{\text{km}}{\text{h}}} \times 72 \frac{\text{km}}{\text{h}}$$

$$0.2778 \times 72 \frac{\text{m}}{\text{s}} \times \frac{\text{h}}{\text{km}} \times \frac{\text{km}}{\text{h}}$$

$$20 \frac{\text{m}}{\text{s}}$$

$$KE = \frac{1}{2} mv^2$$

$$= \frac{1}{2} (2,000 \text{ kg}) \left(20 \frac{\text{m}}{\text{s}} \right)^2$$

$$= \frac{1}{2} (2,000 \text{ kg}) \left(400 \frac{\text{m}^2}{\text{s}^2} \right)$$

$$= \frac{1}{2} \times 2,000 \times 400 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$$

$$= 400,000 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \text{m}$$

$$= 400,000 \text{ N} \cdot \text{m}$$

$$= 400 \text{ kJ}$$

Question #21: How much work is needed to stop a 1,000.0 kg car that is moving straight down the highway at 54.0 km/h?

- (A). 7.5 kJ
(B). 113 kJ
(C). 225 kJ
(D). 1,458 kJ

Question #21: How much work is needed to stop a 1,000.0 kg car that is moving straight down the highway at 54.0 km/h?
(B). 113 kJ

$$\frac{0.2778 \frac{\text{m}}{\text{s}}}{1.0 \frac{\text{km}}{\text{h}}} \times 54.0 \frac{\text{km}}{\text{h}}$$

$$0.2778 \times 54.0 \frac{\text{m}}{\text{s}} \times \frac{\text{h}}{\text{km}} \times \frac{\text{km}}{\text{h}}$$

$$15.0 \frac{\text{m}}{\text{s}}$$

$$KE = \frac{1}{2} mv^2$$

$$= \frac{1}{2} (1,000.0 \text{ kg}) \left(15.0 \frac{\text{m}}{\text{s}} \right)^2$$

$$= \frac{1}{2} (1,000.0 \text{ kg}) \left(225 \frac{\text{m}^2}{\text{s}^2} \right)$$

$$= \frac{1}{2} \times 1,000.0 \times 225 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2}$$

$$= 112,500 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \text{m}$$

$$= 112,500 \text{ N} \cdot \text{m}$$

$$= 113 \text{ kJ}$$

Question #22: If you could "bundle" water, what would be the velocity of a water bundle that falls 100.0 m through the penstock of a hydroelectric dam?

- (A). 44 m/s
(B). 1960 m/s
(C). 200.0 m/s
(D). 980 m/s

Question #22: If you could "bundle" water, what would be the velocity of a water bundle that falls 100.0 m through the penstock of a hydroelectric dam?

- (A). 44 m/s

The velocity of any object in free fall can be found from the velocity of falling object equation, $v = \sqrt{2gh}$. Thus a "bundle" of falling water, or anything else that falls 100.0 feet will have a velocity of

$$\begin{aligned} v &= \sqrt{2gh} \\ &= \sqrt{2 \left(9.8 \frac{\text{m}}{\text{s}^2} \right) (100.0 \text{ m})} \\ &= \sqrt{(2)(9.8)(100.0) \frac{\text{m}}{\text{s}^2} \times \text{m}} \\ &= \sqrt{1960 \frac{\text{m}^2}{\text{s}^2}} \\ &= 44 \frac{\text{m}}{\text{s}} \end{aligned}$$

Question #23: A 1000.0 kg car stops on top of a 50.0 m hill. How much energy was used in climbing the hill?

- (A). 50,000 J
- (B). 9,800 J
- (C). 20 J
- (D). 490 kJ

Question #23: A 1000.0 kg car stops on top of a 50.0 m hill. How much energy was used in climbing the hill?

(D). 490 kJ

How much energy was used by a 1,000.0 kg car climbing a hill 50.0 m high is answered by how much work the car did. In this case $W = Fd$ and the force exerted is the weight of the car, $W = mgh$. Thus,

$$\begin{aligned} W &= mgh \\ &= (1,000.0 \text{ kg}) \left(9.8 \frac{\text{m}}{\text{s}^2} \right) (50.0 \text{ m}) \\ &= 1,000.0 \times 9.8 \times 50.0 \text{ kg} \times \frac{\text{m}}{\text{s}^2} \times \text{m} \\ &= 490,000 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \text{m} \\ &= 490,000 \text{ N} \cdot \text{m} \\ &= 4.9 \times 10^5 \text{ J (or 490 kJ)} \end{aligned}$$

Question #24: A 880 N box is pushed across a level floor for a distance of 5.0 m with a force of 440 N. How much work was done on the box?

- (A). 2200 J
- (B). 4400 J
- (C). 6600 J
- (D). 5500 J

Question #24: A 880 N box is pushed across a level floor for a distance of 5.0 m with a force of 440 N. How much work was done on the box?

(A). 2200 J

$$\begin{aligned} F &= 440 \text{ N} & W &= Fd \\ d &= 5.0 \text{ m} & &= \left(440 \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \right) (5.0 \text{ m}) \\ W &= 880 \text{ N} & &= 2200 \text{ N} \times \text{m} \\ W &= ? & &= 2200 \text{ J} \end{aligned}$$

Question #25: A force of 200 N is needed to push a table across a level classroom floor for a distance of 3 m. How much work was done on the table?

Question #25: A force of 200 N is needed to push a table across a level classroom floor for a distance of 3 m. How much work was done on the table?

(A). 600 J

- (A). 600 J
(B). 200 J
(C). 203 J
(D). 5880 J

$$F = 200 \text{ N}$$

$$d = 3 \text{ m}$$

$$W = ?$$

$$W = Fd$$

$$= \left(200 \frac{\text{Kg} \cdot \text{m}}{\text{s}^2} \right) (3 \text{ m})$$

$$= 200 \times 3 \text{ N} \times \text{m}$$

$$= 600 \text{ J}$$

Heat and Temperature

Questions	Answers with Rationalization
Question #1: Concerning the Celsius and Fahrenheit thermometer scales, (A). the Fahrenheit is more accurate since it has more degrees than the Celsius scale. (B). there is nothing special about either scale. (C). the Celsius is more precise since it has the same degree interval size as the Kelvin scale. (D). the Celsius is less precise since degrees below freezing are negative values.	Question #1: Concerning the Celsius and Fahrenheit thermometer scales... (B). there is nothing special about either scale. The Fahrenheit does have more degrees than the Celsius scale, but this does not make it more accurate or more precise. Having the same degree interval size as Kelvins on the Absolute scale does not make the Celsius scale any more precise or accurate either. The Fahrenheit and Celsius scales are both arbitrary scales with different places on a number line assigned to the boiling and freezing points of water.
Question #2: Numbers on both the Fahrenheit and Celsius scales would have no meaning without (A). conversion equations. (B). Kelvin temperature scale. (C). two fixed points to which they are compared. (D). none of the above.	Question #2: Numbers on both the Fahrenheit and Celsius scales would have no meaning without (C) two fixed points to which they are compared. The two fixed points are arbitrary, which means they were selected by whim or impulse, not by way of reason or a scientific law. The Absolute scale, on the other hand, is based on the relationship between kinetic molecular energy and temperature and is unrelated to and independent of anything else.
Question #3: The two temperature scales with the same interval step size are the (A). Celsius and Fahrenheit. (B). Fahrenheit and Kelvin. (C). Kelvin and Celsius. (D). ... this does not exist.	Question #3: The two temperature scales with the same interval step size are the (C) Kelvin and Celsius. This is how the Kelvin scale was designed.
Question #4: Suppose the volume of gasoline in your gas tank expands with warming temperatures. Do you now have more gasoline? (A). No, you still have the same mass of gasoline. (B). Yes, the volume increased and so has the mass. (C). No, the mass decreased as the volume increased to maintain the density. (D). Yes, the density and the volume increased.	Question #4: Suppose the volume of gasoline in your gas tank expands with warming temperatures. Do you now have more gasoline? (A) No, you still have the same mass of gasoline. The mass stays the same as the volume become greater, resulting in a decreased density. In addition to questions about how this could happen, you know the mass remains the same because if the mass were increasing along with the volume the density would remain the same.
Question #5: Substance A has a higher specific heat than substance B. Which requires the most energy to heat equal masses of A and B to the same temperature? (A). Substance A (B). Substance B.	Question #5: Substance A has a higher specific heat than substance B. Which requires the most energy to heat equal masses of A and B to the same temperature? (A) Substance A. Specific heat is defined as the amount of heat needed to increase the temperature of one gram of a substance one degree Celsius. Thus less heat is required to heat a substance with a low specific heat and

(C). Both require the same amount of heat. (D). Answer depends on the density of each substance. Question #6: With all other factors being equal, the most likely to burn your mouth when taken directly from an oven is a food with (A). higher specific heat. (B). lower specific heat. (C). ... specific heat is not important in this situation. (D). more information needed. Question #7: A large and a small container of water with the same temperature have (A). the same total amounts of internal energy. (B). the same amounts of internal and external energy. (C). different amounts of heat. (D). the same amounts of all forms of energy. Question #8: Anytime a temperature difference occurs, you can expect (A). cold to move to where it is warmer. (B). energy movement from high temperature regions. (C). no energy movement unless it is warm enough, at least above the freezing temperature. (D). energy movement flowing slowly from cold to warmer regions. Question #9: Which of the following quantities is one that cannot be used to measure an amount of heat? (A). Joule. (B). Calorie. (C). Btu. (D). Any of these can be used to measure heat.	more for a substance with a high specific heat. For example, a substance with a specific heat of 0.5 gram/calC° would be warmed 20° C by 10 calories while a substance with a specific heat of 1 gram/calC° would be warmed only 10° C by the same number of calories. Thus the substance with the higher specific heat requires more energy to reach 20° C. Question #6: With all other factors being equal, the most likely to burn your mouth when taken directly from an oven is a food with a (B) lower specific heat. Specific heat is defined as the amount of heat needed to increase the temperature of one gram of a substance one degree Celsius. Thus less heat is required to heat a substance with a low specific heat and more for a substance with a high specific heat. For example, a substance with a specific heat of 0.5 gram/calC° would be warmed 20° C by 10 calories of heat while a substance with a specific heat of 1 gram/calC° would be warmed only 10° C by the same number of calories. Thus the substance with the lower specific heat is going to have a higher temperature from the same heating. Question #7: A large and a small container of water with the same temperature have (C) different amounts of heat. Temperature is the average kinetic energy of the molecules of a substance. Heat is the total molecular energy--meaning both the kinetic and potential energy of the molecules--so, a larger mass of water will contain more heat than a smaller mass of water at the same temperature. Since heat can only be measured as it leaves or enters a substance, you might ask yourself if a large and a small container of water with the same temperature would melt the same number of ice cubes. The large container of water will melt more, of course, so you know it contains a different amount of heat. Question #8: Anytime a temperature difference occurs, you can expect (B) energy movement from high temperature regions. Heat tends to move from a region with higher temperature to a region with a lower temperature just as a ball tends to roll downhill. You must give a ball more energy to move it uphill and more energy is required to pump heat from a cooler region to a warmer one. Question #9: Which of the following quantities is one that cannot be used to measure an amount of heat? (D) Any of these can be used to measure heat.
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Question #10: As you go to higher elevations above sea level the boiling point of water (A). decreases. (B). increases. (C). stays the same. (D). changes with the initial temperature of the water. Question #11: Increasing the rate of heating under a pot of boiling water will (A). increase the temperature of the boiling water. (B). increase the rate of boiling, but not the temperature. (C). increase both the rate of boiling and the temperature of the boiling water. (D). all of the above. Question #12: As a solid goes through a phase change to a liquid, heat is absorbed and the temperature (A). increases. (B). decreases. (C). remains the same. (D). fluctuates. Question #13: The transfer of energy from molecule to molecule is called (A). convection. (B). radiation. (C). conduction. (D). equilibrium. Question #14: No water vapor is added to or removed from a sample of air that is cooling, so the relative humidity of this sample of air will (A). remain the same. (B). be lower. (C). be higher. (D). the answer depends on the temperature. Question #15: Compared to cooler air, warm air can (A). hold more water vapor. (B). less water vapor. (C). the same amount of water vapor. (D). the answer depends on the	Question #10: As you go to higher elevations above sea level the boiling point of water (A) decreases. As you go to higher and higher elevations above sea level you come under less and less pressure. Less pressure means a lower boiling point and more pressure means a higher boiling point. In general, the boiling point decreases 1° C for each 1,000 ft increase in altitude. Question #11: Increasing the rate of heating under a pot of boiling water will (B) increase the rate of boiling, but not the temperature. The temperature of a pot of boiling pot of water remains the same until the water boils away. Boiling is evaporation that occurs throughout a liquid. Increasing the rate of heating will increase the rate of boiling, so evaporation will occur more rapidly. Evaporation is a cooling process, so overall the temperature remains constant. Question #12: As a solid goes through a phase change to a liquid, heat is absorbed and the temperature (C) remains the same. Heat is absorbed or released during a phase change as the latent heat of fusion or the latent heat of vaporization is absorbed or released. The temperature is constant during a phase change as all the energy goes into or comes out of potential energy. Question #13: The transfer of energy from molecule to molecule is called (C) conduction. This is the definition of conduction. Question #14: No water vapor is added to or removed from a sample of air that is cooling, so the relative humidity of this sample of air will (C) be higher. Relative humidity is a measure of how much water vapor is in the air compared to how much could be there at that temperature. Cool air will hold less water vapor than warm air, so cooling the air with no other changes will increase the relative humidity. Question #15: Compared to cooler air, warm air can (A) hold more water vapor. Temperature is a measure of the average kinetic energy of the molecules of a substance. Thus cooler air has slower moving water vapor molecules and slower moving molecules spend more time close to each other than they do in warmer air. Thus in warmer air water vapor molecules are less likely to be attracted and pulled
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exact temperature at the time.

Question #16: A heat pump is able to produce cooler temperatures because the refrigerant is

- (A). a cool liquid that is pumped through the system.
- (B). evaporated in the cool part by reduction of pressure.
- (C). condensed in the cool part by the action of the compressor.
- (D). a working fluid that produces thermal energy from electrical.

Question #17: The average human body temperature is 98.6° F. What is the equivalent temperature on the Celsius scale?

- (A). 22.8° Celsius.
- (B). 37° Celsius.
- (C). 51.2° Celsius.
- (D). 209.48° Celsius.

Question #18: A science article refers to a temperature of 300.0 K. What is the equivalent Fahrenheit temperature?

- (A). 17° F.
- (B). 80.6° F.
- (C). 549° F.
- (D). 572° F.

together than they would in cooler air, so warm air holds more water vapor.

Question #16: A heat pump is able to produce cooler temperatures because the refrigerant is

- (B) evaporated in the cool part by reduction of pressure
- Evaporation cools by removing the latent heat of vaporization as the refrigerant moves from the high pressure to the low pressure part of the heat pump.

Question #17: The average human body temperature is 98.6°F. What is the equivalent temperature on the Celsius scale?

- (B) 37° Celsius.

The equation $T_c = \frac{5}{9} (T_f - 32^\circ)$ describes a relationship between the two temperature scales and is used to convert a Fahrenheit temperature to Celsius. The equation is already solved for the Celsius temperature, T_c . Thus,

$$\begin{aligned} T_c &= \frac{5}{9} (T_f - 32^\circ) \\ &= \frac{5}{9} (98.6^\circ - 32^\circ) \\ &= \frac{333^\circ}{9} \\ &= 37^\circ \text{C} \end{aligned}$$

Question #18: A science article refers to a temperature of 300.0 K. What is the equivalent Fahrenheit temperature?

- (B). 80.6° F.

(a) The equation $T_K = T_C + 273$ describes a relationship between the two temperature scales, and is used to convert Celsius temperature degrees to Kelvins. The equation must be first solved for the Celsius temperature, T_C . Thus,

$$\begin{aligned} T_K &= T_C + 273 \quad \therefore \quad T_C = T_K - 273^\circ \\ &= 300.0 - 273^\circ \\ &= 27.0^\circ \text{C} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad T_F &= \frac{9}{5} T_C + 32^\circ \\ &= \frac{9}{5} 27.0^\circ + 32^\circ \\ &= \frac{243^\circ}{5} + 32^\circ \\ &= 48.6^\circ + 32^\circ \\ &= 80.6^\circ \text{F} \end{aligned}$$

Question #19: How much heat must be supplied to a 1.0 kg iron pan to raise its temperature from 20.0 degree to 100.0° C?

- (A). 8.8 kcal.
- (B). 11 kcal.
- (C). 22 kcal.
- (D). 44 kcal.

Question #20: A bicycle and rider have a combined mass of 120 kg. How many calories of heat are generated in the brakes when the bicycle comes to a stop from a speed of 18 km/hr?

- (A). 6.3 kcal.
- (B). 360 cal.
- (C). 71.7 cal.
- (D). 19,440 cal.

Question #21: A 75.0 kg person consumes a small order of French fries (250.0 Calories) and wishes to "work off" the energy by climbing a 10.0 meter stairway. How many trips

Question #19: How much heat must be supplied to a 1.0 kg iron pan to raise its temperature from 20.0° to 100.0° C?

- (A) 8.8 kcal.

The equation for specific heat is $c = \frac{Q}{m\Delta T}$ where c is the specific heat, Q is the heat in calories, ΔT is the change of temperature, and m is the mass. If the mass is measured in g, the units for specific heat are cal/g°C and if it is measured in kg, the units are kcal/kg°C. Solved for Q , this equation tells you the amount of heat needed to produce a given temperature change is equal to the product of the mass, the specific heat of the substance, and the temperature change, which is what we are looking for in this problem. So, the heat needed is

$$\begin{aligned} Q &= mc\Delta T \\ &= (1.0 \text{ kg}) \left(0.11 \frac{\text{kcal}}{\text{kg} \cdot ^\circ\text{C}} \right) (80.0^\circ\text{C}) \\ &= 1.0 \times 0.11 \times 80.0 \quad \text{kg} \times \frac{\text{kcal}}{\text{kg} \cdot ^\circ\text{C}} \times ^\circ\text{C} \\ &= 8.8 \text{ kcal} \end{aligned}$$

Question #20: A bicycle and rider have a combined mass of 120 kg. How many calories of heat are generated in the brakes when the bicycle comes to a stop from a speed of 18 km/hr?

- (B) 360 cal.

The first step is to convert km/h to m/s by multiplying times 0.2778 m/s per km/h, which gives a velocity of 5.0 m/s. Next, we need to find the energy of the bike and riders. The moving system has kinetic energy of

$$\begin{aligned} KE &= \frac{1}{2} mv^2 \\ &= \frac{1}{2} (120 \text{ kg}) \left(5.0 \frac{\text{m}}{\text{s}} \right)^2 \\ &= \frac{1}{2} (120 \text{ kg}) \left(25 \frac{\text{m}^2}{\text{s}^2} \right) \\ &= \frac{1}{2} \times 120 \times 25 \quad \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} \\ &= 60 \times 25 \quad \frac{\text{kg} \cdot \text{m}}{\text{s}^2} \times \text{m} \\ &= 1,500 \text{ N} \cdot \text{m} \\ &= 1,500 \text{ J} \end{aligned}$$

We can now convert joules to kcal by using the relationship between mechanical energy and heat, which is 1 kilocalorie (kcal) = 4184 joule (J), or 1 kcal/4184 J. Thus,

$$\begin{aligned} &(1,500 \text{ J}) \left(\frac{1 \text{ kcal}}{4184 \text{ J}} \right) \\ &\frac{1,500}{4184} \frac{\text{J} \cdot \text{kcal}}{\text{J}} \\ &0.36 \text{ kcal} \end{aligned}$$

Question #21: A 75.0 kg person consumes a small order of French fries (250.0 Calories) and wishes to "work off" the energy by climbing a 10.0 meter stairway. How many trips up the stairs are needed to use all the energy?

- (B) 142.

up the stairs are needed to use all the energy?

- (A). 418.
- (B). 142.
- (C). 3.3.
- (D). 7.5.

Question #22: A 300.0 watt immersion heater is used to heat 250.0 g of water from 10.0 degree C to 70.0 degree C. How many minutes did this take?

- (A). 3.48 min.
- (B). 4.17 min.
- (C). 75.0 min.
- (D). 150.0 min.

Question #23: How much heat is needed to change 250.0 g of ice at 0° C to water at 0° C?

- (A). 250 kcal
- (B). 450 kcal
- (C). 20.0 kcal
- (D). 550 kcal

The Calorie used by dietitians is a kilocalorie, so 250.0 Cal is 250.0 kcal. The mechanical energy equivalent is 1 kcal = 4184 J, so (250.0 kcal)(4184 J/kcal) = 1,046,000 J. Since $W = Fd$ and the force needed is equal to the weight (mg) of the person, $W = mgh = (75.0 \text{ kg})(9.80 \text{ m/s}^2)(10.0 \text{ m}) = 7350 \text{ J}$ for each stairway climb. A total of 1,046,000 J would require 1,046,000 J/7350 J per climb, or a little more than 142 trips up the stairs.

Question #22: A 300.0 watt immersion heater is used to heat 250.0 g of water from 10.0° to 70.0° C. How many minutes did this take?

(A) 3.48 min.

Since a watt is defined as a joule/s, finding the total energy in joules will tell the time:

$$\begin{aligned} Q &= mc\Delta T \\ &= (250.0 \text{ g}) \left(1.00 \frac{\text{cal}}{\text{g} \cdot ^\circ\text{C}} \right) (60.0^\circ\text{C}) \\ &= 250.0 \times 1.00 \times 60.0 \quad \frac{\text{g}}{1} \times \frac{\text{cal}}{\text{g} \cdot ^\circ\text{C}} \times \frac{^\circ\text{C}}{1} \\ &= 15,000 \text{ cal} \\ &= 15.0 \text{ kcal} \end{aligned}$$

There are 4184 J/kcal, so there are (15.0 kcal) × (4184 J/kcal) = 62,800 J. A 300.0 W heater uses energy at a rate of 300.0 J/s. It would take a 300.0 W heater

$$\frac{62,800 \text{ J}}{300.0 \frac{\text{J}}{\text{s}}} = \frac{62,800}{300.0} \frac{\text{J}}{1} \times \frac{\text{s}}{\text{J}} = 209 \text{ s}$$

to heat the water. This is

$$\frac{209 \text{ s}}{60 \frac{\text{s}}{\text{min}}} = \frac{209}{60} \frac{\text{s}}{1} \times \frac{\text{min}}{\text{s}} = 3.48, \text{ or } 3\frac{1}{2} \text{ min}$$

Question #23: How much heat is needed to change 250.0 g of ice at 0° C to water at 0° C?

(C) 20.0 kcal.

Since the problem specified a solid changing to a liquid without a temperature change, you should recognize that this is a question about a phase change only. The phase change from solid to liquid (or liquid to solid) is concerned with the latent heat of fusion. For water, the latent heat of fusion is given as 80.0 cal/g, and

$$\begin{aligned} m &= 250.0 \text{ g} & Q &= mL_f \\ L_f (\text{water}) &= 80.0 \text{ cal/g} & &= (250.0 \text{ g})(80.0 \frac{\text{cal}}{\text{g}}) \\ Q &= ? & &= 250.0 \times 80.0 \frac{\cancel{\text{g}} \cdot \text{cal}}{\cancel{\text{g}}} \\ & & &= 20,000 \text{ cal} = \boxed{20.0 \text{ kcal}} \end{aligned}$$

Question #24: In an electric freezer, 400.0 g of water at 18.0 degree C is cooled, frozen, and the ice is chilled to -5.00 degree C. How much total heat was removed from the water?

- (A). 39.2 kcal
- (B). 40.2 kcal
- (C). 47.4 kcal
- (D). 92 kcal

Question #24: In an electric freezer, 400.0 g of water at 18.0 degree C is cooled, frozen, and the ice is chilled to -5.00 degree C. How much total heat was removed from the water?

(B) 40.2 kcal.

Step 1:

Cool water from 18.0° C to 0° C.

$$\begin{aligned} Q_1 &= mc\Delta T \\ &= (400.0 \text{ g})(1.00 \frac{\text{cal}}{\text{g}^\circ\text{C}})(18.0^\circ \text{C}) \\ &= (400.0)(1.00)(18.0) \text{ g} \times \frac{\text{cal}}{\text{g}^\circ\text{C}} \times ^\circ\text{C} \\ &= 7.20 \text{ kcal} \end{aligned}$$

Step 2:

Find the energy needed for the phase change of water at 0° C to ice at 0° C.

$$\begin{aligned} Q_2 &= mL_f \\ &= (400.0 \text{ g})(80.0 \frac{\text{cal}}{\text{g}}) \\ &= 32.0 \text{ kcal} \end{aligned}$$

Step 3: Cool the ice from 0° C to ice at -5.00° C.

$$\begin{aligned} Q_3 &= mc\Delta T \\ &= (400.0 \text{ g})(0.50 \frac{\text{cal}}{\text{g}^\circ\text{C}})(5.00^\circ \text{C}) \\ &= 400.0 \times 0.50 \times 5.00 \text{ g} \times \frac{\text{cal}}{\text{g}^\circ\text{C}} \times ^\circ\text{C} \\ &= 1.0 \text{ kcal} \end{aligned}$$

$$\begin{aligned} Q_{\text{Total}} &= Q_1 + Q_2 + Q_3 \\ &= 7.20 \text{ kcal} + 32.0 \text{ kcal} + 1.0 \text{ kcal} \\ &= \boxed{40.2 \text{ kcal}} \end{aligned}$$

Sound

Questions	Answers with Rationalization
Question #1: If you increase the energy that goes into starting a vibration, you will increase the (A). frequency. (B). amplitude. (C). number of cycles per second. (D). wavelength.	Question #1: If you increase the energy that goes into starting a vibration, you will increase the (B). amplitude. The maximum displacement from the rest (equilibrium) position is defined as the amplitude of the vibration. Amplitude can be measured in cm, dm, or m just like any other length measurement.
Question #2: The time required for one cycle of any repeating event is called one (A). hertz. (B). period. (C). frequency. (D). amplitude.	Question #2: The time required for one cycle of any repeating event is called one (B). period. The period is the time required for one complete cycle, such as one complete vibration of a vibrating object.
Question #3: The time for needed one cycle and the frequency of a vibration have a relationship of the frequency varying (A). directly with the time. (B). directly with the square of the time. (C). inversely with the time. (D). inversely with the square of the time.	Question #3: The time for needed one cycle and the frequency of a vibration have a relationship of the frequency varying (C). inversely with the time. The period and frequency are two ways of describing the time involved in a vibration. Since the period (T) is the total time involved in one cycle and the frequency (f) is the number of cycles per second, the relationship is $T = 1/f$, which is an inverse relationship.
Question #4: A longitudinal wave causes particles of a material to move (A). back and forth in the same direction the wave is moving. (B). perpendicular to the direction the wave is moving. (C). in a circular motion in the direction the wave is moving. (D). in a circular motion opposite the direction the wave is moving.	Question #4: A longitudinal wave causes particles of a material to move (A) back and forth in the same direction the wave is moving. A longitudinal wave is a disturbance that causes particles to move closer together or farther apart in the same direction that the wave is moving.
Question #5: A transverse wave causes particles of a material to move (A). back and forth in the same direction the wave is moving. (B). perpendicular to the direction the wave is moving. (C). in a circular motion in the direction the wave is moving. (D). in a circular motion opposite the direction the wave is moving.	Question #5: A transverse wave causes particles of a material to move (B) perpendicular to the direction the wave is moving. A transverse wave is a disturbance that causes motion perpendicular to the direction that the wave is moving.
Question #6: Sound waves travel faster in	Question #6: Sound waves travel faster in

(A). air as compared to solids. (B). liquids as compared to solids. (C). warm air. (D). cool air. Question #7: You hear a higher pitch when a sound wave has a greater (A). amplitude. (B). velocity. (C). frequency. (D). wavelength. Question #8: The extent of displacement of a vibrating tuning fork is related to the resulting sound wave characteristic of (A). frequency. (B). amplitude. (C). wavelength. (D). period. Question #9: Sound waves cannot travel through (A). solid metal. (B). heavy, dense liquids. (C). a vacuum. (D). any of these. Question #10: Both constructive and destructive interference is necessary to produce the sound phenomena known as (A). resonance. (B). refraction. (C). beats. (D). diffusion. Question #11: An efficient transfer of energy that takes place at a natural frequency is known as (A). resonance. (B). beats. (C). the Doppler effect. (D). reverberation. Question #12: During a track and field meet, the time difference between seeing the smoke from a starter's gun and hearing the "bang" would be less (A). on a warmer day. (B). on a cooler day. (C). if a more powerful shell is used.	(C). warm air. Warm air has a greater kinetic energy than cold air, and molecules with a greater kinetic energy will carry information faster than slowly moving molecules. Question #7: You hear a higher pitch when a sound wave has a greater (C) frequency. The human brain interprets higher frequency sounds as higher-pitched sounds and lower frequency sounds as lower-pitched sounds. Question #8: The extent of displacement of a vibrating tuning fork is related to the resulting sound wave characteristic of (B). amplitude. The back and forth vibrations of a tuning fork produce pulses of increased and decreased pressure that move through the air. A graph of the pressure changes shows that the amplitude of these changes is directly related to the displacement of the vibrating tuning fork. Question #9: Sound waves cannot travel through (C) a vacuum. The transmission of a sound wave requires a medium, that is, a solid, liquid, or gas to carry the disturbance so sound does not travel through a vacuum. Question #10: Both constructive and destructive interference is necessary to produce the sound phenomena known as (C). beats. Beats occur when similar two sound waves experience alternating constructive and destructive interference, resulting in a rapidly rising and falling sound level. Interference does not occur in the other listed phenomena. Question #11: An efficient transfer of energy that takes place at a natural frequency is known as (A). resonance. This is a definition of resonance. Question #12: During a track and field meet, the time difference between seeing the smoke from a starter's gun and hearing the "bang" would be less (A). on a warmer day. Sound travels faster on a warm day, so the interval between seeing the smoke and hearing the report would be less.
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(D). if a less powerful shell is used.

Question #13: The driver of a car sounds the horn and hears a continual pitch while an observer hears a lower pitched horn. This means the car

- (A). has a defective horn.
- (B). is moving faster than the speed of sound.
- (C). is moving toward the observer.
- (D). is moving away from the observer.

Question #14: How fast would you expect a 1.27 m sound wave with a frequency of 260 Hz to move?

- (A). 204.7244 m/s
- (B). 330 m/s
- (C). 210 m/s
- (D). 0.0049 m/s

Question #15: The average human ear is most sensitive to 2,500.0 Hz sounds. Assuming sound to move at 334 m/s, what is the wavelength of sounds people are most sensitive?

- (A). 7.5 cm
- (B). 835 cm
- (C). 13.4 cm
- (D). 0.7485 m

Question #13: The driver of a car sounds the horn and hears a continual pitch while an observer hears a lower pitched horn. This means the car

(D). is moving away from the observer.

The apparent shift of frequency is called the Doppler effect, which results in a higher frequency as a sounding source approaches and a lower shift as it moves away.

Question #14: How fast would you expect a 1.27 m sound wave with a frequency of 260 Hz to move?

(B). 330 m/s

A question of how fast a 1.27 m sound wave with a frequency of 260 Hz will move is about the relationship between the velocity, wavelength, and frequency of a wave. The relationship is called the wave equation, $v = \lambda f$, where v is the velocity of a sine wave, λ (lambda) is the length of one wave, and f is the frequency of the wave. The wave equation is already solved for velocity, so

$$\begin{aligned} v &= \lambda f \\ &= (1.27 \text{ m}) \left(260 \frac{1}{\text{s}} \right) \\ &= 1.27 \times 260 \text{ m} \times \frac{1}{\text{s}} \\ &= 330 \frac{\text{m}}{\text{s}} \end{aligned}$$

Question #15: The average human ear is most sensitive to 2,500.0 Hz sounds. Assuming sound to move at 334 m/s, what is the wavelength of sounds people are most sensitive?

(C) 13.4 cm

The question asks about the wavelength of a sound with a known frequency (2,500.0 Hz) and velocity (334 m/s). The relationship between the velocity, wavelength, and frequency of a wave is called the wave equation, $v = \lambda f$, where v is the velocity of a sine wave, λ (lambda) is the length of one wave, and f is the frequency of the wave. The wave equation must first be solved for the wavelength, and

$$\begin{aligned} v &= \lambda f \therefore \lambda = \frac{v}{f} \\ &= \frac{334 \frac{\text{m}}{\text{s}}}{2500.0 \frac{1}{\text{s}}} \\ &= \frac{334}{2500.0} \frac{\text{m}}{\cancel{\text{s}}} \times \frac{\cancel{\text{s}}}{1} \\ &= 0.134 \text{ m} \\ &= 13.4 \text{ cm} \end{aligned}$$

Question #16: What is the frequency of water waves if a new crest is observed to move by a bridge every 2 seconds?

- (A). 0.5 Hz
- (B). 1/15 Hz
- (C). 2 Hz
- (D). 5 Hz

Question #16: What is the frequency of water waves if a new crest is observed to move by a bridge every 2 seconds?

(A). 0.5 Hz

One complete wave every two s means that $T = 2 \text{ s}$ since this is the time required for one cycle. The relationship between the period and frequency of a recurring cycle is described by the equation

$$T = \frac{1}{f}$$

where T is the period and f is the frequency. Thus the frequency of a vibrating system with a period of 2 s would be

$$\begin{aligned} T &= \frac{1}{f} \therefore f = \frac{1}{T} \\ &= \frac{1}{2 \text{ s}} \\ &= \frac{1}{2} \frac{1}{\text{s}} \\ &= 0.5 \frac{1}{\text{s}} \\ &= 0.5 \text{ Hz} \end{aligned}$$

Question #17: What is the velocity of sound in ft/s when the air temperature is 30.0° C?

- (A). 1087 ft/s
- (B). 1089 ft/s
- (C). 1119 ft/s
- (D). 1147 ft/s

Question #17: What is the velocity of sound in feet per second when the air temperature is 30.0° C?

(D). 1147 ft/s

$$\begin{aligned} v_0 &= 1087 \text{ ft/s} & v_{T_p} &= v_0 + \left(\frac{2.0 \text{ ft/s}}{^{\circ}\text{C}} \right) (T_p) \\ T_p &= 30.0^{\circ} \text{ C} & &= 1087 \text{ ft/s} + \left(\frac{2.0 \text{ ft/s}}{^{\circ}\text{C}} \right) (30^{\circ} \text{ C}) \\ v_{T_p} &= ? & &= 1087 + 2.0 \times 30.0 \text{ ft/s} + \left(\frac{2.0 \text{ ft/s}}{^{\circ}\text{C}} \right) (30^{\circ} \text{ C}) \\ & & &= 1087 + 60. \text{ ft/s} + \left(\frac{\text{ft/s}}{^{\circ}\text{C}} \right) (^{\circ}\text{C}) \\ & & &= 1147 \text{ ft/s} \end{aligned}$$

Question #18: An echo is heard exactly 1.00 s after a sound when the air temperature is 30.0° C. How many feet away is the reflecting surface?

- (A). 575 ft
- (B). 1150 ft
- (C). 1147 ft
- (D). 1087 ft

Question #19: If the air temperature is 90.00° F, how many seconds will elapse between seeing lightning and hearing the thunder if the lightning strikes one mile (5280.0 feet) away?

- (A). 1269 s
- (B). 4.162 s
- (C). 4.587 s
- (D). 4.858 s

Question #20: What is the frequency of a tuning fork that resonates with an air column in a 24.0 cm closed tube with an air temperature of 20.0° C?

- (A). 343 Hz
- (B). 357 Hz
- (C). 345 Hz

Question #18: An echo is heard exactly 1.00 s after a sound when the air temperature is 30.0° C. How many feet away is the reflecting surface?

(A) 575 ft

First, find the velocity of sound at 30°C:

$$\begin{aligned}
 v_0 &= 1087 \text{ ft/s} & v_{T_p} &= v_0 + \left(\frac{2.0 \text{ ft/s}}{^{\circ}\text{C}} \right) (T_p) \\
 T_p &= 30.0^{\circ} \text{ C} & &= 1087 \text{ ft/s} + \left(\frac{2.0 \text{ ft/s}}{^{\circ}\text{C}} \right) (30^{\circ} \text{ C}) \\
 v_{T_p} &= ? & &= 1087 + 2.0 \times 30.0 \text{ ft/s} + \left(\frac{2.0 \text{ ft/s}}{^{\circ}\text{C}} \right) (30^{\circ} \text{ C}) \\
 & & &= 1087 + 60. \text{ ft/s} + \left(\frac{\text{ft/s}}{^{\circ}\text{C}} \right) (^{\circ}\text{C}) \\
 & & &= 1147 \text{ ft/s}
 \end{aligned}$$

Now that you have the velocity and the time, you can find distance:

$$\begin{aligned}
 t &= 1.00 \text{ s} & v &= \frac{d}{t} \therefore d = vt \\
 v &= 1147 \text{ ft/s} & &= \left(1147 \frac{\text{ft}}{\text{s}} \right) (1.00 \text{ s}) \\
 d &= ? & &= 1147 \times 1.00 \frac{\text{ft}}{\text{s}} \times \text{s} \\
 & & &= 1147 \text{ ft} \\
 & & &= 1150 \text{ ft}
 \end{aligned}$$

Since the sound pulse must travel from the source $\frac{1150 \text{ ft}}{2} = \boxed{575 \text{ ft}}$ to the reflecting surface, then back to the source:

Question #19: If the air temperature is 90.00° F, how many seconds will elapse between seeing lightning and hearing the thunder if the lightning strikes one mile (5280.0 feet) away?

(C). 4.587 s

$$\begin{aligned}
 T_c &= \frac{5}{9} (T_f - 32^{\circ}) & v_{T_P} &= v_0 + \left(\frac{2.00 \text{ ft/s}}{^{\circ}\text{C}} \right) (T_P) \\
 &= \frac{5}{9} (90.00^{\circ} - 32^{\circ}) & &= 1087 \frac{\text{ft}}{\text{s}} + \left(\frac{2.00 \text{ ft/s}}{^{\circ}\text{C}} \right) (32^{\circ} \text{ C}) \\
 &= \frac{5}{9} (58^{\circ}) & &= 1087 \frac{\text{ft}}{\text{s}} + 64 \frac{\text{ft}}{\text{s}} \\
 &= \frac{290^{\circ}}{9} & &= 1151 \frac{\text{ft}}{\text{s}} \\
 &= 32^{\circ} \text{ C}
 \end{aligned}$$

$$\begin{aligned}
 v &= \frac{d}{t} \therefore t = \frac{d}{v} \\
 &= \frac{5280.0 \text{ ft}}{1151 \frac{\text{ft}}{\text{s}}} \\
 &= \frac{5280.0 \text{ ft}}{1151 \frac{\text{ft}}{\text{s}}} \times \frac{\text{s}}{\text{ft}} \\
 &= 4.587 \text{ s}
 \end{aligned}$$

Question #20: What is the frequency of a tuning fork that resonates with an air column in a 24.0 cm closed tube with an air temperature of 20.0° C?

(B). 357 Hz

(D). 329 Hz

The wavelength can be found from the relationship between the length of the air column and λ , specifically, a tube closed at one end must have a minimum of $1/4$ wavelength to have a node at the closed end and a resonant antinode at the open end. Thus, $\lambda = 4L = 4(0.240 \text{ m}) = 0.960 \text{ m}$. The velocity can be determined from the given air temperature:

$$\begin{aligned}v_{T_P} &= v_0 + \left(\frac{0.60 \text{ m/s}}{^\circ\text{C}}\right)(T_P) \\&= 331 \text{ m/s} + \left(\frac{0.60 \text{ m/s}}{^\circ\text{C}}\right)(20.0^\circ\text{C}) \\&= 331 \text{ m/s} + 12 \text{ m/s} \\&= 343 \text{ m/s}\end{aligned}$$

Now that we have the wavelength and the velocity we can use the wave equation to find the frequency:

$$\begin{aligned}v &= \lambda f \therefore f = \frac{v}{\lambda} \\&= \frac{343 \text{ m/s}}{0.960 \text{ m}} \\&= \frac{343}{0.960} \frac{\text{m}}{\text{s}} \times \frac{1}{\text{m}} \\&= 357 \frac{1}{\text{s}} \\&= \boxed{357 \text{ Hz}}\end{aligned}$$

Question #21: What are the fundamental frequency and the frequency of the first overtone of a 70.0 cm closed organ pipe?

- (A). 123/368 Hz
(B). 343/1029 Hz
(C). 118/355 Hz
(D). 123/369 Hz

Question #21: What are the fundamental frequency and the frequency of the first overtone of a 70.0 cm closed organ pipe?

(A). 123/368 Hz

The fundamental frequency can be found from the relationship

$$f_n = \frac{nv}{4L}$$

where $n = 1$ is the fundamental and $n = 3, 5, 7$, and so on are the possible harmonics. Assuming room temperature to be 20.0°C since you have to assume some temperature to find the velocity, the velocity of sound is 343 m/s . Thus for the fundamental frequency ($n = 1$):

$$\begin{aligned}f &= \frac{nv}{4L} \\&= \frac{(1)(343 \text{ m/s})}{(4)(0.700 \text{ m})} \\&= \frac{343}{2.80} \frac{\text{m}}{\text{s}} \times \frac{1}{\text{m}} \\&= 122.5 \frac{1}{\text{s}} \\&= 123 \text{ Hz}\end{aligned}$$

The first overtone above the fundamental ($n = 3$) is therefore:

$$\begin{aligned}f &= \frac{nv}{4L} \\&= \frac{(3)(343 \text{ m/s})}{(4)(0.700 \text{ m})} \\&= \frac{1029}{2.80} \frac{\text{m}}{\text{s}} \times \frac{1}{\text{m}} \\&= 367.5 \frac{1}{\text{s}} \\&= 368 \text{ Hz}\end{aligned}$$

Question #22: What is the fundamental frequency of a 0.5 m string if wave speed on the string is 400 m/s?

- (A). 400 Hz
(B). 331 Hz
(C). 200 Hz
(D). 131 Hz

Question #22: What is the fundamental frequency of a 0.5 m string if wave speed on the string is 400 m/s?

(A). 400 Hz

The fundamental frequency can be found from the relationship $f_n = \frac{nv}{2L}$, where $n = 1$ is the fundamental, and

$$\begin{aligned} f_n &= \frac{nv}{2L} \\ &= \frac{(1)(400 \text{ m/s})}{(2)(0.5 \text{ m})} \\ &= \frac{400}{1} \frac{\text{m}}{\text{s}} \times \frac{1}{\text{m}} \\ &= 400 \frac{1}{\text{s}} \\ &= 400 \text{ Hz} \end{aligned}$$

Question #23: A tuning fork is found to resonate with an air column closed at one end when the tube is 60.0 cm long. If the air is 20.0° C, the speed of sound in air is 343 m/s. What is the frequency of the tuning fork?

- (A). 143 Hz
- (B). 286 Hz.
- (C). 138 Hz
- (D). 572 Hz

Question #23: A tuning fork is found to resonate with an air column closed at one end when the tube is 60.0 cm long. If the air is 20.0° C, the speed of sound in air is 343 m/s. What is the frequency of the tuning fork?

(A). 143 Hz

The relationship between frequency, velocity, and wavelength of a sound formed in a closed tube is $f_n = \frac{nv}{4L}$, where $n = 1$ is the fundamental, and

$$\begin{aligned} f &= \frac{nv}{4L} \\ &= \frac{(1)(343 \text{ m/s})}{(4)(0.600 \text{ m})} \\ &= \frac{343}{2.40} \frac{\text{m}}{\text{s}} \times \frac{1}{\text{m}} \\ &= 143 \frac{1}{\text{s}} \\ &= 143 \text{ Hz} \end{aligned}$$

Electricity and Magnetism

Questions	Answers with Rationalization
Question #1: How does an electron acquire a negative charge? (A). From an imbalance of subatomic particles. (B). By induction or contact with charged objects. (C). From the friction of certain objects rubbing together. (D). Charge is a fundamental property of an electron.	Question #1: How does an electron acquire a negative charge? (D). Charge is a fundamental property of an electron. The negative charge on an electron is as fundamental to the electron as gravitational attraction is to masses. You cannot separate gravity from masses and you cannot separate charge from an electron.
Question #2: Which of the following is most likely to acquire an electrostatic charge? (A). Objects that are electrical conductors. (B). Objects that are electrical nonconductors. (C). Conductors and nonconductors are equally likely to acquire electrostatic charges. (D). Objects that are grounded.	Question #2: Which of the following is most likely to acquire an electrostatic charge? (B). Objects that are electrical nonconductors. Electric charges on objects result from the gain or loss of electrons. Electrons do not move easily through nonconductors so electrons can be added and the charge tends to remain, giving the object an electrostatic charge.
Question #3: A quantity of electric charge is measured in a unit called a (an) (A). coulomb. (B). volt. (C). watt. (D). amp.	Question #3: A quantity of electric charge is measured in a unit called a (an) (A). coulomb. The coulomb is the metric unit for measuring electric charge.
Question #4: The unit that describes the potential difference that occurs when a certain amount of work is used to move a certain quantity of charge is called a (an) (A). ohm. (B). volt. (C). amp. (D). watt.	Question #4: The unit that describes the potential difference that occurs when a certain amount of work is used to move a certain quantity of charge is called a (an) (B) volt. In electrical matters, the potential difference that is created by doing one joule of work in moving one coulomb of charge is defined to be one volt. The volt is a measure of potential difference between two points.
Question #5: Which of the following units are measures of time rates? (A). Amp and volt. (B). Coulomb and joule. (C). Volt and watt. (D). Amp and watt.	Question #5: Which of the following units are measures of time rates? (D) Amp and watt. An amp is a measure of the rate at which an electric current flows, measured in the quantity of charge that moves through a conductor in a unit of time, or coulombs/second. A watt is a measure of the rate of using energy, measured in the quantity of energy that is used in a unit of time, or joules/second.
Question #6: An electric current is measured in units of (A). coulomb (B). volts.	Question #6: An electric current is measured in units of (C). amps. An amp is a measure of the rate at which an electric current flows,

(C). amps. (D). watts. Question #7: In an electric current the electrons are moving (A). at a very slow rate. (B). at near the speed of light. (C). faster than the speed of light. (D). at a subspace speed known as "Warp 8." Question #8: If you multiply amps times volts the answer will be in units of (A). resistance. (B). work. (C). current. (D). power. Question #9: Which of the following is a measure of electrical work? (A). kW (B). C (C). kWh (D). C/s Question #10: If an electric charge is somehow suddenly neutralized, the electric field that surrounds it will (A). immediately cease to exist. (B). collapse inward at the speed of light. (C). continue to exist until neutralized. (D). move off into space until it finds another charge. Question #11: If you reverse the direction of an electric current in a wire the magnetic field around the wire will (A). be oriented as it was before. (B). become oriented with an opposite north direction. (C). flip to become aligned parallel to the length of the wire. (D). cease to exist. Question #12: A step-up transformer steps up the (A). power. (B). current. (C). voltage. (D). all of these.	measured in the quantity of charge that moves through a conductor in a unit of time, or coulombs/second. Question #7: In an electric current the electrons are moving (A) at a very slow rate. In a direct current the electrons are moved along a conductor slowly as a result of the imposed electric field. The electric field, on the other hand, travels through through conductor at nearly the speed of light. Question #8: If you multiply amps times volts the answer will be in units of (D). power. An amp is coulombs/second and a volt is joules/coulomb. When multiplied, this gives joules/second. A joule/second is known as a watt, a unit of power. Question #9: Which of the following is a measure of electrical work? (C) kWh Since Power = Work/time, then Work = Power times time. A Watt-hour (or kilowatt-hour) is therefor a unit of work. The electric utility measures and charges for usage in kilowatt-hours because they need to know how much work you have done with their power. Question #10: If an electric charge is somehow suddenly neutralized, the electric field that surrounds it will (B). collapse inward at the speed of light. This can be observed by noting what happens to the magnetic field around the primary coil of a direct current transformer. When a direct current is applied to the primary coil a magnetic field grows around the primary and then becomes stationary. An induced voltage occurs in the secondary only for an instant when the magnetic field moves across the secondary coil. The current is induced in the secondary again, but in the opposite direction, when the current in the primary is turned off and the field collapses back across the secondary. Question #11: If you reverse the direction of an electric current in a wire the magnetic field around the wire will (B). become oriented with an opposite north direction. Magnetic field lines are created around a current carrying wire. These lines are closed concentric circles that are at right angles to the length of the wire. The direction of the magnetic field is determined by the direction of the current. Question #12: A step-up transformer steps up the (C). voltage. A transformer steps up or steps down the voltage of an alternating current and the ratio of input and output voltage is determined by the number of loops in the primary and secondary coils. Increasing the voltage decreases the current and assuming no losses, the power input will equal the power output.
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Question #13: What current would flow between two points in an electric circuit with a potential difference of 120 V and a resistance of 30 ohm?

- (A). 3 A
- (B). 4 A
- (C). 40 A
- (D). 80 A

Question #14: A 1300 watt hair blow dryer is designed to operate on 120 volts. How much electrical current does the dryer require?

- (A). 10.83 A
- (B). 11 A
- (C). 1180 A
- (D). 156 A

Question #15: A step-up transformer has 5 loops on its primary coil and 20 loops on its secondary coil. If the primary coil is supplied with an alternating current at 120 volts, what is the voltage in the secondary?

- (A). 4 V
- (B). 30 V
- (C). 120 V
- (D). 480 V

Question #13: What current would flow between two points in an electric circuit with a potential difference of 120 V and a resistance of 30 ohm?

(B). 4 A

Voltage (V)	120 V
Current (I)	? A
Resistance (R)	30 Ω

Ohm's law describes a relationship between these three variables and Ohm's law is $V = IR$ where V is the voltage, I is the current, and R is the resistance.

$$\begin{aligned}
 V &= IR \quad \therefore I = \frac{V}{R} \\
 &= \frac{120 \text{ V}}{30 \frac{\text{V}}{\text{A}}} \\
 &= \frac{120 \text{ V}}{30} \frac{\text{V}}{1} \times \frac{\text{A}}{\text{V}} \\
 &= 4 \text{ A}
 \end{aligned}$$

Question #14: A 1300 watt hair blow dryer is designed to operate on 120 volts. How much electrical current does the dryer require?

(B). 11 A

Power (P)	1300 W
Potential (V)	120 V
Current (I)	? A

The equation $P = IV$ tells you a relationship between power, amps, and volts where I is the amp of a current, V is the voltage, and P is the power of that current. Thus

$$\begin{aligned}
 P &= IV \quad \therefore I = \frac{P}{V} \\
 &= \frac{1300 \frac{\text{J}}{\text{s}}}{120 \frac{\text{J}}{\text{C}}} \\
 &= \frac{1300}{120} \frac{\text{J}}{\text{s}} \times \frac{\text{C}}{\text{J}} \\
 &= 10.8 \frac{\text{C}}{\text{s}} \\
 &= 11 \text{ A}
 \end{aligned}$$

Question #15: A step-up transformer has 5 loops on its primary coil and 20 loops on its secondary coil. If the primary coil is supplied with an alternating current at 120 volts, what is the voltage in the secondary?

(D). 480 V

Volts on the primary (VP)	120 V
Number of loops on the primary (NP)	5 loops
Volts on the secondary (VS)	? V
Number of loops on the secondary (NS)	20 loops

The voltage created in a length of wire in the armature of a generator can be found from volts per wire loop $\frac{V_p}{N_p} = \frac{V_s}{N_s}$ where is VP is the volts on the primary, NP is the number of loops on the primary coil, VS is the volts on the secondary, and NS is the number of loops on the secondary coil.

$$\begin{aligned}\frac{V_p}{N_p} &= \frac{V_s}{N_s} \quad \therefore V_s = \frac{V_p N_s}{N_p} \\ &= \frac{(120 \text{ V})(20 \text{ loops})}{5 \text{ loops}} \\ &= \frac{120 \times 20}{5} \quad \frac{\text{V} \cdot \text{loops}}{\text{loops}} \\ &= 480 \text{ V}\end{aligned}$$

Question #16: A small radio operates on 3.00 V and has a resistance of 15.0 ohm. At what rate does the radio use electric energy?

- (A). 0.200 A
- (B). 5.00 A
- (C). 0.600 W
- (D). 45.0 W

Question #16: A small radio operates on 3.00 V and has a resistance of 15.0 ohm. At what rate does the radio use electric energy?

(C). 0.600 W

Note that there are two separate electrical units which are rates: (1) the amp (coulomb/s), and (2) the watt (joule/s). The question asked for a rate of using energy. Energy is measured in joules, so you are looking for the power of the radio in watts. To find watts ($P = IV$), you will need to calculate the current (I) since it is not given. The current can be obtained from the relationship of Ohm's law:

$$\begin{aligned}I &= \frac{V}{R} \\ &= \frac{3.00 \text{ V}}{15.0 \frac{\text{V}}{\text{A}}} \\ &= 0.200 \text{ A}\end{aligned}$$

$$\begin{aligned}P &= IV \\ &= (0.200 \text{ C/s})(3.00 \text{ J/C}) \\ &= \boxed{0.600 \text{ watt}}\end{aligned}$$

Light

Questions	Answers with Rationalization
Question #1: An image that does not have light rays coming from the image, but is the result of your brain's interpretations of light rays is called a (A). real image. (B). imagined image. (C). virtual image. (D). phony image.	Question #1: An image that does not have light rays coming from the image, but is the result of your brain's interpretations of light rays is called a (C). virtual image. A virtual image is the result of your eyes' and brain's interpretations of light rays, not actual light rays originating from an image. Light rays do originate from a real image, such as the one on a movie screen.
Question #2: Light traveling at some angle as it moves from water into the air is refracted away from the normal as it enters the air, so the fish you see under water is actually (A). above the refracted image. (B). below the refracted image. (C). beside the refracted image. (D). in the same place as the refracted image.	Question #2: Light traveling at some angle as it moves from water into the air is refracted away from the normal as it enters the air, so the fish you see under water is actually (B). below the refracted image. A light ray moving from water to air will be refracted away from the normal.
Question #3: When viewed straight down (90° to the surface), a fish under water is (A). above the image (away from you). (B). below the image (closer to you). (C). beside the image. (D). in the same place as the image.	Question #3: When viewed straight down (90° to the surface), a fish under water is (D). in the same place as the image. An incident ray that is perpendicular to the surface is not refracted at all. However, the fish will appear larger than it really is since other light rays will be refracted away from the normal.
Question #4: Any part of the electromagnetic spectrum, including the colors of visible light, can be measured in units of (A). wavelength. (B). frequency. (C). energy. (D). any of these.	Question #4: Any part of the electromagnetic spectrum, including the colors of visible light, can be measured in units of (D). any of these.
Question #5: The ratio of the speed of light in a vacuum to the speed of light in some transparent materials is called (A). the critical angle. (B). total internal reflection. (C). the law of reflection. (D). the index of refraction.	Question #5: The ratio of the speed of light in a vacuum to the speed of light in some transparent materials is called (D). the index of refraction. This is a definition of the index of refraction.
Question #6: Which characteristic of waves of visible light is most similar to waves of microwaves?	Question #6: Which characteristic of waves of visible light is most similar to waves of microwaves?

(A). speed (B). wavelength (C). frequency (D). amplitude Question #7: The shortest of all the radio waves is the (A). FM broadcast (B). AM broadcast (C). microwave (D). shortwave Question #8: Which of the following can only be explained by a wave model of light? (A). reflection (B). refraction (C). interference (D). photoelectric effect Question #9: The sky appears to be blue when the sun is high in the sky because (A). blue is the color of air, water, and other fluids in large amounts. (B). red light is scattered more than blue light. (C). blue light is scattered more than the other colors. (D). none of the above Question #10: Max Planck found wave theory-shattering evidence that light of a certain color emitted from vibrating molecules would have (A). a continuous range of energy amounts. (B). a variable and unpredictable energy amount. (C). a fixed amount of energy. (D). no energy at all. Question #11: When you see a rainbow, you know for sure that (A). the sun must be directly, or almost directly overhead. (B). that rain is falling between where you are and where the sun is located at the time. (C). you are between the sun in one direction and a rainshower in the other direction. (D). that the sun, you, and the location of the rainfall must be at right angles.	(A). speed Question #7: The shortest of all the radio waves is the (C). microwave. Waves shorter than microwaves fall in the zone of infrared radiation. Question #8: Which of the following can only be explained by a wave model of light? (C). interference Question #9: The sky appears to be blue when the sun is high in the sky because (C). blue light is scattered more than the other colors. Question #10: Max Planck found wave theory-shattering evidence that light of a certain color emitted from vibrating molecules would have (C). a fixed amount of energy. Question #11: When you see a rainbow, you know for sure that (C). you are between the sun in one direction and a rainshower in the other direction.
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Question #12: An optically formed duplicate of an object that has light actually coming from the duplicate is
 (A). a real image.
 (B). found only in science fiction.
 (C). possible only if a laser is used.
 (D). a virtual image.

Question #13: If you want to see your whole image in a mirror, from head to foot, you will need a flat mirror that is
 (A). the same height as you.
 (B). at least half as tall as you are.
 (C). is twice your height.
 (D). the same height as a door.

Question #14: The virtual image seen in a flat mirror is
 (A). inside-out.
 (B). up-side-down.
 (C). left and right reversed.
 (D). all of these.

Question #15: A slide projector uses a convex lens system, with the film between one and two focal lengths of the lens. Why is the image not inverted?
 (A). It is upright because it is a real image.
 (B). It is upright because the film is put in upside down.
 (C). It is upright because it is a virtual image.
 (D). None of the above.

Question #16: If the Sun is 1.50×10^8 km from the Earth, how many minutes are required for sunlight to reach the Earth?
 (A). 0.0 min
 (B). 1,440 min
 (C). 500.0 min
 (D). 8.3 min

Question #12: An optically formed duplicate of an object that has light actually coming from the duplicate is
 (A). a real image.

Question #13: If you want to see your whole image in a mirror, from head to foot, you will need a flat mirror that is
 (B). at least half as tall as you are.

Question #14: The virtual image seen in a flat mirror is
 (C). left and right reversed.

Question #15: A slide projector uses a convex lens system, with the film between one and two focal lengths of the lens. Why is the image not inverted?
 (B). It is upright because the film is put in upside down.

Question #16: If the Sun is 1.50×10^8 km from the Earth, how many minutes are required for sunlight to reach the Earth?
 (D). 8.3 min

Listing the known and unknown quantities:

$$\begin{aligned} d &= 1.50 \times 10^8 \text{ km} \\ &= 1.50 \times 10^{11} \text{ m} \\ c &= 3.00 \times 10^8 \text{ m / s} \\ t &= ? \end{aligned}$$

and thus:

$$\begin{aligned} v &= \frac{d}{t} \quad \therefore \quad t = \frac{d}{v} \\ &= \frac{1.50 \times 10^{11} \text{ m}}{3.00 \times 10^8 \frac{\text{m}}{\text{s}}} &= \frac{5.00 \times 10^2 \text{ s}}{60.0 \frac{\text{s}}{\text{min}}} \\ &= \frac{1.50 \times 10^{11}}{3.00 \times 10^8} \frac{\text{m}}{1} \times \frac{\text{s}}{\text{m}} &= \frac{5.00 \times 10^2}{60.0} \frac{\text{s}}{1} \times \frac{\text{s}}{\text{min}} \\ &= 5.00 \times 10^2 \text{ s} &= 8.33 \text{ min} \end{aligned}$$

Question #17: How many hours are required for a radio signal from a space probe near the planet Pluto, 6.00×10^9 km away to reach the Earth?

- (A). 0.134 hr
- (B). 8.33 min
- (C). 5.56 hr
- (D). 0.00 s

Question #18: A right ray is reflected from a mirror with an angle 10° to the normal. What was the angle of incidence?

- (A). 80°
- (B). 10°
- (C). 35°
- (D). 170°

Question #19: Light travels through a transparent substance at 2.20×10^8 m/s. What is the substance?

- (A). ethyl alcohol
- (B). water
- (C). a vacuum
- (D). solid ice

Question #20: What is the speed of light in a diamond?

- (A). 3.020×10^8 m/s
- (B). 7.26×10^8 m/s
- (C). 1.24×10^8 m/s
- (D). none of the above.

Question #17: How many hours are required for a radio signal from a space probe near the planet Pluto, 6.00×10^9 km away to reach the Earth?

(C). 5.56 hr

Listing the known and unknown quantities:

$$\begin{aligned}d &= 6.00 \times 10^9 \text{ km} \\&= 6.00 \times 10^{12} \text{ m} \\c &= 3.00 \times 10^8 \text{ m/s} \\t &= ?\end{aligned}$$

and thus:

$$\begin{aligned}v &= \frac{d}{t} \quad \therefore t = \frac{d}{v} \\&= \frac{6.00 \times 10^{12} \text{ m}}{3.00 \times 10^8 \frac{\text{m}}{\text{s}}} &= \frac{2.00 \times 10^4 \text{ s}}{3600.0 \frac{\text{s}}{\text{h}}} \\&= \frac{6.00 \times 10^{12}}{3.00 \times 10^8} \frac{\text{m}}{1} \times \frac{\text{s}}{\text{m}} &= \frac{2.00 \times 10^4}{3.600 \times 10^3} \frac{\text{s}}{1} \times \frac{\text{h}}{\text{s}} \\&= 2.00 \times 10^4 \text{ s} &= 5.56 \text{ h}\end{aligned}$$

Question #18: A right ray is reflected from a mirror with an angle 10° to the normal. What was the angle of incidence?

(B). 10°

The law of Reflection is $\theta_i = \theta_r$, where θ_i is the angle between the incident ray and the normal and θ_r is the angle between the reflected ray and the normal. Thus $\theta_i = \theta_r$, so $\theta_i = 10^\circ$.

Question #19: Light travels through a transparent substance at 2.20×10^8 m/s. What is the substance?

(A). ethyl alcohol

Listing the known and unknown quantities:

$$\begin{aligned}v &= 2.20 \times 10^8 \text{ m/s} \\c &= 3.00 \times 10^8 \text{ m/s} \\n &= ?\end{aligned} \quad \begin{aligned}n &= \frac{c}{v} \\&= \frac{3.00 \times 10^8 \frac{\text{m}}{\text{s}}}{2.20 \times 10^8 \frac{\text{m}}{\text{s}}} \\&= 1.36\end{aligned}$$

According to reference table, a substance with an index of refraction of 1.36 is ethyl alcohol.

Question #20: What is the speed of light in a diamond?

(C). 1.24×10^8 m/s

The index of refraction is found from $n = \frac{c}{v}$, where n is the index of refraction of a transparent material, c is the speed of light in a vacuum, and v is the speed of light in the material ($c = 3.00 \times 10^8$ m/s). The index of refraction of a diamond is found in a reference table ($n = 2.42$).

$$\begin{aligned} n &= 2.42 & n &= \frac{c}{v} \quad \therefore v = \frac{c}{n} \\ c &= 3.00 \times 10^8 \text{ m/s} & &= \frac{3.00 \times 10^8 \frac{\text{m}}{\text{s}}}{2.42} \\ v &= ? & &= 1.24 \times 10^8 \frac{\text{m}}{\text{s}} \end{aligned}$$

Question #21: What is the energy of a photon of red light with a frequency of 4.00×10^{14} Hz?

- (A). 1.66×10^{-48} J
- (B). 2.65×10^{-19} J
- (C). 6.63×10^{-34} J
- (D). 4.00×10^{14} J

Question #21: What is the energy of a photon of red light with a frequency of 4.00×10^{14} Hz?

(B). 2.65×10^{-19} J

Listing the known and unknown quantities:

Frequency	$f = 4.00 \times 10^{14}$ Hz
Planck's constant	$h = 6.63 \times 10^{-34}$ Js
Energy	$E = ?$

The relationship between the frequency (f) and energy (E) of a photon is found in equation, $E = hf$.

$$\begin{aligned} E &= hf \\ &= (6.63 \times 10^{-34} \text{ J} \cdot \text{s}) \left(4.00 \times 10^{14} \frac{1}{\text{s}} \right) \\ &= (6.63 \times 10^{-34}) (4.00 \times 10^{14}) \text{ J} \cdot \text{s} \times \frac{1}{\text{s}} \\ &= 2.65 \times 10^{-19} \frac{\text{J} \cdot \text{s}}{\text{s}} \\ &= 2.65 \times 10^{-19} \text{ J} \end{aligned}$$

Question #22: A light wave has a frequency of 4.90×10^{14} Hz. What color would you observe?

- (A). orange
- (B). blue
- (C). green
- (D). yellow

Question #22: A light wave has a frequency of 4.90×10^{14} Hz. What color would you observe?

(A). orange

Listing the known and unknown quantities:

Frequency	$f = 4.90 \times 10^{14} \text{ Hz}$
Speed of light	$c = 3.00 \times 10^8 \text{ m/s}$
Wavelength	$\lambda = ?$

First, you can find the color of a particular frequency of light by using the relationship between the wavelength (λ), frequency (f), and speed of light in a vacuum (c), $c = \lambda f$ to find the wavelength.

$$\begin{aligned}
 \text{(A)} \quad c &= \lambda f, \therefore \lambda = \frac{c}{f} \\
 &= \frac{3.00 \times 10^8 \frac{\text{m}}{\text{s}}}{4.90 \times 10^{14} \frac{1}{\text{s}}} \\
 &= \frac{3.00 \times 10^8 \frac{\text{m}}{\text{s}}}{4.90 \times 10^{14} \frac{1}{\text{s}}} \times \frac{\text{s}}{1} \\
 &= 6.12 \times 10^{-7} \text{ m}
 \end{aligned}$$

(B) According to a reference table, this is the wavelength of a photon of orange light.

Question #23: What is the energy of a microwave photon of wavelength 1.00 mm?

- (A). $3.00 \times 10^{11} \text{ J}$
 (B). $1.99 \times 10^{-22} \text{ J}$
 (C). $2.21 \times 10^{-45} \text{ J}$
 (D). 1.00 J

Question #23: What is the energy of a microwave photon of wavelength 1.00 mm?

(B). $1.99 \times 10^{-22} \text{ J}$

Step 1: The relationship between the wavelength (λ), frequency (f), and speed of light in a vacuum (c), is found in equation, $c = \lambda f$. Step 2: The relationship between the frequency (f) and energy (E) of a photon is found in equation, $E = hf$.

(First step)

$$\begin{aligned}
 c &= \lambda f, \therefore f = \frac{c}{\lambda} \\
 &= \frac{3.00 \times 10^8 \frac{\text{m}}{\text{s}}}{1.00 \times 10^{-3} \text{ m}} \\
 &= \frac{3.00 \times 10^8 \frac{\text{m}}{\text{s}}}{1.00 \times 10^{-3} \frac{\text{m}}{\text{s}}} \times \frac{1}{\text{m}} \\
 &= 3.00 \times 10^{11} \frac{1}{\text{s}} \\
 &= 3.00 \times 10^{11} \text{ Hz}
 \end{aligned}$$

(Second step)

$$\begin{aligned}
 E &= hf \\
 &= (6.63 \times 10^{-34} \text{ J} \cdot \text{s}) \left(3.00 \times 10^{11} \frac{1}{\text{s}} \right) \\
 &= (6.63 \times 10^{-34}) (3.00 \times 10^{11}) \text{ J} \cdot \text{s} \times \frac{1}{\text{s}} \\
 &= 1.99 \times 10^{-22} \frac{\text{J} \cdot \text{s}}{\text{s}} \\
 &= 1.99 \times 10^{-22} \text{ J}
 \end{aligned}$$

Atomic Structure

Questions	Answers with Rationalization
Question #1: Which of the following cannot be broken down to anything simpler? (A). water (B). table salt (C). silver (D). sugar	Question #1: Which of the following cannot be broken down to anything simpler? (C). silver. Water can be broken down to oxygen and hydrogen, table salt can be broken down to sodium and chlorine, and sugar can be broken down to carbon, hydrogen, and oxygen. Silver cannot be broken down to anything simpler because it is an element, as are oxygen, hydrogen, sodium, chlorine, and carbon.
Question #2: Elements combine in fixed mass ratios to form compounds. This must mean that elements (A). are made up of continuous matter without subunits. (B). are composed of discrete units called atoms. (C). have unambiguous atomic numbers. (D). are always chemically active.	Question #2: Elements combine in fixed mass ratios to form compounds. This must mean that elements (B). are composed of discrete units called atoms. If matter were continuous there would be no reason for one amount to combine with another amount. Matter is made up of discrete units called atoms that combine in a fixed weight ratios.
Question #3: The first part of an atom to be discovered was the (A). proton. (B). neutron. (C). electron. (D). nucleus.	Question #3: The first part of an atom to be discovered was the (C). electron. The English physicist J. J. Thomson discovered the electron, a fundamental part of an atom, in 1897.
Question #4: The electron was discovered through experiments with (A). electricity. (B). light. (C). radio waves. (D). radioactivity.	Question #4: The electron was discovered through experiments with (A). electricity. The English physicist J. J. Thomson was working with a high-voltage electrical source connected to two metal plates in an evacuated glass tube when a greenish beam was observed to move from the cathode to the anode. Working with this beam between charged metal plates and a strong magnetic field, Thomson found it was made of negatively charged particles, or electrons.
Question #5: The nucleus was discovered through experiments with (A). electricity. (B). light. (C). radio waves. (D). radioactivity.	Question #5: The nucleus was discovered through experiments with (D). radioactivity. Rutherford and his co-workers studied alpha particle scattering from a thin gold foil. The alpha particles struck a detecting screen, producing a flash of visible light. Measurements of the angles between the flashes, the foil, and the radioactive source of the alpha particles showed that the particles were scattered in all directions, including straight back from the foil. These measurements gave Rutherford a means of estimating the size of the nucleus.
Question #6: The planetary model of an atom, with the nucleus playing the role of the Sun and the electrons playing the	Question #6: The planetary model of an atom, with the nucleus playing the role of the Sun and the electrons playing the role of planets, is unacceptable because (B). an electron is accelerating and would lose energy. According to understandings about the relationship between charged

role of planets, is unacceptable because

- (A). the electrical attraction between a proton and an electron is too weak.
- (B). an electron is accelerating and would lose energy.
- (C). the nuclear attraction between a proton and an electron is too strong.
- (D). none of these because the planetary model is acceptable.

Question #7: Most of the volume of an atom is occupied by

- (A). electrons.
- (B). protons.
- (C). neutrons.
- (D). empty space

Question #8: The atomic number of an atom identifies the number of

- (A). protons.
- (B). neutrons.
- (C). quantum orbits.
- (D). excited states.

Question #9: The Bohr model of the atom was able to explain the Balmer series because

- (A). larger orbits required electrons to have more negative energy in order to match the angular momentum.
- (B). differences between the energy levels of the orbits matched the difference between energy levels of the line spectra.
- (C). electrons were allowed to exist only in allowed orbits and nowhere else.
- (D). none of the above

Question #10: The idea of matter waves, as reasoned by de Broglie, describes a wavelike behavior of any

- (A). particle, moving or not.
- (B). particle that is moving.
- (C). charged particle that is moving.
- (D). particle that is stationary.

particles and electromagnetic radiation, an accelerating electric charge should emit electromagnetic radiation such as light. If an electron gave off light, it would lose energy. The energy loss would mean that the electron could not maintain its orbit, and would be pulled into the nucleus and the atom would collapse. Therefore the planetary model of an atom is unacceptable because atoms continue to exist.

Question #7: Most of the volume of an atom is occupied by

(D). empty space

Rutherford was able to estimate the radius of the nucleus from his experiments with alpha particle scattering. The radius of the nucleus was found to be approximately 10^{-13} cm. Since the radius of the atom was found to be on the order of 10^{-8} cm, this means the electrons are moving around the nucleus at a distance 100,000 times the radius of the nucleus, meaning the volume of an atom is mostly empty space.

Question #8: The atomic number of an atom identifies the number of

(A). protons.

The atomic number identifies the number of protons in the nucleus of an atom. A neutral atom also has negatively charged electrons that are equal in number to the protons.

Question #9: The Bohr model of the atom was able to explain the Balmer series because

(B). differences between the energy levels of the orbits matched the difference between energy levels of the line spectra.

Question #10: The idea of matter waves, as reasoned by de Broglie, describes a wavelike behavior of any

(B). particle that is moving.

Question #11: According to the equation de Broglie derived to describe matter waves, doubling the velocity of an electron would result in
(A). less momentum.
(B). a greater mass.
(C). a smaller wavelength.
(D). an unchanged wavelength and mass.

Question #12: A hydrogen atom has an electron in the sixth excited state so the principal quantum number of this electron is
(A). 7.
(B). 6.
(C). 5.
(D). 4.

Question #13: One reason the Bohr model of the atom failed was because it did not explain why
(A). accelerating electrons do not emit electromagnetic radiation.
(B). moving electrons have a greater mass.
(C). electrons in the orbits of an atom have negative energies.
(D). electrons in greater orbits of an atom have greater velocities.

Question #14: A hydrogen atom is in a ground state when its electron
(A). has moved away from the atom to the ground.
(B). has stopped moving.
(C). is moving back and forth through the nucleus.
(D). remains at the lowest energy level.

Question #15: An atom of hydrogen emits a photon when its electron
(A). jumps from a lower-energy orbit to a higher-energy orbit.
(B). jumps from a higher-energy orbit to a lower-energy orbit.
(C). combines with a proton.
(D). combines with a neutron.

Question #11: According to the equation de Broglie derived to describe matter waves, doubling the velocity of an electron would result in
(C). a smaller wavelength.

Question #12: A hydrogen atom has an electron in the sixth excited state so the principal quantum number of this electron is
(A). 7.

Question #13: One reason the Bohr model of the atom failed was because it did not explain why
(A). accelerating electrons do not emit electromagnetic radiation.

Question #14: A hydrogen atom is in a ground state when its electron
(D). remains at the lowest energy level.

Question #15: An atom of hydrogen emits a photon when its electron
(B). jumps from a higher-energy orbit to a lower-energy orbit.

Question #16: What is the energy of a photon of black light (ultraviolet) that has a frequency of 2.00×10^{16} Hz?

- (A). 1.33×10^{-17} J
(B). 13.26×10^{18} J
(C). 1.33×10^{51} J
(D). 3.05×10^{-19} J

Question #17: What is the de Broglie wavelength of an electron with a velocity of 2.00×10^7 m/s?

- (A). 1.82×10^{-23} m
(B). 1.21×10^{-10} m
(C). 3.64×10^{-11} m
(D). 3.32×10^{-41} m

Question #18: What is the frequency of a photon emitted when an electron in a hydrogen atom jumps from $n = 3$ to $n = 2$?

- (A). 3.02×10^{-19} Hz
(B). 4.56×10^{14} Hz
(C). 7.29×10^{14} Hz
(D). 6.2×10^{14} Hz

Question #19: What is the electron configuration for potassium (atomic number 19)?

- (A). $1s^2 2p^6 3s^4 3p^6 4s^2$
(B). $1s^4 2p^6 3s^2 3p^6 4s^2$
(C). $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
(D). $1s^2 2s^2 2p^6 3s^2 3p^8$

Question #16: What is the energy of a photon of black light (ultraviolet) that has a frequency of 2.00×10^{16} Hz?

- (A). 1.33×10^{-17} J

$$\begin{aligned} E &= hf \\ &= (6.63 \times 10^{-34} \text{ Js}) \left(2.00 \times 10^{16} \frac{1}{\text{s}} \right) \\ &= (6.63 \times 10^{-34}) (2.00 \times 10^{16}) \text{ Js} \times \frac{1}{\text{s}} \\ &= 1.33 \times 10^{-17} \text{ J} \end{aligned}$$

Question #17: What is the de Broglie wavelength of an electron with a velocity of 2.00×10^7 m/s?

- (C). 3.64×10^{-11} m

$$\begin{aligned} \lambda &= \frac{h}{mv} \\ &= \frac{6.63 \times 10^{-34} \text{ Js}}{(9.11 \times 10^{-31} \text{ kg}) \left(2.00 \times 10^7 \frac{\text{m}}{\text{s}} \right)} \\ &= \frac{6.63 \times 10^{-34} \text{ Js}}{1.82 \times 10^{-23} \frac{\text{kg m}}{\text{s}}} \\ &= 3.64 \times 10^{-11} \frac{\frac{\text{kg m}^2}{\text{s}^2} \cdot \text{s}}{\frac{\text{kg m}}{\text{s}}} \\ &= 3.64 \times 10^{-11} \text{ m} \end{aligned}$$

Question #18: What is the frequency of a photon emitted when an electron in a hydrogen atom jumps from $n = 3$ to $n = 2$?

- (B). 4.56×10^{14} Hz

$$\text{For } n = 3, E_H = -2.42 \times 10^{-19} \text{ J}$$

$$\text{For } n = 2, E_L = -5.44 \times 10^{-19} \text{ J}$$

$$\begin{aligned} hf &= E_H - E_L \therefore f = \frac{E_H - E_L}{h} \\ &= \frac{(-2.42 \times 10^{-19} \text{ J}) - (-5.44 \times 10^{-19} \text{ J})}{6.63 \times 10^{-34} \text{ Js}} \\ &= \frac{3.02 \times 10^{-19} \text{ J}}{6.63 \times 10^{-34} \text{ Js}} \\ &= 4.56 \times 10^{14} \frac{1}{\text{s}} \\ &= 4.56 \times 10^{14} \text{ Hz} \end{aligned}$$

Question #19: What is the electron configuration for potassium (atomic number 19)?

- (C). $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

First, note that an atomic number of 19 means a total of nineteen electrons. Second, refer to the order of filling matrix in Figure 9.21. Remember that only two electrons can occupy an orbital, but there are three orientations of the p orbital, for a total of 6 electrons. Starting at the lowest energy level, 2 electrons go in 1s, making $1s^2$; then 2 go in 2s, making $2s^2$. That is a total of 4 electrons so far. Next $2p^6$, $3s^2$, and $3p^6$ use 14 more electrons for a total of 18 so far. The remaining electron goes into the next sublevel, $4s^1$ and all the electrons have been used, giving a configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$.

Question #20: What is the energy of a photon of red light with a frequency of 4.00×10^{14} Hz?

- (A). 1.66×10^{-48} J
- (B). 2.65×10^{-19} J
- (C). 6.63×10^{-34} J
- (D). 4.00×10^{14} J

Question #20: What is the energy of a photon of red light with a frequency of 4.00×10^{14} Hz?

(B). 2.65×10^{-19} J

Listing the known and unknown quantities:

Frequency $f = 4.00 \times 10^{14}$ Hz

Planck's constant $h = 6.63 \times 10^{-34}$ Js

Energy $E = ?$

The relationship between the frequency (f) and energy (E) of a photon is found in equation, $E = hf$.

$$E = hf$$

$$= (6.63 \times 10^{-34} \text{ J} \cdot \text{s}) \left(4.00 \times 10^{14} \frac{1}{\text{s}} \right)$$

$$= (6.63 \times 10^{-34}) (4.00 \times 10^{14}) \text{ J} \cdot \text{s} \times \frac{1}{\text{s}}$$

$$= 2.65 \times 10^{-19} \frac{\text{J} \cdot \text{s}}{\text{s}}$$

$$= 2.65 \times 10^{-19} \text{ J}$$

Elements and The Periodic Table

Questions	Answers with Rationalization
Question #1: Compared to metals, nonmetals are (A). more brittle as a solid. (B). better electrical conductors. (C). more ductile as a solid. (D). better conductors of heat.	Question #1: Compared to metals, nonmetals are (A). more brittle as a solid. Metals have physical properties of a metallic luster, high heat and electrical conductivity, malleability, and ductility. Nonmetals, on the other hand, has physical properties of not having a metallic luster, a poor conductor of heat and electricity, and when solid, is a brittle material that cannot be pounded or pulled into new shapes.
Question #2: Under ordinary, room temperature conditions, the greatest number of elements are (A). gases. (B). liquids. (C). metallic solids. (D). nonmetallic plasmas.	Question #2: Under ordinary, room temperature conditions, the greatest number of elements are (C). metallic solids. In general, elements located in the left two-thirds of the periodic table are metals, and the nonmetals are on the right side of the table. About 80 percent of all the elements are metals.
Question #3: A solution of sugar dissolved in water is (A). a heterogeneous mixture. (B). a homogeneous mixture. (C). an alloy. (D). a pure substance.	Question #3: A solution of sugar dissolved in water is (B). a homogeneous mixture. A mixture has unlike parts and a composition that varies from sample to sample. A homogeneous mixture is the same throughout but a heterogeneous mixture has physically distinct parts with different properties. A solution is a homogeneous mixture because it is the same throughout a given sample, but one solution can vary from the next. A solution of sugar in water does not have a fixed, definite composition. An alloy, by the way, is a homogeneous mixture of two or more metals, with the atoms of one metal replacing or occupying interstitial positions between the atoms of the other metal.
Question #4: Which of the following represents a physical change? (A). Electricity is used to generate oxygen and hydrogen from water. (B). Calcium carbonate is dissolved by stomach acid. (C). Solid ice is melted into liquid water. (D). Natural gas is burned as a heat source.	Question #4: Which of the following represents a physical change? (C). Solid ice is melted into liquid water. A physical change does not alter the identity of matter and involves physical properties only since no new substances are formed. Examples of physical changes include evaporation, condensation, melting, freezing, and dissolving, as well as crushing or bending. When ice melts into liquid water it changes to a different phase of matter but is still recognized as water. The other items are chemical changes, which alters the identity of matter, producing new substances with different properties.
Question #5: Carbon cannot be broken down into anything simpler by chemical means, so carbon must be (A). a heterogeneous mixture. (B). a homogeneous mixture. (C). an element. (D). a compound.	Question #5: Carbon cannot be broken down into anything simpler by chemical means, so carbon must be (C). an element. An element is a pure substance that cannot be broken down into anything simpler by chemical or physical means. A compound, on the other hand, is a pure substance that can be decomposed by a chemical change into simpler substances with a fixed mass ratio.
Question #6: How many naturally occurring elements are found on the earth in significant quantities?	Question #6: How many naturally occurring elements are found on the earth in significant quantities?

(A). 112 (B). 92 (C). 89 (D). 32 Question #7: About 75 percent of the earth's solid surface is made up of (A). silicon and oxygen. (B). nitrogen and oxygen. (C). hydrogen and oxygen. (D). hydrogen, oxygen, and carbon. Question #8: About 99 percent of the earth's atmospheric air is made up of (A). silicon and oxygen. (B). nitrogen and oxygen. (C). hydrogen and oxygen. (D). hydrogen, oxygen, and carbon. Question #9: Isotopes are atoms of an element with identical chemical properties but with different (A). numbers of protons. (B). masses. (C). numbers of electrons. (D). atomic numbers. Question #10: The masses of all isotopes are based on a comparison to the mass a particular isotope of (A). hydrogen. (B). carbon. (C). oxygen. (D). uranium. Question #11: The sum of the number of protons and neutrons in the nucleus of an atom is called the (A). atomic number. (B). atomic mass. (C). atomic weight.	(C). 89 The naturally-occurring elements generally range from hydrogen (atomic number 1) to uranium (atomic number 92) with the following exceptions: Technetium (#43) and promethium (#61) are not found anywhere on earth; francium (#87) and astatine (#85) undergo radioactive decay quickly, so they do not exist in any significant quantities; and, plutonium (#94) does occur naturally in small amounts. Thus, there are 92-4+1, or 89 naturally occurring elements that occur in significant quantities and 23 short-lived artificial elements for a total of 112 that are known at the present time. Question #7: About 75 percent of the earth's solid surface is made up of (A). silicon and oxygen. Only 8 elements make up about 99 percent of the solid surface of the earth. Oxygen is most abundant, making up about 50 percent of the weight of the earth's crust. Silicon makes up more than 20 percent, so these two nonmetals alone make up about 75 percent of the earth's solid surface. The other 6 elements, in order of abundance, are aluminum (8.1%), iron (5.0%), calcium (3.6%), sodium (2.8%), potassium (2.6%), and magnesium (2.1%). Question #8: About 99 percent of the earth's atmospheric air is made up of (B). nitrogen and oxygen. Air is mostly nitrogen (about 78%) and oxygen (about 21%), with traces of five other elements and compounds. Question #9: Isotopes are atoms of an element with identical chemical properties but with different (B). masses. The isotopes of a given element have the same number of protons, which identifies them as atoms of a particular element with certain chemical properties, but different number of neutrons, which makes a difference in the masses of the atoms. Question #10: The masses of all isotopes are based on a comparison to the mass a particular isotope of (B). carbon. The mass of all isotopes is based on a comparison to the mass of an atom of carbon-12, which is assigned a mass of exactly 12.00 units called atomic mass units (u). Question #11: The sum of the number of protons and neutrons in the nucleus of an atom is called the (D). mass number. Mass numbers are used to identify isotopes, and the number is the sum of the number of protons and neutrons in a nucleus. The mass number is also the closest whole number to the actual atomic mass of an isotope. Only carbon-12 has an atomic mass with a whole number, by definition. The mass number is usually written as a number following the
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(D). mass number. Question #12: The weighted average of the masses of the stable isotopes of an element as they occur in nature is called the (A). atomic number. (B). atomic mass. (C). atomic weight. (D). mass number. Question #13: The modern periodic law is based on (A). atomic number. (B). atomic mass. (C). atomic weight. (D). chemical activity. Question #14: Each family, or group of elements in a vertical column of the periodic table has elements with chemical characteristics that are (A). exactly the same. (B). similar. (C). different. (D). exactly opposite. Question #15: Which of the following belongs to the alkali metal family of elements? (A). sodium (B). calcium (C). chlorine (D). neon Question #16: Which of the following belongs to the halogen family of elements? (A). sodium (B). calcium (C). chlorine (D). neon Question #17: Which of the following belongs to the noble gas family of elements? (A). sodium (B). calcium (C). chlorine (D). neon	name of the element (as carbon-12) or as a superscript to the top left on the chemical symbol of that element. The atomic number is written as a subscript to the bottom left of that symbol. Question #12: The weighted average of the masses of the stable isotopes of an element as they occur in nature is called the (C). atomic weight. Atomic weight is the name given to the weighted average of the masses of stable isotopes of an element as they occur in nature. Question #13: The modern periodic law is based on (A). atomic number. The atomic number, that is, the number of protons in the nucleus and the number of electrons around the nucleus is the significant, essential basis for the modern periodic table. Question #14: Each family, or group of elements in a vertical column of the periodic table has elements with chemical characteristics that are (B). similar. A vertical column of elements is called a family (or group) of elements and elements in families have similar properties, but this is more true of some families than others. Question #15: Which of the following belongs to the alkali metal family? (A). sodium All of the elements in group IA have an outside electron configuration of one electron. With the exception of hydrogen, the IA elements are shiny, low-density metals that are so soft you can cut them easily with a knife. These IA metals are called the alkali metals because they react violently with water to form an alkaline solution. Question #16: Which of the following belongs to the halogen family of elements? (C). chlorine The elements in group VIIA have an outside configuration of seven electrons, needing only one more electron to completely fill the outer shell. These elements are called the halogens. Question #17: Which of the following belongs to the noble gas family of elements? (D). neon The elements in group VIIIA have orbitals that are filled to capacity in the outside shells. These elements are colorless, odorless gases that almost never react with other elements to form compounds. They are sometimes called the noble gases because they are chemically inert.
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Question #18: An atom of an element belonging to the alkali metal family has

- (A). one outer shell electron.
- (B). two outer shell electrons.
- (C). all outer shell electrons but one.
- (D). all outer shell electrons.

Question #19: An atom of an element belonging to the halogen family has

- (A). one outer shell electron.
- (B). two outer shell electrons.
- (C). all outer shell electrons but one.
- (D). all outer shell electrons.

Question #20: An atom of an element belonging to the noble gas family has

- (A). one outer shell electron.
- (B). two outer shell electrons.
- (C). all outer shell electrons but one.
- (D). all outer shell electrons.

Question #18: An atom of an element belonging to the alkali metal family has

- (A). one outer shell electron.

Question #19: An atom of an element belonging to the halogen family has

- (C). all outer shell electrons but one.

Question #20: An atom of an element belonging to the noble gas family has

- (D). all outer shell electrons.

Compounds and Chemical Change

Questions	Answers with Rationalization
Question #1: The smallest unit of an element that can exist alone or in combination with other elements is the (A). atom. (B). molecule. (C). ion. (D). electron.	Question #1: The smallest unit of an element that can exist alone or in combination with other elements is the (A). atom. An element is defined as a pure substance that cannot be broken down to anything simpler by chemical or chemical means. Atoms can be reasoned to exist from the way elements combine in fixed-weight ratios (see chapter 9). An atom can be considered smallest unit of an element that can exist alone or in combination with other elements.
Question #2: Ozone is a form of oxygen that is (A). monatomic. (B). diatomic. (C). triatomic. (D). tetratomic.	Question #2: Ozone is a form of oxygen that is (C). triatomic. Ozone is O₃, a molecule of three oxygen atoms bound together. Since "tri-" means "three," ozone is a triatomic molecule.
Question #3: Chemical energy is stored within molecules as (A). thermal energy. (B). internal activation energy. (C). internal potential energy. (D). kinetic energy of the nucleus.	Question #3: Chemical energy is stored within molecules as (C). internal potential energy. A chemical reaction is a change in matter in which different chemical substances are created by forming or breaking bonds. In general, bonds are formed when atoms of elements are bound together to form a new substance and chemical bonds are broken when a compound is decomposed into simpler substances. Energy may be absorbed when a new substance is formed and this energy is stored as internal potential energy. For example, green plants use the process of photosynthesis to make new substances. These new substances have internal potential energy that is released when these substances are burned. Changes in internal potential energy like this are called chemical energy.
Question #4: The outer electrons of an atom that participate in chemical bonding are called the ___?___ electrons. (A). reactant. (B). valence. (C). net. (D). product.	Question #4: The outer electrons of an atom that participate in chemical bonding are called the ___?___ electrons. (B). valence. The outer electrons that participate in chemical bonding are called valence electrons. The inner electrons are in stable, fully occupied orbitals and do not participate in chemical bonds.
Question #5: Which of the following represents the most stable outer orbital arrangement of electrons after a chemical reaction? (A). 1 electron. (B). 3 electrons. (C). 6 electrons. (D). 8 electrons.	Question #5: Which of the following represents the most stable outer orbital arrangement of electrons after a chemical reaction? (D). 8 electrons. The noble gases have filled outer orbitals and do not normally form compounds. Half-filled and filled orbitals are particularly stable arrangements and atoms have a tendency to seek such stable, filled orbitals. For the representative elements this tendency is called the octet rule (actually a generalization since there are exceptions). The octet rule states that atoms attempt to acquire an outer orbital with eight electrons through chemical reactions.
Question #6: The element sodium (Na) has how many valence electrons?	Question #6: The element sodium (Na) has how many valence electrons? (A). 1 The element sodium (symbol Na, atomic number 11) has an electron

(A). 1 (B). 2 (C). 7 (D). 8 Question #7: In a chemical reaction the element sodium (Na) will (A). lose an electron. (B). lose two electrons. (C). gain an electron. (D). neither gain nor lose electrons. Question #8: Ionic bonds are formed when (A). electrons are transferred from the excited to the ground state. (B). electrons are transferred from the ground state to the excited state. (C). electrons are transferred between atoms. (D). atoms share electrons. Question #9: Covalent bonds are formed when (A). electrons are transferred from the excited to the ground state. (B). electrons are transferred from the ground state to the excited state. (C). electrons are transferred between atoms. (D). atoms share electrons. Question #10: The number of atoms in a molecule of ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$, is (A). 4. (B). 10. (C). 14. (D). 15. Question #11: When the atoms of a metal combine chemically with the atoms of a nonmetal, the nonmetal ions tend to (A). become positive ions. (B). remain neutral. (C). become negative ions. (D). share electrons.	configuration of $1s^2 2s^2 2p^6 3s^1$ so it has one valence electron (the $3s^1$). This is true of all the elements in group IA. Question #7: In a chemical reaction the element sodium (Na) will (A). lose an electron. Sodium (symbol Na, atomic number 11) has an electron configuration of $1s^2 2s^2 2p^6 3s^1$. Following the octet rule, sodium will lose the $3s^1$ electron to form a stable, filled outer orbital arrangement (the two 2s electrons plus the six 2p electrons equals a total of eight). Question #8: Ionic bonds are formed when (C). electrons are transferred between atoms. An ionic bond is the chemical bond of electrostatic attraction between negative and positive ions. Ions are formed when electrons are transferred between atoms, so this results in an ionic bond. Question #9: Covalent bonds are formed when (D). atoms share electrons. A covalent bond is a chemical bond that forms when atoms share a pair of electrons. Question #10: The number of atoms in a molecule of ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$, is (D). 15. The parenthesis and subscript tells us that two polyatomic ammonium ions are present for a total of 10 atoms (8 hydrogen plus 2 nitrogen), along with one sulfate polyatomic ion for a total of 5 atoms (4 oxygen and 1 sulfur). So, there is a total of 10 plus 5, or 15 atoms in one molecule of ammonium sulfate. Question #11: When the atoms of a metal combine chemically with the atoms of a nonmetal, the nonmetal ions tend to (C). become negative ions. Nonmetals tend to become negative ions because they tend to add electrons in order to satisfy the octet rule.
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Question #12: A crystal grain of potassium chloride is made up of
(A). K^+ and Cl^- ions.
(B). K^+ and Cl^- ions.
(C). molecules of KCl.
(D). atoms of K and Cl.

Question #13: The correct symbol for a magnesium ion is
(A). Mg^+
(B). Mg^-
(C). Mg^{2+}
(D). Mg^{2-}

Question #14: The formula of a compound formed when aluminum (Al) combines with oxygen (O) is
(A). AlO
(B). Al_3O_2
(C). Al_2O_2
(D). Al_2O_3

Question #15: The sulfate ion is $(SO_4)^{2-}$ so the formula for calcium sulfate would be
(A). $CaSO_4$
(B). Ca_2SO_4
(C). $Ca_2(SO_4)_2$
(D). $Ca_2(SO_4)_3$

Question #16: The number of oxygen atoms in a molecule of copper(II) phosphate, $Cu_3(PO_4)_2$, is

Question #12: A crystal grain of potassium chloride is made up of (B). K^+ and Cl^- ions.
Potassium (K) is an alkali metal in family IA, and chlorine (Cl) is a halogen in family VIIA. Since potassium is a metal and chlorine a nonmetal, you would expect potassium and chlorine atoms to react, forming a compound with ionic bonds. The oppositely charged ions attract one another, and when pulled together, they form an ionic solid with the ions arranged in an orderly geometric structure. This results in a crystalline solid that is typical of salts such as potassium chloride.

Question #13: The correct symbol for a magnesium ion is (C). Mg^{2+}
Magnesium (Symbol Mg, atomic number 12) is an alkaline metal in family IIA and has an electron configuration of $1s^2 2s^2 2p^6 3s^2$. Following the octet rule, magnesium will lose the $3s^2$ electrons to form a stable, filled outer orbital arrangement (the two 2s electrons plus the six 2p electrons equals a total of eight). Therefore, magnesium, as the other alkaline metal elements will become an ion with a charge of plus two. The symbol for this ion is Mg^{2+} .

Question #14: The formula of a compound formed when aluminum (Al) combines with oxygen (O) is (D). Al_2O_3
Aluminum (symbol Al, atomic number 13) is in family IIA and has an electron configuration of $1s^2 2s^2 2p^6 3s^2 3p^1$. Following the octet rule, aluminum will lose the $3s^2 3p^1$ electrons to form a stable, filled outer orbital arrangement (the two 2s electrons plus the six 2p electrons equals a total of eight). Therefore, aluminum will become an ion with a charge of plus two. Oxygen (symbol O, atomic number 8) is in family VIA and has an electron configuration of $1s^2 2s^2 2p^4$. Again following the octet rule, oxygen will gain 2 more electrons to fill the 2p orbitals, forming a stable, filled outer orbital arrangement. Therefore, oxygen would acquire a noble gas configuration by accepting two electrons. Three oxygen atoms, each acquiring two electrons, are needed to balance the six electrons lost by two aluminum atoms (two are needed because atoms cannot deal with half electrons). The ratio of aluminum atoms to oxygen atoms in the compound is 2:3 and the formula for aluminum oxide is Al_2O_3 .

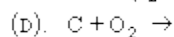
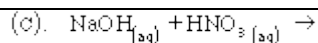
Question #15: The sulfate ion is $(SO_4)^{2-}$ so the formula for calcium sulfate would be (A). $CaSO_4$
Calcium (Ca) is an alkaline metal in family IIA and has an electron configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$. Following the octet rule, magnesium will lose the $4s^2$ electrons to form a stable, filled outer orbital arrangement (the two 3s electrons plus the six 3p electrons equals a total of eight). Therefore, magnesium, as the other alkaline metal elements will become an ion with a charge of plus two. That the sulfate polyatomic ion has a charge of 2- is given, so one polyatomic ion is needed to balance the two electrons lost by calcium and the formula is $CaSO_4$.

Question #16: The number of oxygen atoms in a molecule of copper(II) phosphate, $Cu_3(PO_4)_2$, is

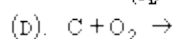
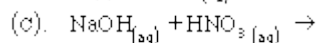
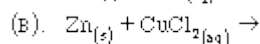
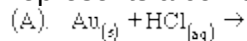
(A). 2 (B). 4 (C). 6 (D). 8 Question #17: If the absolute difference in electronegativity between two bonded atoms is 2.0, the type of bond is (A). ionic. (B). polar covalent. (C). covalent. (D). none. Question #18: Ionic compounds are usually composed of (A). two metals. (B). two nonmetals. (C). a metal and a nonmetal. (D). a metal and a semiconductor. Question #19: If the correct formula for sodium bicarbonate is NaHCO_3, then the formula for calcium bicarbonate would be (A). CaHCO_3 (B). $\text{Ca}(\text{HCO}_3)_2$ (C). Ca_2HCO_3 (D). CaHCO_2 Question #20: The formula for the covalent compound of tetraphosphorus trisulfide would be (A). P_3S_3 (B). P_3S_4 (C). P_4S_3 (D). P_4S_4	(D). 8 According to the formula there are two phosphate polyatomic ions, each with 4 oxygen atoms. Thus, there are a total of 8 oxygen atoms. Question #17: If the absolute difference in electronegativity between two bonded atoms is 2.0, the type of bond is (A). ionic. Electronegativity is the comparative ability of atoms of an element to attract bonding electrons and the assigned numerical values for electronegativities are given in Figure 11.10. An absolute difference in electronegativity of 1.7 or greater means that an ionic bond is to be expected. Question #18: Ionic compounds are usually composed of (C). a metal and a nonmetal. In general, elements in families IA and IIA of the periodic table tend to form positive ions by losing electrons and the elements in families VIA and VIIA tend to form negative ions by gaining electrons, so metals and nonmetals tend to form ionic compounds. Elements of families IIIA and VA have less of a tendency to form ionic compounds, except for those in higher periods. Question #19: If the correct formula for sodium bicarbonate is NaHCO_3, then the formula for calcium bicarbonate would be (B). $\text{Ca}(\text{HCO}_3)_2$ The element sodium (symbol Na, atomic number 11) has an electron configuration of $1s^2 2s^2 2p^6 3s^1$ so it has one valence electron (the $3s^1$). This is true of all the elements in group IA. The bicarbonate ion has a negative charge of 1. Therefore, one sodium ion with a charge of plus 1 and one polyatomic bicarbonate ions with a charge of minus 1 will balance. The formula NaHCO_3 confirms this arrangement. Calcium, (Ca) on the other hand, is an alkaline metal in family IIA and has an electron configuration of $1s^2 2s^2 2p^6 3s^2 3p^4 4s^2$. Following the octet rule, magnesium will lose the $4s^2$ electrons to form a stable, filled outer orbital arrangement (the two 3s electrons plus the six 3p electrons equals a total of eight). Therefore, magnesium, as the other alkaline metal elements will become an ion with a charge of plus two. Two bicarbonate polyatomic ions, each acquiring one electron, are needed to balance the two electrons lost by calcium. The ratio of calcium ions to bicarbonate polyatomic ions is 1:2 and the formula is $\text{Ca}(\text{HCO}_3)_2$. Question #20: The formula for the covalent compound of tetraphosphorus trisulfide would be (C). P_4S_3 Tetra- is a prefix that means 4 and tri- is a prefix that means 3. Therefore, there are 4 phosphorous atoms and 3 sulfur atoms so the formula must be P_4S_3.
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Chemical Formulas and Equations

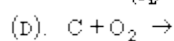
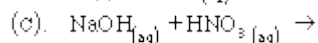
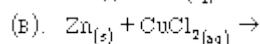
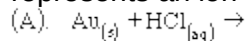
Questions	Answers with Rationalization
Question #1: The formula weight of $C_6H_{12}O_6$ is (A). 180 u (B). 72 u (C). 29 u (D). 24 u Question #2: The number of sulfur atoms in $Al_2(SO_4)_3$ is (A). 3 (B). 4 (C). 7 (D). 12 Question #3: The formula for a compound is given as C_2H_2. This is a (an) (A). empirical formula. (B). molecular formula. (C). structural formula. (D). redundant formula. Question #4: What number should be in front of H_2SO_4 in the equation below? $2Al + ? H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2$ (A). 1 (B). 2 (C). 3 (D). 4 Question #5: The equation below is balanced by placing what number in front of the carbon dioxide? $2C_2H_5OH + 6O_2 \rightarrow ? CO_2 + 6H_2O$ (A). 4 (B). 6 (C). 10 (D). 14 Question #6: Which of the following equations is balanced? (A). $4P + 10 O_2 \rightarrow 2 P_2O_5$ (B). $SO_2 + O_2 \rightarrow S_2O_5$ (C). $2Al + 3HCl \rightarrow 2AlCl_3 + 3H_2$ (D). $3Mg(OH)_2 + 2H_3PO_4 \rightarrow Mg_3(PO_4)_2 + 6H_2O$ Question #7: Which of the following represents a replacement reaction? (A). $Au_{(s)} + HCl_{(aq)} \rightarrow$ (B). $Zn_{(s)} + CuCl_2_{(aq)} \rightarrow$	Question #1: The formula weight of $C_6H_{12}O_6$ is (A). 180 6 of C = $6 \times 12 \text{ u} = 72 \text{ u}$ 12 of H = $12 \times 1 \text{ u} = 12 \text{ u}$ 6 of O = $6 \times 16 \text{ u} = 96 \text{ u}$ $72 + 12 + 96 = 180 \text{ u}$ Question #2: The number of sulfur atoms in $Al_2(SO_4)_3$ is (A). 3 The formula tells us there are 3 SO_4 polyatomic ions bound to 2 aluminum atoms. There is 1 sulfur atom in a SO_4 polyatomic ion, so there are 3 sulfur atoms in 3 SO_4 polyatomic ions. Question #3: The formula for a compound is given as C_2H_2. This is a (an) (B). molecular formula. C_2H_2 is a covalent compound, so the formula might be molecular. Since it is not the simplest whole number ratio (which would be CH), then the formula is molecular. Question #4: What number should be in front of H_2SO_4 in the equation below? $2Al + ? H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2$ (C). 3 There are 3 SO_4 polyatomic ions and there are 6 hydrogen atoms on the right side. A coefficient of 3 in front of the H_2SO_4 will balance both the SO_4 and the hydrogen atoms. Question #5: The equation below is balanced by placing what number in front of the carbon dioxide? $2C_2H_5OH + 6O_2 \rightarrow ? CO_2 + 6H_2O$ (A). 4 There are 4 C and a total of $2 + 12 = 14$ O in the reactants. There are 6 O in the water product, leaving a need for 4 C and $14 - 6$, or 8 O in the products. Placing a coefficient of 4 in front of the carbon dioxide will balance the C and O atoms. Question #6: Which of the following equations is balanced? (A). $4P + 10 O_2 \rightarrow 2 P_2O_5$ (B). $SO_2 + O_2 \rightarrow S_2O_5$ (C). $2Al + 3HCl \rightarrow 2AlCl_3 + 3H_2$ (D). $3Mg(OH)_2 + 2H_3PO_4 \rightarrow Mg_3(PO_4)_2 + 6H_2O$ Inventory the number of each kind of atom on both sides of each equation and you will find that only (D) is balanced. Question #7: Which of the following represents a replacement reaction? (A). $Au_{(s)} + HCl_{(aq)} \rightarrow$ (B). $Zn_{(s)} + CuCl_2_{(aq)} \rightarrow$



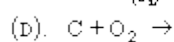
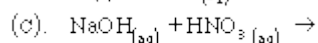
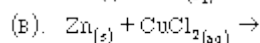
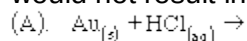
Question #8: Which of the following represents a combination reaction?



Question #9: Which of the following represents an ion exchange reaction?



Question #10: Which of the following would not result in a reaction?



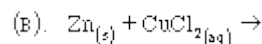
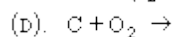
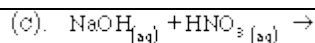
Question #11: If the atomic mass of oxygen is 16 u and Avogadro's number is 6.02×10^{23} particles per mole, then one mole of ozone, O_3 , contains

(A). 6.02×10^{23} molecules.

(B). 12.04×10^{23} molecules.

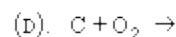
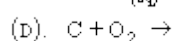
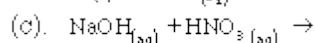
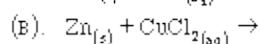
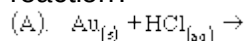
(C). 18.06×10^{23} molecules.

(D). 9.63×10^{24} molecules.



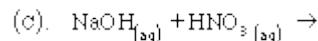
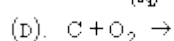
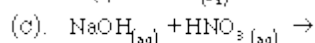
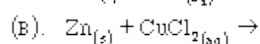
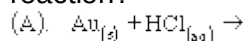
A replacement reaction follows the general form of $\text{XY} + \text{Z}$ yields $\text{XZ} + \text{Y}$ or $\text{XY} + \text{A}$ yields $\text{AY} + \text{X}$. Equation (B) is the only one that approaches this general form. Zinc is above copper in the activity series shown in figure 12.13 (page 289), so you can predict that a replacement reaction would occur.

Question #8: Which of the following represents a combination reaction?



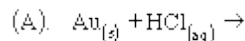
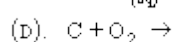
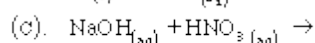
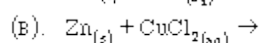
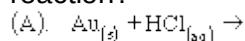
A combination follows the general form of $\text{X} + \text{Y}$ yields XY .

Question #9: Which of the following represents an ion exchange reaction?



In general, an ion exchange reaction has the form $\text{AX} + \text{BY}$ yields $\text{AY} + \text{BX}$ where one of the products removes ions from the solution as a precipitate, a gas, or water. This is a reaction between an acid and a base, which forms a salt and water.

Question #10: Which of the following would not result in a reaction?



Reacting solid gold with a water solution of hydrochloric acid might be an ion exchange reaction, but gold is below hydrogen in the activity series so no reaction occurs.

Question #11: If the atomic mass of oxygen is 16 u and Avogadro's number is 6.02×10^{23} particles per mole, then one mole of ozone, O_3 , contains

(A). 6.02×10^{23} molecules.

An amount of a substance that contains Avogadro's number of atoms, ions, molecules, or any other chemical unit is defined as a mole of the substance. Thus a mole of O_3 contains Avogadro's number of molecules, which is 6.02×10^{23} molecules.

Question #12: How many moles of hydrogen atoms are present in 1 mole of H_2O? (A). 1 (B). 2 (C). 3 (D). 4 Question #13: The mass of 6.02×10^{23} atoms of gold (Au) is (A). 79 g (B). 197 g (C). 475.6×10^{23} g (D). $1,186 \times 10^{23}$ g Question #14: How many moles of oxygen atoms are present in 1 mole of H_2O? (A). 1/2 (B). 1 (C). 2 (D). 3 Question #15: The formula weight of carbon dioxide, CO_2 is (A). 28 u (B). 44 u (C). 56 u (D). 72 u	Question #12: How many moles of hydrogen atoms are present in 1 mole of H_2O? (B). 2 Every mole contains Avogadro's number of particles and 2 times Avogadro's number of hydrogen atoms combines with Avogadro's number of oxygen atoms to form Avogadro's number of water molecules. There are therefore two moles of hydrogen in each mole of H_2O. Question #13: The mass of 6.02×10^{23} atoms of gold (Au) is (B). 197 g The mole concept and Avogadro's number provide a relationship between numbers and masses. The atomic weight of gold is 197 u, so the gram-atomic weight of gold is 197 grams. A gram-atomic weight is one mole of an element, so the mass of 6.02×10^{23} atoms of gold is 197 grams. Question #14: How many moles of oxygen atoms are present in 1 mole of H_2O? (B). 1 Every mole contains Avogadro's number of particles. Since 2 times Avogadro's number of hydrogen atoms combines with Avogadro's number of oxygen atoms to form Avogadro's number of water molecules. There is therefore one mole of oxygen in each mole of H_2O. Question #15: The formula weight of carbon dioxide, CO_2 is (B). 44 u 1 of C = $1 \times 12 \text{ u} = 12 \text{ u}$ 2 of O = $2 \times 16 \text{ u} = 32 \text{ u}$ $32 + 12 = 44 \text{ u}$
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Water and Solutions

Questions	Answers with Rationalization
Question #1: Which of the following is a solution? (A). A cup of tea with a teaspoon of dissolved sugar. (B). The brass in a candle stick. (C). Air in the atmosphere. (D). All of the above.	Question #1: Which of the following is a solution? (D). All of the above. A solution is defined as a homogeneous mixture of ions or molecules of two or more substances. Solutions are not limited to solids dissolved in liquids. There are three phases of matter (solid, liquid, gas), so it is possible to have nine kinds of solutions. Thus, response A asks if the brass in a candle stick is a solution. Yes, because brass is an alloy of copper and zinc with other metals in varying lesser amounts, a solution of a solid dissolved in a solid. Response B asks if air in the atmosphere is a solution. Yes, because air is oxygen, carbon dioxide, and other gases dissolved in nitrogen gas, a solution of gases dissolved in a gas.
Question #2: The part of a solution that is present in the larger amount is the (A). solute. (B). solvent. (C). clear part. (D). colorant.	Question #2: The part of a solution that is present in the larger amount is the (B). solvent. The amounts of the components of a solution are identified by the general terms solvent and solute. The solvent is the component present in the larger amount. The solute is the component that dissolves in the solvent.
Question #3: The solvent plus the ___ ? ___ make up a solution. (A). pigment (B). stain (C). solute (D). colorant	Question #3: The solvent plus the ___ ? ___ make up a solution. (C). solute The amounts of the components of a solution are identified by the general terms solvent and solute. The solvent is the component present in the larger amount. The solute is the component that dissolves in the solvent.
Question #4: A solution with a relatively large amount of solute is said to be (A). dilute. (B). a weak solution. (C). concentrated. (D). an amalgam.	Question #4: A solution with a relatively large amount of solute is said to be (C). concentrated. The relative amounts of solute and solvent are described by the concentration of a solution. In general, a solution with a large amount of solute is concentrated, and a solution with much less solute is dilute.
Question #5: A concentration of 1 ppb expressed as money means 1 cent in every (A). \$1,000 (B). \$10,000,000 (C). \$100,000,000 (D). \$1,000,000,000	Question #5: A concentration of 1 ppb expressed as money means 1 cent in every (B). \$10,000,000 Concentration ratios that describe small concentrations of solute are sometimes reported as a ratio of parts per million (ppm) or parts per billion (ppb). A concentration of 1 ppb expressed as money means 1 cent in every \$10,000,000, which is one billion cents.
Question #6: To convert a solution concentration expressed in ppm to the equivalent percent concentration you simply divide by (A). 1,000 (B). 10,000 (C). 100,000 (D). 1,000,000	Question #6: To convert a solution concentration expressed in ppm to the equivalent percent concentration you simply divide by (B). 10,000 Percent means parts per 100 and ppm means parts per 1,000,000 and the difference in the two expressions of concentration is a matter of 10,000. Thus you divide by 10,000 to convert a concentration expressed in ppm to the equivalent percent concentration.

Question #7: Rubbing alcohol that is a 12% by volume concentration has 12 mL of pure isopropyl alcohol mixed with
(A). 88 mL of water.
(B). 100 mL of water.
(C). 88 mL of solution.
(D). 100 mL of solution.

Question #8: Hydrogen peroxide that is 3% by weight concentration has 3 grams of pure hydrogen peroxide mixed with
(A). 97 grams of solution.
(B). 100 grams of solution.
(C). 97 grams of water.
(D). 100 grams of water.

Question #9: Seawater with a salinity of 5‰ has 5 g of salt dissolved in
(A). 995 g of pure water.
(B). 1,000 g of pure water.
(C). 95 g of seawater.
(D). 100 g of seawater.

Question #10: A solution is saturated when
(A). there is not enough solvent.
(B). no more solute can be dissolved.
(C). the solute is dissolving at the same rate it is coming out of solution.
(D). none of the above.

Question #11: Water has an unusually high latent heat and some unusual density changes occur with temperature changes because
(A). water has a density of 1 kg/L
(B). of hydrogen bonding.
(C). water has a greater density than ice.
(D). a molecule has 2 hydrogen atoms but only 1 oxygen.

Question #7: Rubbing alcohol that is a 12% by volume concentration has 12 mL of pure isopropyl alcohol mixed with
(A). 88 mL of water.
The concentration term of percent by volume is defined as the volume of solute in 100 volumes of solution. This concentration term is just like any other percentage ratio, that is, part divided by the whole times 100 percent. The distinction is that the part is a volume of solute and the whole is the volume of solution. Thus to have 100 volumes of a 12% solution of alcohol you would need 88 mL of water to add to the 12 mL of alcohol in order to have 100 mL of solution.

Question #8: Hydrogen peroxide that is 3% by weight concentration has 3 grams of pure hydrogen peroxide mixed with
(C). 97 grams of water.
The concentration term of percent by weight is defined as the weight of solute in 100 weight units of solution. This concentration term is just like any other percentage ratio, that is, part divided by the whole times 100 percent. The distinction is that the part is a weight of solute and the whole is the weight of solution. Thus to have 100 weight units of a 3% solution by weight of hydrogen peroxide you would need 97 grams of water to add to the 3 grams of hydrogen peroxide in order to have 100 grams of solution. (Note that a gram is a unit of mass, not a unit of weight. However, weight is proportional to mass in a given location so mass units such as grams are sometimes used to calculate a percent by weight.)

Question #9: Seawater with a salinity of 5‰ has 5 g of salt dissolved in
(A). 995 g of pure water.
Salinity is defined as the mass of salts dissolved in 1,000 grams of solution.

Question #10: A solution is saturated when
(C). the solute is dissolving at the same rate it is coming out of solution.
A saturated solution is understood to be a state of equilibrium that exists between dissolving solute and solute coming out of solution.

Question #11: Water has an unusually high latent heat and some unusual density changes occur with temperature changes because
(B). of hydrogen bonding.
A hydrogen bond is a weak to moderate bond between the hydrogen end of a polar molecule and the negative end of a second polar molecule. In water hydrogen bonding occurs between a hydrogen atom of one molecule and the oxygen end of another water molecule. This bonding is responsible for some of the physical properties of water. These properties include its unusually high heat of fusion and heat of vaporization as well as its unusual density changes that occur with changes in temperature.

Question #12: As water cools from 4° to 0°C its density (A). increases. (B). decreases. (C). does not change. (D). might or might not change, depending on other variables. Question #13: Which of the following are properties of basic solutions? They (A). turn blue litmus paper red. (B). taste sour. (C). feel slippery. (D). They react with active metals producing hydrogen gas. Question #14: Which of the following are properties of acidic solutions? They (A). turn blue litmus paper red. (B). taste sour. (C). react with active metals, releasing hydrogen gas. (D). all of these are true. Question #15: A bottle of whiskey contains 40% alcohol by volume. This means that the whiskey contains 40 mL of alcohol (A). in every 100 mL of whiskey. (B). mixed with 100 mL of water. (C). mixed with 60 mL of whiskey. (D). in every 140 mL of whiskey. Question #16: The water hardness in an area is reported as 700 ppm total dissolved solids. This is the same concentration as (A). 0.007% (B). 0.07% (C). 0.7% (D). 7% Question #17: According to the modern definition, NH_3 is (A). an acid because it contains hydrogen. (B). a base because it can accept	Question #12: As water cools from 4 ° to 0 ° its density (B). decreases. The density of water increases as it is cooled down to 4°C, then hydrogen bonding between the oxygen atom and two hydrogen atoms of other water molecules results in an open, hexagonal structure. The open space of the hexagonal channel in this structure results in ice being less dense than water. The shape of the channel also suggests why snowflakes always have six sides. Question #13: Which of the following are properties of basic solutions? They (C). feel slippery Basic solutions feel slippery on the skin. They have a caustic action on plant and animal tissue, converting tissue into soluble materials. A strong base reacts with fat to make soap and glycerine. This accounts for the slippery feeling on the skin. Question #14: Which of the following are properties of acidic solutions? They (D). all of these are true. Acids in water have the characteristic properties of (1) having a sour taste such as the taste of citrus fruits; (2) having the ability to change the color of certain substances such as the color of litmus from blue to red; (3) reacting with active metals, releasing hydrogen gas, and; (4) the ability to neutralize bases, forming water and salts from the reaction. Question #15: A bottle of whiskey contains 40% alcohol by volume. This means that the whiskey contains 40 mL of alcohol (A). in every 100 mL of whiskey. The concentration term of percent by volume is defined as the volume of solute in 100 volumes of solution. This concentration term is just like any other percentage ratio, that is, part divided by the whole times 100 percent. The distinction is that the part is a volume of solute and the whole is the volume of solution. Thus to have 100 volumes of a 40% solution of alcohol you would need 60 mL of water to add to the 40 mL of alcohol in order to have 100 mL of solution. In other words, there would be 40 mL of alcohol in every 100 mL of whiskey. The 40% whiskey would be labeled as 80 proof since proof is the percent times two. Question #16: The water hardness in an area is reported as 700 ppm total dissolved solids. This is the same concentration as (B). 0.07% Percent means parts per 100 and ppm means parts per 1,000,000 and the difference in the two expressions of concentration is a matter of 10,000. Thus you divide by 10,000 to convert a concentration expressed in ppm to the equivalent percent concentration. Thus 700 divided by 10,000 is 0.07%. Question #17: According to the modern definition, NH_3 is (B). a base because it can accept H^+ ions from water. The modern concept of an acid considers the properties of acids in terms of the hydronium ion, H_3O^+. A hydronium ion is a water molecule to which an H^+ ion is attached. Since a hydrogen ion is a hydrogen atom
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H⁺ ions from water. (C). an acid because it raises the H⁺ ion concentration. (D). a base because it lowers the pH. Question #18: If the force of attraction between the ions in a solid is very strong, you would expect the solid to have (A). low solubility in water. (B). high solubility in water. (C). low solubility in a non-polar solvent. (D). high solubility in a non-polar solvent. Question #19: A solution with a pH of 5 (A). is twice as acidic as one with a pH of 4. (B). is half as acidic as a solution with a pH of 4. (C). is ten times as acidic as a solution with a pH of 4. (D). is one tenth as acidic as a solution with a pH of 4. Question #20: Which of the following solutions is likely to have a pH greater than 7? (A). sodium chloride (B). ammonia (C). carbonic acid (D). pure water	without its single electron, it could be considered as an ion consisting of a single proton. Thus the H⁺ ion can be called a proton. An acid is defined as any substance that is a proton donor when dissolved in water, increasing the hydronium ion concentration. A base is defined as any substance that is a proton acceptor when dissolved in water, increasing the hydroxide ion concentration. Question #18: If the force of attraction between the ions in a solid is very strong, you would expect the solid to have (A). low solubility in water. An ionic solid dissolves in water because the number of water molecules around the surface is greater than the number of other ions of the solid. The attraction between polar water molecules and an ion enables the water molecules to pull ions away from the crystal, a process called dissolving. If the force of attraction between the ions in the crystal is greater than the force of attraction between the polar water molecules and the ions the solid will have a low solubility. Question #19: A solution with a pH of 5 (D). is one tenth as acidic as a solution with a pH of 4. A neutral solution has a pH of 7, acidic solutions have values below 7 and smaller numbers mean greater acidic properties. Bases have values greater than 7 and larger numbers mean greater basic properties. The pH scale is logarithmic and a pH of 4 is ten times as acidic than a pH of 5. Conversely, a pH of 5 represents one tenth the acidity as a pH of 4. Question #20: Which of the following solutions is likely to have a pH greater than 7? (B). ammonia First, recall that an acid is defined as any substance that is a proton donor when dissolved in water, increasing the hydronium ion concentration. A base is defined as any substance that is a proton acceptor when dissolved in water, increasing the hydroxide ion concentration. Second, recall that a neutral solution has a pH of 7, acidic solutions have values below 7 and smaller numbers mean greater acidic properties. Bases have values greater than 7 and larger numbers mean greater basic properties. Finally, note that pure water and a sodium chloride solution are neither proton donors or acceptors, so they both have a pH of 7. Carbonic acid is an acid, so it would have a pH less than 7. Ammonia has basic properties, so its pH would be greater than 7.
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Organic Chemistry

Questions	Answers with Rationalization
Question #1: Organic chemistry is the study of (A). any compound from any living thing. (B). carbon-containing compounds that were formed by living things. (C). any compound with carbon as the principal element. (D). none of the above.	Question #1: Organic chemistry is the study of (C). any compound with carbon as the principal element. Carbon is the essential element of organic matter and organic chemistry is defined as the study of compounds in which carbon is the principal element, whether the compound was formed by living things or not.
Question #2: Carbon can form many different compounds because it can (A). make a molecule in the shape of a cube, tetrahedron, or cylinder. (B). combine with other carbon atoms in addition to other elements. (C). combine with more metals than other elements. (D). combine with more nonmetals than other elements.	Question #2: Carbon can form many different compounds because it can (B). combine with other carbon atoms in addition to other elements. The carbon atom has a valence of four, and can combine with one, two, three, or four other carbon atoms in addition to a wide range of other kinds of atoms. The number of possible molecular combinations is almost limitless, which explains why there are so many organic compounds.
Question #3: A hydrocarbon is an organic compound consisting of (A). water and carbon. (B). any number of elements as long as they include carbon and hydrogen. (C). carbon and hydrogen. (D). carbon, hydrogen, and oxygen.	Question #3: A hydrocarbon is an organic compound consisting of (C). carbon and hydrogen. A hydrocarbon is an organic compound consisting of only two elements and as the name implies, these elements are hydrogen and carbon.
Question #4: A hydrocarbon with two double covalent bonds between carbon atoms is a (an) (A). alkane. (B). alkene. (C). alkyne. (D). aromatic hydrocarbon.	Question #4: A hydrocarbon with two double covalent bonds between carbon atoms is a (an) (B). alkene. The alkanes are hydrocarbons with single covalent bonds between carbon atoms, the alkenes are hydrocarbons with one or more double covalent carbon-to-carbon bonds, and alkynes are hydrocarbons with one or more carbon-to-carbon triple bonds. Aromatic hydrocarbons are hydrocarbons based on the benzene ring structure, which has carbon-to-carbon bonds somewhere between single and double.
Question #5: Organic compounds called isomers have (A). the same molecular formulas but different physical structures. (B). different molecular formulas with the same physical structures. (C). different molecular formulas	Question #5: Organic compounds called isomers have (A). the same molecular formulas but different physical structures. Compounds with the same molecular formulas with different structures are called isomers. Butane, for example, can occur as a straight chain isomer called normal, or n-butane, and it can occur as a branched isomer called isobutane. The isomers of a particular hydrocarbon, such as butane, have different physical and chemical properties because they have different structures.

with the same chemical properties.

(D). none of the above.

Question #6: The number of carbon atoms in a molecule of octane is

(A). 1

(B). 2

(C). 4

(D). 8

Question #7: Organic compounds called aromatic hydrocarbons are compounds that

(A). have a wonderful odor.

(B). are based on the benzene ring structure.

(C). occur in nature with all carbon bonds saturated.

(D). occur in nature with all carbon bonds unsaturated.

Question #8: A gasoline mixture of hydrocarbons that burns very rapidly has

(A). a low octane number.

(B). a high octane number.

(C). many branched chains.

(D). smaller hydrocarbon molecules.

Question #9: An organic molecule with a general formula of ROH is a (an)

(A). ether.

(B). ester.

(C). alcohol.

(D). organic acid.

Question #10: An organic molecule with a general formula of RCOOH is a (an)

(A). ester.

(B). organic acid.

(C). ketone.

Question #6: The number of carbon atoms in a molecule of octane is (D). 8

The prefix octa- means 8, so there are 8 carbon atoms in a molecule of octane. Some other prefixes and their meaning are mono- (1), di- (2), tri- (3), tetra- (4), penta- (5), hexa- (6), hepta- (7), nona- (9), and deca- (10).

Question #7: Organic compounds called aromatic hydrocarbons are compounds that

(B). are based on the benzene ring structure.

Aromatic hydrocarbons are hydrocarbons based on the benzene ring structure, which has carbon-to-carbon bonds somewhere between single and double. These organic compounds are called aromatic hydrocarbons because of their aromatic character.

Question #8: A gasoline mixture of hydrocarbons that burns very rapidly has

(A). a low octane number.

Gasoline distilled from crude oil consists mostly of straight-chain molecules not suitable for use as an automotive fuel because it burns too rapidly, producing more of an explosion than a smooth burn.

Branched chain molecules burn more slowly, without the pinging or knocking explosions. The burning rate of gasoline is described by the octane number scale, which is based on a branched isomer of octane assigned an octane number of 100. Straight chain heptane burns very rapidly and is given an octane rating of zero. A mixture of 87 percent branched isomer of octane and 13 percent heptane would have an octane rating of 87. Performance will not be improved by using a higher octane in a tuned automobile engine that does not ping or knock with 87 octane gasoline.

Question #9: An organic molecule with a general formula of ROH is a (an)

(C). alcohol.

An alcohol is an organic compound formed by replacing one or more hydrogen on an alkane with a hydroxyl functional group (OH). This remains attached to a hydrocarbon group (R), giving the compound a set of properties that are associated with alcohols and the general formula of ROH. The name of the hydrocarbon group R determines the name of the alcohol. If the hydrocarbon group in ROH is ethyl, for example, the alcohol is called ethyl alcohol.

Question #10: An organic molecule with a general formula of RCOOH is a (an)

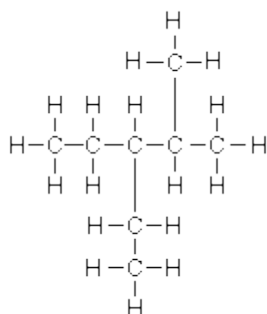
(B). organic acid.

Organic acids are acids that were derived from organisms and they all contain the carboxyl functional group (COOH) and have a general formula of RCOOH.

(D). aldehyde. Question #11: The characteristic odor and taste of fruit such as bananas, oranges, and pineapples comes from certain (A). ketones. (B). ethers. (C). aldehydes. (D).esters. Question #12: The human body breaks down starches to (A). monosaccharides. (B). simple sugars. (C). glucose. (D). any of the above. Question #13: The direct reserve source of energy in the muscles of a human is (A). glycol. (B). glycerol. (C). glycogen. (D). dextrose. Question #14: All proteins are made up of a side chain and (A). alpha-amino acid. (B). amine. (C). a nitrogen atom. (D). peptide. Question #15: Fats and oils are esters formed from three fatty acids and glycerol into a (an) (A). alpha-amino acid. (B). polysaccharide. (C). triglyceride. (D). disaccharide. Question #16: Which of the following is a polymer? (A). Cellulose. (B). Polyethylene. (C). Wool. (D). All are polymers.	Question #11: The characteristic odor and taste of fruit such as bananas, oranges, and pineapples comes from certain (D).esters. Esters (RCOOR') are common in both plants and animals, giving fruits and flowers their characteristic odor and taste. Amyl acetate, for example, is the chemical responsible for what you identify as the flavor of banana. Natural flavors are complex mixtures of esters along with other organic compounds. Question #12: The human body breaks down starches to (D). any of the above. Starches are complex carbohydrates known as polysaccharides, which are polymers made up of simpler monosaccharide units joined together in straight or branched chains. The human digestive process breaks these polymers apart into the basic monosaccharide units. Monosaccharides serve as food and are also known as simple sugars because they are mostly 6-carbon molecules such as glucose. Glucose is found in the sap of plants and in the human bloodstream where it is called blood sugar. All sugars and starches are broken down to glucose, which is transported by the blood stream for use by the body. Question #13: The direct reserve source of energy in the muscles of a human is (C). glycogen. Glycogen is a starch-like polysaccharide that is synthesized by the human body and stored in the muscles and liver. Glycogen, like starch, is a very high molecular weight polysaccharide but it is more highly branched. These highly branched polysaccharides serve as a direct reserved source of energy in the muscles. In the liver they serve as a reserve source to maintain the blood sugar level. Question #14: All proteins are made up of a side chain and (A). alpha-amino acid. Proteins are formed from amino acids, which have a carbon atom (called alpha) with attached amino group (NH₂), carboxylic group (COOH), and a side chain (R) that determines the properties of the protein. Question #15: Fats and oils are esters formed from three fatty acids and glycerol into a (an) (C). triglyceride. Fats and oils are esters formed from glycerol (1,2,3-trihydroxypropane) and three long-chain carboxylic acids (fatty acids). This ester is called a triglyceride and its structural formula is shown in Figure 14.24. Question #16: Which of the following is a polymer? (D). All are polymers. Polymers are huge, chainlike molecules made of hundreds or thousands of smaller, repeating molecular units called monomers. Polymers occur naturally in plants and animals. Cellulose, for example, is a natural plant polymer made of glucose monomers. Wool and hair are natural animal polymers made of protein monomers. Synthetic polymers are manufactured from a wide variety of substances. Polyethylene is a
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familiar polymer used for vegetable bags, dry cleaning bags, grocery bags, and plastic squeeze bottles. Polyethylene is a polymer produced by a polymerization reaction of ethylene, which is derived from petroleum.

Question #17: The IUPAC name for the molecule below is



- (A). octane.
- (B). 3-ethyl-4-methylpentane.
- (C). methylheptane.
- (D). 3-ethyl-2-methylpentane.

Question #18: An alkane with 3 carbon atoms would have how many hydrogen atoms in the molecule?

- (A). 4
- (B). 6
- (C). 8
- (D). 10

Question #19: The R in R-COOH or R-C=O stands for

- (A). a reactive atom.
- (B). a separate functional group.
- (C). any hydrocarbon group.
- (D). the right side of the molecule.

Question #20: When wine "goes bad," the ethanol is converted into

- (A). CH_3COOH
- (B). CH_3OCH_3
- (C). $\text{CH}_3\text{CH}_2\text{OH}$
- (D). CH_3OH

Question #17: The IUPAC name for the molecule below is (D). 3-ethyl-2-methylpentane.

The longest chain has 5 carbon atoms, so the base name of the molecule is pentane. The location of other groups of atoms attached to the base chain are identified by counting carbon atoms from the left or right, selecting the direction that results in the smallest numbers. Counting from the right the second and third carbon atoms are identified as having attachments. The hydrocarbon groups attached to the base is a methyl group on the second carbon atom and an ethyl group on the third carbon atom. These are listed in alphabetic order, so the name is 3-ethyl-2-methylpentane.

Question #18: An alkane with 3 carbon atoms would have how many hydrogen atoms in the

- (C). 8

The general alkane formula of $\text{C}_n\text{H}_{2n+2}$ shows that a 3 carbon atoms will have an n of 3 so the number of hydrogen atoms is 2 times 3, or 6 plus 2 more for a total of 8.

Question #19: The R in R-COOH or R-C=O stands for

- (C). any hydrocarbon group.

The R (or R') in a functional group formula stands for one or more of the hydrocarbon groups.

Question #20: When wine "goes bad," the ethanol is converted into

- (A). CH_3COOH

Acetic acid (CH_3COOH) forms from the oxidation of ethanol. An oxidized bottle of wine contains acetic acid in place of the alcohol, which gives the wine a vinegar taste. This is why a small amount is served for tasting before a wine is served.

Nuclear Reactions

Questions	Answers with Rationalization
Question #1: Which of the following would be attracted toward a positively charged sheet of metal? (A). alpha particle (B). beta particle (C). gamma ray (D). none of the above	Question #1: Which of the following would be attracted toward a positively charged sheet of metal? (B). beta particle An alpha particle is the nucleus of a helium atoms with two protons and two neutrons, so it has a positive charge. A beta particle is an electron, so it has a negative charge. A gamma ray is a photon of very short wavelength and it has no charge. The beta particle would be attracted to a positively charged sheet of metal since opposite charges attract.
Question #2: Which of the following would be attracted toward a negatively charged sheet of metal? (A). alpha particle (B). beta particle (C). gamma ray (D). none of the above	Question #2: Which of the following would be attracted toward a negatively charged sheet of metal? (A). alpha particle An alpha particle is the nucleus of a helium atoms with two protons and two neutrons, so it has a positive charge. A beta particle is an electron, so it has a negative charge. A gamma ray is a photon of very short wavelength and it has no charge. The alpha particle would be attracted to a negatively charged sheet of metal since opposite charges attract.
Question #3: The rate of radioactive decay is increased by (A). increased heat. (B). increased pressure. (C). the use of a nuclear catalyst. (D). none of the above.	Question #3: The rate of radioactive decay is increased by (D). none of the above. Radioactive decay is a purely natural process and cannot be controlled or influenced by any outside variables.
Question #4: The emission of a beta particle from a nucleus results in (A). a decrease in the atomic number. (B). an increase in the atomic number. (C). no change in the atomic number. (D). none of the above.	Question #4: The emission of a beta particle from a nucleus results in (B). an increase in the atomic number. Beta emission is the expulsion of a high speed electron from a disintegrating nucleus. The emission of a beta particle increases the number of protons in a nucleus. It is as if a neutron changed to a proton by emitting the electron. So, emission of a beta particle increases the number of protons by one, but the mass number remains the same. The mass of the expelled electron is so small that it is ignored.
Question #5: The emission of a gamma ray from a nucleus results in (A). a decrease in the atomic number. (B). an increase in the atomic number. (C). no change in the atomic number. (D). none of the above.	Question #5: The emission of a gamma ray from a nucleus results in (D). none of the above. A gamma ray is a high-energy burst of electromagnetic radiation (a photon) from an excited nucleus. The nucleus changes from an excited state to a lower energy state and there is no change in the number of nucleons or the mass number.
Question #6: After emission and moving 2 to 12 cm through the air an alpha particle most likely becomes	Question #6: After emission and moving 2 to 12 cm through the air an alpha particle most likely becomes (A). an ordinary helium atom. Alpha emission is the expulsion of a helium nucleus, with 2 protons and 2 neutrons, from an unstable, disintegrating nucleus. The helium

- (A). an ordinary helium atom.
- (B). an electron.
- (C). increased energy in the impacted material.
- (D). none of the above

Question #7: An element in a radioactive decay series will continue to disintegrate into various radioactive elements until it becomes a stable isotope of

- (A). uranium.
- (B). lead.
- (C). bismuth.
- (D). lawrencium.

Question #8: The decay rate for a given number of nuclei of each radioactive isotope is observed to be

- (A). an identifying characteristic of that isotope.
- (B). specific for each radioactive element.
- (C). dependent on external conditions of temperature, pressure, and chemical state.
- (D). totally random and not dependent on any external condition.

Question #9: The mass of a given nucleus is always ___ ? ___ the sum of the masses of the individual particles of which it is made.

- (A). less than
- (B). more than
- (C). the same as
- (D). sometimes less, sometimes more, but never the same as

Question #10: When applied to $E = mc^2$ the mass defect of a given nucleus is found to be the

- (A). energy released when the nucleus formed.
- (B). energy required to break the nucleus into individual particles.
- (C). same as the binding energy.
- (D). any of the above.

nucleus eventually acquires two electrons to become an ordinary helium atom.

Question #7: An element in a radioactive decay series will continue to disintegrate into various radioactive elements until it becomes a stable isotope of

- (B). lead.

There are three naturally occurring radioactive decay series. One begins with thorium-232 and ends with lead-208, another begins with uranium-235 and ends with lead-207, and the third series begins with uranium-238 and ends with lead-206. Thus all radioactive decay series go through a series of decay reactions until they reach a stable lead nucleus.

Question #8: The decay rate for a given number of nuclei of each radioactive isotope is observed to be

- (A). an identifying characteristic of that isotope.

The radioactive decay constant is a specific constant for a particular isotope, and each isotope has its own decay constant. This is often described in terms of its half-life.

Question #9: The mass of a given nucleus is always ___ ? ___ the sum of the masses of the individual particles of which it is made.

- (A). less than

The mass of a nucleus is always less than the sum of the masses of the individual particles of which it is made. The difference is called the mass defect of the nucleus, and the explanation for the defect is found in $E = mc^2$. When nucleons join to make a nucleus, energy is released as the more stable nucleus is formed. The energy equivalent released when a nucleus is formed is the same as the binding energy, the energy required to break the nucleus into individual protons and neutrons.

Question #10: When applied to $E = mc^2$ the mass defect of a given nucleus is found to be the

- (D). any of the above.

The mass of a nucleus is always less than the sum of the masses of the individual particles of which it is made. The difference is called the mass defect of the nucleus, and the explanation for the defect is found in $E = mc^2$. When nucleons join to make a nucleus, energy is released as the more stable nucleus is formed. The energy equivalent released when a nucleus is formed is the same as the binding energy, the energy required to break the nucleus into individual protons and neutrons. The binding energy of the nucleus of any isotope can be calculated from the mass defect of the nucleus, which also tells you the amount of energy released when the nucleus formed.

Question #11: The nucleus of the greatest stability is found in the isotope of the element

- (A). aluminum.
- (B). iron.
- (C). hydrogen.
- (D). lead.

Question #12: Radiation can be a hazard to living organisms because it

- (A). produces ionization along its path of travel.
- (B). disrupts chemical bonds.
- (C). generates free polyatomic ions.
- (D). all of the above.

Question #13: In general, the public receives how much radiation exposure each year?

- (A). none
- (B). more than 500 millirem.
- (C). between 100 and 500 millirem.
- (D). about 130 rem.

Question #14: A Geiger counter is able to provide an indirect measure of radioactivity because radiation has a property of

- (A). ionization.
- (B). making matter glow in the dark.
- (C). fogging photographic film.
- (D). attracting electrons.

Question #15: The use of which unit would indicate radioactivity is being measured at its source?

- (A). rad
- (B). rem
- (C). curie
- (D). roentgen

Question #16: One of the first observable effects of over exposure to very low level radioactivity is

- (A). loss of hair.
- (B). changes in the blood count.

Question #11: The nucleus of the greatest stability is found in the isotope of the element

- (B). iron.

The energy equivalent released when a nucleus is formed is the same as the binding energy, the energy required to break the nucleus into individual protons and neutrons. The ratio of binding energy to the number of protons and neutrons is a measure of the stability of a nucleus. The greatest ratio of binding energy per number of protons and neutrons occurs near an isotope of iron with a mass number of 56, then decreases for both more massive and less massive nuclei.

Question #12: Radiation can be a hazard to living organisms because it

- (D). all of the above.
- Radioactivity can be a hazard to living organisms because it produces ionization along its path of travel. This ionization can (1) disrupt chemical bonds in essential macromolecules such as DNA and (2) produce molecular fragments, which are free polyatomic ions that can interfere with enzyme action and other essential cell functions.

Question #13: In general, the public receives how much radiation exposure each year?

- (C). between 100 and 500 millirem.

Natural radioactivity is part of the environment and people receive between 100 and 500 millirems each year from natural sources. The actual amount received varies with certain activities and situations that increase the yearly exposure to radiation. In general, the exposure for the average person is about 130 millirems per year.

Question #14: A Geiger counter is able to provide an indirect measure of radioactivity because radiation has a property of

- (A). ionization.

When radiation passes through the probe of a Geiger counter it ionizes some of the gas atoms in the tube, releasing free electrons. The electrons are accelerated by an electric field in the tube and an avalanche of free electrons creates a pulse of current that is amplified and measured. More radiation means more avalanches, so the pulses are an indirect means of measuring radiation. Each avalanche makes a pop or click when the circuit includes a speaker or earphone.

Question #15: The use of which unit would indicate radioactivity is being measured at its source?

- (C). curie

The activity of a radioactive source is a measure of the number of nuclear disintegrations per unit of time. The unit of activity at the source is called a curie, which is defined as 3.70×10^{10} nuclear disintegrations per second.

Question #16: One of the first observable effects of over exposure to very low level radioactivity is

- (B). changes in the blood count.

- (C). glowing in the dark.
(D). leukemia.

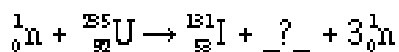
Question #17: The U-238 isotope is most likely to emit

- (A). an alpha particle.
(B). a beta particle.
(C). a gamma ray.
(D). It is not possible to predict.

Question #18: This type of radiation is released when Rn-224 decays to Po-220:

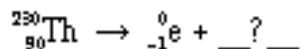
- (A). alpha.
(B). beta.
(C). gamma.
(D). all of these.

Question #19: Which of the following correctly balances the following nuclear fission reaction?



- (A). ${}_{39}^{104}\text{Y}$
(B). ${}_{36}^{105}\text{Kr}$
(C). ${}_{39}^{102}\text{Y}$
(D). ${}_{36}^{102}\text{Kr}$

Question #20: Which of the following correctly balances this decay reaction?



- (A). ${}_{91}^{230}\text{Th}$
(B). ${}_{89}^{230}\text{Ac}$
(C). ${}_{88}^{231}\text{Ac}$
(D). ${}_{91}^{230}\text{Pa}$

Question #17: The U-238 isotope is most likely to emit

- (A). an alpha particle.

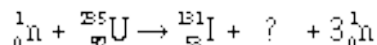
All nuclei with an atomic number greater than 83 are outside the upper right limit of the band of stability. Emission of an alpha particle reduces the number of protons by two and the number of neutrons by two, moving the nucleus more toward the band of stability. Any nucleus that lies beyond the upper right part of the band of stability is an alpha emitter, as is U-238.

Question #18: This type of radiation is released when Rn-224 decays to Po-220:

- (A). alpha.

All nuclei with an atomic number greater than 83 are outside the upper right limit of the band of stability. Emission of an alpha particle reduces the number of protons by two and the number of neutrons by two, moving the nucleus more toward the band of stability. Any nucleus that lies beyond the upper right part of the band of stability is an alpha emitter, as is Rn-224. Furthermore, since Rn-224 lost 4 atomic mass units to become Po-220, you know that an alpha particle must have been emitted.

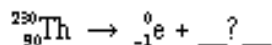
Question #19: Which of the following correctly balances the following nuclear fission reaction?



- (C). ${}_{39}^{102}\text{Y}$

From the subscripts you can see that the atomic number of the new isotope formed is $92 - 53$, or 39 and the mass number is $(1 + 235) = (131 + ? + 3)$, or 102 .

Question #20: Which of the following correctly balances this decay reaction?



- (D). ${}_{91}^{230}\text{Pa}$

The emission of a beta particle increases the number of protons in a nucleus. It is as if a neutron changed to a proton by emitting the electron. So, emission of a beta particle increases the number of protons by one, but the mass number remains the same. The mass of the expelled electron is so small that it is ignored. Therefore, the atomic number is increased by 1 to 91 and the element is now Pa and the mass number remains 230.

Rocks and Minerals

Questions	Answers with Rationalization
Question #1: Quartz is a mineral because it is (A). a white rock. (B). natural, inorganic, and has a crystalline structure. (C). an element. (D). composed of more than one element.	Question #1: Quartz is a mineral because it is (B). natural, inorganic, and has a crystalline structure. In the earth sciences, a mineral is defined as a naturally occurring, inorganic solid element or compound with a crystalline structure. The crystal structure might be on the microscopic scale and is not necessarily obvious to the unaided eye.
Question #2: Granite is a rock because it is (A). made up of several minerals. (B). natural, inorganic, and has a crystalline structure. (C). made up of several different elements. (D). found on the earth's surface.	Question #2: Granite is a rock because it is (A). made up of several minerals. A rock is a solid aggregation of one or more minerals that have been cohesively brought together by a rock-forming process. There are many possibilities of different kinds of rocks that could exist from many different variations of mineral mixtures.
Question #3: One of the following mineral properties can be determined by observing, rather than doing something to a sample. (A). streak (B). fracture (C). luster (D). hardness	Question #3: One of the following mineral properties can be determined by observing, rather than doing something to a sample. (C). luster A test for fracture requires that you break the sample, the streak test requires that you rub a sample across a piece of tile, and the test for hardness requires that you scratch the sample. Luster, on the other hand, describes how the sample looks.
Question #4: When hit with a hammer, most minerals will show (A). streak. (B). fracture. (C). cleavage. (D). hardness.	Question #4: When hit with a hammer, most minerals will show (B). fracture. Fracture results in an irregular surface rather than a well-defined, smooth plane. Most minerals do not show cleavage when broken, but do show fracture.
Question #5: An igneous rock that formed by rapid cooling will probably have crystals that are (A). large. (B). small. (C). middle-sized. (D). mixed large and small sizes.	Question #5: An igneous rock that formed by rapid cooling will probably have crystals that are (B). small. Rapid cooling does not result in sufficient time for larger crystals to form, so igneous rocks that form by rapid cooling will have a fine texture with small crystals.
Question #6: Materials that settle out of water are known as (A). mud. (B). sand. (C). sediments. (D). hard water.	Question #6: Materials that settle out of water are known as (C). sediments. Sediments are accumulations of silt, sand, or gravel that settle out the atmosphere or out of water. The word "sediment" means the act of settling, or to settle.

Question #7: In which type of rock would you expect to find fossils? (A). granite (B). obsidian (C). shale (D). marble Question #8: A rock made up of cemented sand grains is (A). conglomerate. (B). quartzite. (C). shale. (D). sandstone. Question #9: Which of the following does not belong with the others? (A). gypsum (B). sandstone (C). marble (D). limestone Question #10: The elements that occur most frequently in the earth's crust are (A). carbon and oxygen. (B). hydrogen and oxygen. (C). magnesium and iron. (D). silicon and oxygen. Question #11: One of the following cooled too quickly to allow crystals to form. (A). obsidian (B). quartz (C). halite (D). mica Question #12: Heat and pressure can change slate to the metamorphic rock schist. What is probably formed when more heat and pressure acts on schist? (A). shale (B). slate (C). gneiss (D). marble	Question #7: In which type of rock would you expect to find fossils? (C). shale Obsidian and granite are igneous rocks, formed under conditions that would probably destroy any material or sign of former life. Marble is a metamorphic rock formed when calcite crystals were recrystallized from limestone, again a condition that would probably destroy any fossils. Shale, on the other hand, is a sedimentary rock that formed from sediments of silt or clay in a process that would help preserve evidence of former life. Question #8: A rock made up of cemented sand grains is (D). sandstone. Sandstone, as the name implies, is a sedimentary rock formed from sand that has been consolidated into solid rock. Sand is a sediment with a size range of 1/16 to 2 mm. Conglomerate is a sedimentary rock formed from sediment with size ranges over 256 mm. Shale is a sedimentary rock formed from sediments with a size range less than 1/16 mm. Quartzite is a metamorphic rock formed from recrystallized quartz-rich sandstone. Question #9: Which of the following does not belong with the others? (C). marble Marble does not belong with the others because it is a metamorphic rock and all the others are sedimentary rocks. Question #10: The elements that occur most frequently in the earth's crust are (D). silicon and oxygen. Oxygen is the most abundant element in the earth's crust, making up about 50 percent by weight. Silicon is the second most abundant element in the crust, making up about 25 percent. Together, oxygen and silicon make up about 75 percent by weight of the elements that make up the crust. Question #11: One of the following cooled too quickly to allow crystals to form. (A). obsidian Obsidian has the same chemical composition as granite, but it cooled so rapidly that no crystals were formed. It is a translucent volcanic glass, not a mineral. Question #12: Heat and pressure can change slate to the metamorphic rock schist. What is probably formed when more heat and pressure acts on schist? (C). gneiss Slate is a metamorphic rock formed as a result of high temperature and pressure acting on the sedimentary rock shale. Slate is fine-grained with no crystals visible to the unaided eye, but alignment of the microscopic crystals results in a tendency of slate to split into flat sheets. Greater heat and pressure can cause more metamorphic change, resulting in larger crystals and increased foliation. The metamorphic rock schist can be produced by slate by further metamorphism. In schist the cleavage
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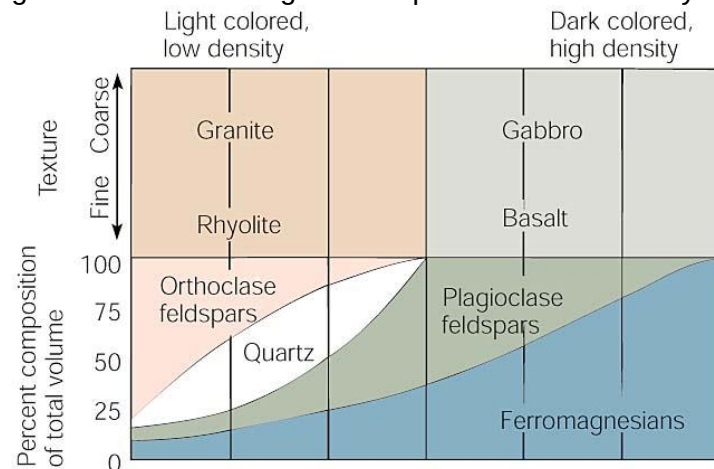
surfaces and coarser mica crystals are now visible to the unaided eye. Still further metamorphism of schist may break down the mica crystals and produce alternating bands of light and dark minerals. These bands are characteristic of the metamorphic rock gneiss. Slate, schist, and gneiss are three examples of a continuous transition that can take place from the metamorphism of shale.

Question #13: Magma that cools slowly beneath the earth's surface produces
 (A). ore minerals.
 (B). intrusive igneous rocks.
 (C). fine-grained metamorphic rocks.
 (D). extrusive igneous rocks.

Question #14: A fine-grained igneous rock made of mostly orthoclase feldspar, quartz crystals, and a few ferromagnesian crystals is probably
 (A). rhyolite.
 (B). granite.
 (C). gabbro.
 (D). basalt.

Question #13: Magma that cools slowly beneath the earth's surface produces
 (B). intrusive igneous rocks.
 Magma that cools slowly deep below the surface produces coarse-grained intrusive igneous rocks. Since magma below the surface loses heat slowly the atoms have sufficient time to produce large crystals, giving the intrusive rocks a coarse-grained texture.

Question #14: A fine-grained igneous rock made of mostly orthoclase feldspar, quartz crystals, and a few ferromagnesian crystals is probably
 (A). rhyolite.
 The igneous rock classification scheme is based on mineral composition and texture. A combination of orthoclase feldspars, quartz, and some ferromagnesian would be light colored, and low density rocks with the same mineral composition as granite. However, granite is coarse-grained and the fine-grained equivalent is named rhyolite.



Question #15: Igneous rocks are classified by
 (A). grain size and color.
 (B). density and texture.
 (C). mineral composition and texture.
 (D). mineral composition and hardness.

Question #16: Sedimentary rocks are classified by
 (A). sediment size or chemical composition.
 (B). mineral composition and color.
 (C). sediment size and hardness.

Question #15: Igneous rocks are classified by
 (C). mineral composition and texture.
 The igneous rock classification scheme is based on mineral composition and texture. For example, a dark colored, high density blend of plagioclase feldspars and ferromagnesian with a fine-grained texture is called basalt.

Question #16: Sedimentary rocks are classified by
 (A). sediment size or chemical composition.
 There are two sources of sediments that provide the raw materials for sedimentary rocks, (1) weathered rock fragments, and (2) dissolved rock materials. Weathered rock fragments are called clastic sediments, and these accumulate from rocks that are in various stages of being broken down. Sedimentary rocks that are formed from clastic sediments are named according to the sizes of the sediments. Dissolved rock materials

(D). sediment size alone. Question #17: Metamorphic rocks are classified according to (A). degree of foliation or recrystallization. (B). grain size or sediments color. (C). chemical composition or sediment size. (D). extent of melting. Question #18: What process in the rock cycle are you performing if you pack a snowball so tightly it becomes an iceball? (A). weathering, sedimentation (B). melting (C). deformation, recrystallization (D). none of the above. Question #19: What process in the rock cycle are you performing if you put salt on ice on your sidewalk? (A). weathering, sedimentation (B). melting (C). deformation, recrystallization (D). none of the above. Question #20: What process in the rock cycle are you performing if you mix your garbage in your garden. (A). weathering, sedimentation (B). melting (C). deformation, recrystallization (D). none of the above.	form chemical sediments that are deposited through (a) chemical precipitation from solution, (b) crystallization from evaporating water, or (c) as biological sediments. Question #17: Metamorphic rocks are classified according to (A). degree of foliation or recrystallization. Metamorphic changes caused by heat and pressure depend on the mineral composition of the parent rock and the extent of the pressure, temperature, and hot solutions that may or may not be present to induce chemical changes. Pressure on parent rocks with flat crystal flakes tends to align the flakes in parallel sheets. This new crystal alignment is called foliation, which gives a metamorphic rock the property of breaking along the planes between the aligned mineral grains. The extent of foliation is determined by the extent of the metamorphic changes. Some metamorphic rocks are nonfoliated because they consist mainly of one mineral, and the grains are not aligned into sheets. These are formed by a recrystallization of the parent rock. Question #18: What process in the rock cycle are you performing if you pack a snowball so tightly it becomes an iceball? (C). deformation, recrystallization The rock cycle concept views an igneous, a sedimentary, or a metamorphic rock as a temporary stage in the ongoing transformation of rocks to new types. Making an iceball by using heat and pressure sounds much like the deformation and recrystallization rock cycle process that takes place when metamorphic rocks are formed. Question #19: What process in the rock cycle are you performing if you put salt on ice on your sidewalk? (B). melting The rock cycle concept views an igneous, a sedimentary, or a metamorphic rock as a temporary stage in the ongoing transformation of rocks to new types. Melting ice sounds much like the melting process that takes place in the rock cycle to form igneous rocks. Question #20: What process in the rock cycle are you performing if you mix your garbage in your garden. (A). weathering, sedimentation The rock cycle concept views an igneous, a sedimentary, or a metamorphic rock as a temporary stage in the ongoing transformation of rocks to new types. Mixing garbage in your garden sounds like the rock cycle process of sedimentation and weathering.
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Inside the Earth

Questions	Answers with Rationalization
Question #1: Seismic waves moving through the earth's crust suggest (A). oceanic crust of rocks like basalt; continental crust of rocks like granite. (B). continental crust of quartzite. (C). oceanic crust of sedimentary rock. (D). both oceanic and continental crust of rocks like basalt.	Question #1: Seismic waves moving through the earth's crust suggest (A). oceanic crust of rocks like basalt; continental crust of rocks like granite. Seismic studies have found that the earth's crust is a thin skin that covers the entire earth, existing below the oceans as well as making up the continents. The crust beneath the oceans is much thinner than the continental crust and seismic waves move through the oceanic crust faster than they do through continental crust. This is explained by the two types of crust being made up of different kinds of igneous rock: Continental crust is less dense, granite-type rock while the oceanic crust is more dense, basalt-type of rock.
Question #2: The Mohorovicic discontinuity is the seismic boundary between (A). crust and mantle. (B). asthenosphere and lithosphere (C). mantle and core. (D). mantle and asthenosphere	Question #2: The Mohorovicic discontinuity is the seismic boundary between (A). crust and mantle. The boundary between the crust and the mantle was discovered in 1909 by the Yugoslavian scientist Mohorovicic. The boundary is a zone one or two km thick where seismic P-waves increase in velocity because of changes in the composition of the materials. Seismic waves increase in velocity at the boundary because the composition on both sides is different. The mantle is richer in ferromagnesian minerals and poorer in silicon than the crust.
Question #3: Which type of seismic wave does not pass through a fluid? (A). body wave (B). P-wave (C). S-wave (D). surface wave	Question #3: Which type of seismic wave does not pass through a fluid? (C). S-wave An S-wave is a sideways (or transverse) wave in which the disturbance vibrates materials from side to side, perpendicular to the direction of wave movement. Both P- and S-waves pass easily through solid rock. A P-wave can also pass through a fluid but a S-wave cannot because you can compress a fluid (P-wave) but you cannot shear a fluid (S-wave).
Question #4: The asthenosphere is located (A). between the crust and the mantle. (B). in the mantle. (C). between the crust and the lithosphere. (D). in the crust.	Question #4: The asthenosphere is located (B). in the mantle. The asthenosphere is a thin zone in the mantle, from a depth of 130 km to 160 km. The asthenosphere is plastic, mobile, and yields to stresses. The rocks above and below the asthenosphere are rigid, solid, and brittle. The rocks above, which includes the crust, is called the lithosphere.
Question #5: The size and shape of the earth's core can be measured by information from the (A). earth's magnetic field. (B). S-wave shadow zone. (C). P-wave shadow zone. (D). nature of meteorites.	Question #5: The size and shape of the earth's core can be measured by information from the (C). P-wave shadow zone. Seismic P-waves spread throughout the earth from a large earthquake. These waves are measured by seismic recording stations all around the world except between 103° and 142° of arc from the earthquake. This is the P-wave shadow zone, explained by P-waves being refracted by the core, leaving a shadow. The paths of P-waves can be accurately calculated, so the size and shape of the earth's core can also be accurately calculated.

Question #6: Part of the earth's core is believed to be liquid as indicated by information from the
(A). P-wave shadow zone.
(B). S-wave shadow zone.
(C). earth's magnetic field.
(D). nature of meteorites.

Question #7: The earth's crust
(A). makes up about one-third of the mass of the earth.
(B). is only found in continents.
(C). exists below the oceans and make up the continents.
(D). supports the asthenosphere, which floats in the crust.

Question #8: The earth's magnetic field
(A). occasionally undergoes magnetic reversal, flipping the poles.
(B). is and always has been stable and with a constant orientation.
(C). will cease to exist forever within the next 2,000 years.
(D). none of the above.

Question #9: The least dense rocks are found in
(A). continental crust.
(B). oceanic crust.
(C). the mantle.
(D). the core.

Question #10: The asthenosphere is
(A). the uppermost layer of the mantle.
(B). a plastic layer in which the lithosphere is imbedded.
(C). the boundary between the mantel and the core.
(D). the solid, dense layer below the moho.

Question #11: Which of the following statements about seismic waves is not true?
(A). In a P-wave the material vibrates back and forth in the

Question #6: Part of the earth's core is believed to be liquid as indicated by information from the
(B). S-wave shadow zone.
Seismic S-waves spread throughout the earth from a large earthquake. These waves are measured by seismic recording stations all around the world except they are not recorded in the entire region more than 103 away from the epicenter. The S-wave shadow zone is formed because S-waves cannot travel through the earth's core. This, and other seismic data indicate that the outer part is liquid, or at least it acts like a liquid.

Question #7: The earth's crust
(C). exists below the oceans and make up the continents.
Seismic studies have found that the earth's crust is a thin skin that covers the entire earth, existing below the oceans as well as making up the continents. The crust beneath the oceans is much thinner than the continental crust and seismic waves move through the oceanic crust faster than they do through continental crust. This is explained by the two types of crust being made up of different kinds of igneous rock: Continental crust is less dense, granite-type rock while the oceanic crust is more dense, basalt-type of rock.

Question #8: The earth's magnetic field
(A). occasionally undergoes magnetic reversal, flipping the poles.
There is nothing static about the earth's magnetic poles. Geophysical studies have found that the magnetic poles are moving slowly around the geographic poles. Studies have also found that the earth's magnetic field occasionally undergoes magnetic reversal. During a magnetic reversal, the north magnetic pole and the south magnetic pole exchange positions. The present magnetic field orientation has persisted for the past 700,000 years and is now decreasing in strength as it prepares for another reversal.

Question #9: The least dense rocks are found in
(A). continental crust.
Continental crust is less dense, granite-type rock, while the oceanic cruse is more dense, basaltic rock. Both types of crust behave as if they were floating on the mantle, which is more dense than either type of crust.

Question #10: The asthenosphere is
(B). a plastic layer in which the lithosphere is imbedded.
The asthenosphere is a thin zone in the mantle, from a depth of 130 km to 160 km. The asthenosphere is plastic, mobile, and yields to stresses. The rocks above and below the asthenosphere are rigid, solid, and brittle. The rocks above, which includes the crust, is called the lithosphere. The lithosphere is made up of comparatively rigid plates that are moving, floating in the upper mantle like giant tabular ice sheets floating in the ocean.

Question #11: Which of the following statements about seismic waves is not true?
(B). Both P-waves and S-waves travel readily through molten rock.
A P-wave is a pressure (or longitudinal) wave in which the disturbance vibrates materials back and forth in a direction parallel to the direction of

same direction as the direction of wave motion. (B). Both P-waves and S-waves travel readily through molten rock. (C). P-waves move more rapidly than S-waves. (D). Seismic waves are reflected at boundaries between layers of rocks with different densities. Question #12: Which of the following statements about the earth's core is true? (A). Once molten, the core has gradually solidified as it cooled. (B). P-waves cannot travel through this layer of the earth. (C). The core makes up about 1/3 of the earth's volume. (D). the core is mostly made of iron. Question #13: The theory of plate tectonics describes sea-floor spreading as occurring at a (A). divergent boundary. (B). subduction zone. (C). convergent boundary. (D). transform boundary. Question #14: Earthquakes in California that occur along the San Andreas fault are associated with a (A). subduction zone. (B). transform boundary. (C). convergent boundary. (D). divergent boundary. Question #15: Volcanic mountains on the land and along a coast usually means a (A). ocean-ocean plate convergence. (B). continent-continent plate convergence. (C). ocean-continent plate	wave movement. A S-wave is a sideways (or transverse) wave in which the disturbance vibrates materials from side to side, perpendicular to the direction of wave movement. Both P- and S-waves pass easily through solid rock. A P-wave can also pass through a fluid but a S-wave cannot because you can compress a fluid (P-wave) but you cannot shear a fluid (S-wave). Since molten rock is a liquid S-waves cannot travel through molten rock. Question #12: Which of the following statements about the earth's core is true? (D). the core is mostly made of iron. The core makes up about 15 percent of the earth's total volume and about one-third of its mass. This data, plus information from the nature of meteorites indicates that the earth's core is mostly made of iron or a nickel and iron alloy. Question #13: The theory of plate tectonics describes sea-floor spreading as occurring at a (A). divergent boundary. Divergent boundaries occur between two plates moving away from each other. Magma forms as the plates separate, decreasing pressure on the mantle below. This molten material from the asthenosphere rises, cools, and adds new crust to the edges of the separating plates. The new crust tends to move horizontally from both sides of the divergent boundary, usually known as an oceanic ridge. A divergent boundary is thus a new crust zone. Question #14: Earthquakes in California that occur along the San Andreas fault are associated with a (B). transform boundary. Transform boundaries occur between two plates sliding by each other. Crust is neither created nor destroyed at transform boundaries as one plate slides horizontally past another along a long, vertical fault. The movement is neither smooth nor equal along the length of the fault, however, as short segments move independently with sudden jerks that are separated by periods without motion. The Pacific Plate, for example, is moving slowly to the northwest, sliding past the North American Plate. The San Andreas Fault is one boundary along the California coastline. Vibrations from plate movements along this boundary are the famous California earthquakes. Question #15: Volcanic mountains on the land and along a coast usually means a (C). ocean-continent plate convergence. Continent-oceanic plate convergence produces a characteristic set of geologic features as the oceanic plate of denser basaltic material is subducted beneath the less dense granite-type continental plate. The subduction zone is marked by an oceanic trench, deep-seated earthquakes, and volcanic mountains on the continent). The trench is
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convergence. (D). none of the above. Question #16: New crustal material is produced where according to the theory of plate tectonics? (A). divergent plate boundaries (B). subduction zones (C). ocean-continent plate convergence (D). ocean-ocean plate convergence Question #17: The average rate of sea-floor spreading that occurs along the Mid-Atlantic ridge is about (A). 2.5 cm/yr (B). 23 m/yr (C). 0.0003 cm/yr (D). 234 m/yr Question #18: When the lithosphere of one place is subducted beneath a second plate, the subducted material does not become (A). new mantle material. (B). lava that is erupted from volcanoes. (C). a body of granite below the surface. (D). part of the outer molten core	formed from the down-bending associated with subduction and the volcanic mountains from subducted and melted crust that rise up through the overlying plate to the surface. The earthquakes are associated with the movement of the subducted crust under the overlying crust. Question #16: New crustal material is produced where according to the theory of plate tectonics? (A). divergent plate boundaries Divergent boundaries occur between two plates moving away from each other. Magma forms as the plates separate, decreasing pressure on the mantle below. This molten material from the asthenosphere rises, cools, and adds new crust to the edges of the separating plates. The new crust tends to move horizontally from both sides of the divergent boundary, usually known as an oceanic ridge. A divergent boundary is thus a new crust zone. Most new crust zones are presently on the seafloor, producing sea-floor spreading. Question #17: The average rate of sea-floor spreading that occurs along the Mid-Atlantic ridge is about (A). 2.5 cm/yr The average rate of spreading along the Mid-Atlantic Ridge is about 2.5 centimeters per year (about an inch). This may seem slow, but the process has been going on for millions of years and has caused a tiny inlet of water between the continents of Europe, Africa, and the Americas to grow into the vast Atlantic Ocean that exists today. Question #18: When the lithosphere of one place is subducted beneath a second plate, the subducted material does not become (D). part of the outer molten core The creation of new crust at a divergent boundary means that old crust must be destroyed somewhere else at the same rate, or else the earth would have a continuously expanding diameter. Old crust is destroyed by returning to the asthenosphere at convergent boundaries. The collision produces an elongated belt of down-bending called a subduction zone. The lithosphere of one plate, which contains the crust, is subducted beneath the second plate and partly melts, becoming part of the mantle. The more dense components of this subducted crust may become igneous materials that remain in the mantle. Some of it may eventually migrate to a spreading ridge to make new crust again. The less dense components may return to the surface as a silicon, potassium, and sodium-rich lava, forming volcanoes on the upper plate, or it may cool below the surface to form a body of granite.
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Building Earth's Surface

Questions	Answers with Rationalization
Question #1: The basic concept needed to understand the principle of uniformity is (A). immense spans of geologic time. (B). the lack of sudden or catastrophic events. (C). an earth that has not changed since it was formed. (D). that catastrophic events formed all features on the earth's surface.	Question #1: The basic concept needed to understand the principle of uniformity is (A). immense spans of geologic time. The basic concept in understanding the principle of uniformity is the concept of immense spans of geologic time. Immense spans of time with slow, incomprehensible change taking place is difficult to comprehend since it cannot be observed or experienced in a lifetime. Thus, understanding the principle of uniformity requires a mental model. This model is based on the observable events that build up the surface and wear it down and on an understanding of geologic time.
Question #2: Rocks that are stressed by compressional forces, then return to their original shape have undergone (A). fracture strain. (B). plastic strain. (C). elastic strain. (D). mobile strain.	Question #2: Rocks that are stressed by compressional forces, then return to their original shape have undergone (C). elastic strain. The adjustment to stress is called strain and there are three types: elastic, plastic, and fracture. In elastic strain, rock units recover their original shape after the stress is released. In plastic strain, rock units are molded or bent under stress and do not return to their original shape after the stress is released. In fracture strain rock units crack or break as the name suggests.
Question #3: Rocks that are stressed by pulling forces, then bending in a way that they do not return to their original shape have undergone (A). elastic strain. (B). plastic strain. (C). fracture strain. (D). mobile strain.	Question #3: Rocks that are stressed by pulling forces, then bending in a way that they do not return to their original shape have undergone (B). plastic strain. The adjustment to stress is called strain and there are three types: elastic, plastic, and fracture. In elastic strain, rock units recover their original shape after the stress is released. In plastic strain, rock units are molded or bent under stress and do not return to their original shape after the stress is released. In fracture strain rock units crack or break as the name suggests.
Question #4: Anticlines, synclines, and domes are evidence of (A). fracture strain (B). elastic strain. (C). plastic strain. (D). mobile strain.	Question #4: Anticlines, synclines, and domes are evidence of (C). plastic strain. Anticlines, synclines, and domes are regional structures that were created from plastic deformation of flat, horizontal layers of sedimentary rocks. When the folding occurred the rock layers were in a ductile condition, probably under considerable confining pressure from deep burial. The anticlines, synclines, or domes are under very different conditions when they can be seen at the surface.
Question #5: Normal faulting is associated with (A). elastic strain. (B). plastic strain. (C). pulling-apart stress. (D). compressional stress.	Question #5: Normal faulting is associated with (C). pulling-apart stress. A normal fault is one in which an upper block of rock (the hanging wall) has moved downward relative to a block of rock below (the footwall). This is normal in the sense that you would expect an upper block to slide down a lower block along a slope. Normal faulting results from a pulling-apart stress that might be associated with diverging plates.

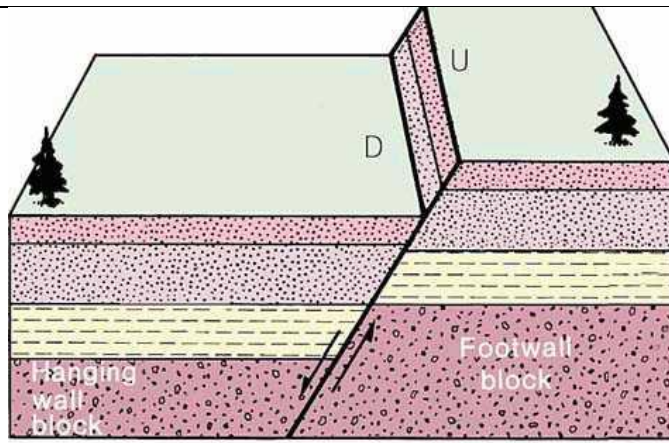
Question #6: Reverse or thrust faulting is probably the result of

- (A). pulling-apart stress.
- (B). compressional stress.
- (C). elastic stress.
- (D). plastic strain.

Question #7: About 15% of all the earthquakes that occur in the world do not have a shallow focus and occur

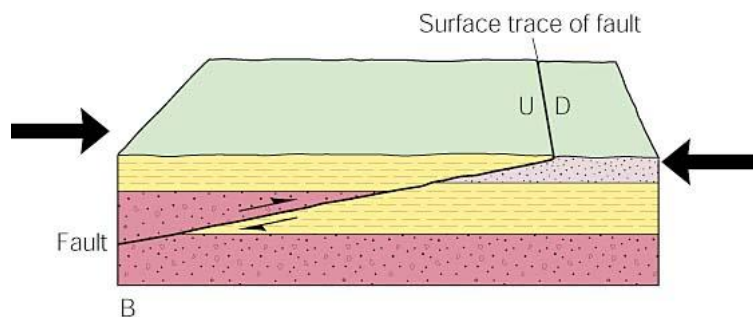
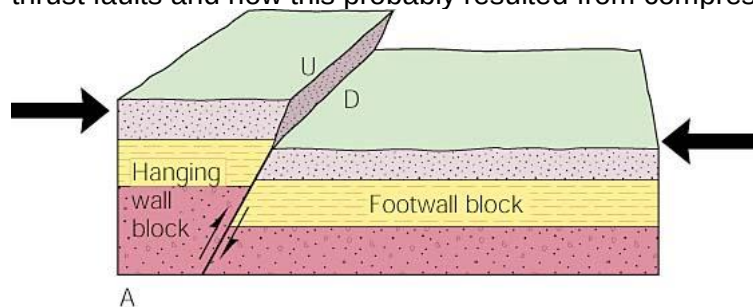
- (A). around the Mediterranean Ocean.
- (B). in the State of California, U.S.A.
- (C). in a narrow zone around the Pacific Ocean.
- (D). equally distributed around the world.

Question #8: Most earthquakes that occur worldwide are



Question #6: Reverse or thrust faulting is probably the result of (B). compressional stress.

In a reverse fault the hanging wall block has moved upward relative to the footwall block. The hanging and footwall relationship in reverse and thrust faults and how this probably resulted from compressive stress.



Question #7: About 15% of all the earthquakes that occur in the world do not have a shallow focus and occur (C). in a narrow zone around the Pacific Ocean.

About 85 percent of all earthquakes are of a shallow focus, occurring in the top 70 km (about 45 miles) of the surface, 12 percent are intermediate-focus earthquakes -- 70 to 350 km deep (45 and 220 mi) -- that occur in the upper part of the mantle, and 3 percent are deep-focus earthquakes that occur in the lower part of the upper mantle. There is a worldwide pattern to the distribution of earthquakes, as most occur in long narrow belts, although they do occasionally occur elsewhere. Of all the intermediate-depth earthquakes in the world, 9 out of 10 occur in a narrow zone, or belt, which encircles the rim of the Pacific Ocean. Essentially all the earth's deep-depth earthquakes also occur within this particular belt.

Question #8: Most earthquakes that occur worldwide are

- (A). near the surface along a fault.
(B). deep, away from plate boundaries.
(C). deep, but along converging plate boundaries.
(D). near the surface away from plate boundaries.

Question #9: In California the boundary between the North American Plate and the Pacific Plate is known as

- (A). Juan de Fuca zone.
(B). San Andreas fault.
(C). Cascade volcano zone
(D). Pacific Plate Beltway.

Question #10: Most earthquakes are explained by

- (A). the cooling and contracting of the earth.
(B). the movement of rock blocks along faults.
(C). an expanding earth surface from the creation of new crust.
(D). rock slides in deep, subterranean caverns.

Question #11: The place on the earth's surface directly above the place where seismic waves originate is the

- (A). origin.
(B). focus.
(C). seismoplace.

- (A). near the surface along a fault.

About 85 percent of all earthquakes are of a shallow focus, occurring in the top 70 km (about 45 miles) of the surface, and along a fault plane. You might expect more earthquakes near the earth's surface since the rocks here are brittle, and those deeper are more ductile from increased temperature and pressure.

Question #9: In California the boundary between the North American Plate and the Pacific Plate is known as

- (B). San Andreas fault.

Shallow-focus earthquakes are typical of those that occur at the boundary of the North American Plate, which is moving against the Pacific Plate. In California, the boundary between these two plates is known as the San Andreas fault. The San Andreas fault runs north-south for some 1,300 km (800 miles) through California, with the Pacific Plate moving on one side and the North American Plate moving on the other. The two plates are tightly pressed against each other, and friction between the rocks along the fault prevents them from moving easily. Stress continues to build along the entire fault as one plate attempts to move along the other. Some elastic deformation does occur from the stress, but eventually the rupture strength of the rock (or the friction) is overcome. The stressed rock, now released of the strain, snaps suddenly into new positions in the phenomenon known as elastic rebound. The rocks are displaced to new positions on either side of the fault, and the vibrations from the sudden movement are felt as an earthquake. The elastic rebound and movement tend to occur along short segments of the fault at different times rather than along long lengths. Thus, the resulting earthquake tends to be a localized phenomenon rather than a regional one.

Question #10: Most earthquakes are explained by

- (B). the movement of rock blocks along faults.

An earthquake is a quaking, shaking, vibrating, or upheaval of the ground, a result of the sudden release of energy that comes from stress on rock beneath the earth's surface. There are limits as to how much stress rock can take before it fractures. When it does fracture, the sudden movement of blocks of rock produces vibrations that move out as waves throughout the earth. These vibrations are called seismic waves. It is strong seismic waves that people feel as a shaking, quaking, or vibrating during an earthquake. Seismic waves are generated when a huge mass of rock breaks and slides into a different position. Major earthquakes occur along existing fault planes or when a new fault is formed by the fracturing of rock. In either case, most earthquakes occur along a fault plane when there is displacement of one side relative to the other.

Question #11: The place on the earth's surface directly above the place where seismic waves originate is the

- (D). epicenter.

The actual place where seismic waves originate beneath the surface is called the focus of the earthquake. The focus is considered to be the center of the earthquake and the place of initial rock movement on a fault. The point on the earth's surface directly above the focus is called the earthquake epicenter.

(D). epicenter.

Question #12: All seismic waves leave the focus of an earthquake at the same time, but some distance away the ___?___ arrive first.

- (A). P-waves
- (B). S-waves
- (C). surface waves
- (D). none of the above.

Question #13: The time lag between the arrival of S- and P-waves is needed from a minimum of how many recording stations to locate the source of an earthquake?

- (A). 4
- (B). 3
- (C). 2
- (D). 1

Question #14: A very large ocean wave generated by an earthquake, landslide, or volcanic explosion is known as a (an)

- (A). tidal wave.
- (B). tsunami.
- (C). swell wave.
- (D). amplitude wave.

Question #15: The magnitude of an earthquake is usually reported by numbers and each higher number means

- (A). 10 times more energy.
- (B). an increasing level of awareness by people at the epicenter.
- (C). 10 times more movement and 30 times more energy.

Question #12: All seismic waves leave the focus of an earthquake at the same time, but some distance away the ___?___ arrive first.

(A). P-waves

Seismic S- and P-waves leave the focus of an earthquake at essentially the same time. As they travel away from the focus, they gradually separate because the P-waves travel faster than the S-waves.

Question #13: The time lag between the arrival of S- and P-waves is needed from a minimum of how many recording stations to locate the source of an earthquake?

(B). 3

To locate an epicenter, at least three recording stations measure the time lag between the arrival of the P-waves and the slower S-waves. The difference in the speed between the two waves is a constant. Therefore, the farther they travel, the greater the time lag between the arrival of the faster P-waves and the slower S-waves. By measuring the time lag and knowing the speed of the two waves, it is possible to calculate the distance to their source. However, the calculated distance provides no information about the direction or location of the source of the waves. The location is found by first using the calculated distance as the radius of a circle drawn on map. The place where the circles from the three recording stations intersect is the location of the source of the waves.

Question #14: A very large ocean wave generated by an earthquake, landslide, or volcanic explosion is known as a (an)

(B). tsunami.

Tsunami is a Japanese term now used to describe the very large ocean waves that can be generated by an earthquake, landslide, or volcanic explosion. Such large waves were formerly called "tidal waves." Since the large, fast waves were not associated with tides or tidal forces in any way, the term tsunami is preferred. A tsunami, like other ocean waves, is produced by some strong disturbance in the seafloor, travels at speeds of 725 km/hr (450 mi/hr), and produces a wave height of 15 to 30 m (50 to 100 ft) when it breaks on the shore. Because of its great wavelength, a tsunami does not just break on the shore, then withdraw. Depending on the sea-floor topography, the water from a tsunami may continue to rise for 5 to 10 minutes, flooding the coastal region before the wave withdraws. A gently sloping seafloor and a funnel-shaped bay can force tsunamis to great heights as they break on the shore.

Question #15: The magnitude of an earthquake is usually reported by numbers and each higher number means

(C). 10 times more movement and 30 times more energy.

The energy of the vibrations, or motion of the land associated with an earthquake is called its magnitude. Earthquake magnitude is often reported by the media using the Richter scale. This scale assigns a number that increases with the magnitude of an earthquake. The numbers have meaning about the severity of the ground-shaking vibrations, and the energy released by the earthquake. Each higher

(D). a doubling of the amplitude of a swinging pendulum. Question #16: One of the following was not formed by complex folding resulting from compressional forces. (A). Cascade Range (B). Appalachian Range (C). Rocky Mountain Range (D). Himalayan Range Question #17: The Teton Mountains of Wyoming and the Sierra Nevadas of California are classic examples of (A). fault block mountains. (B). compressional folding. (C). volcanic intrusion. (D). complex folding. Question #18: The volcanoes of the Cascade Mountain Range are (A). shield volcanoes. (B). low profile lava flows. (C). cinder cone volcanoes. (D). composite volcanoes.	number indicates about 10 times more ground movement and about 30 times more energy released than the preceding number. Question #16: One of the following was not formed by complex folding resulting from compressional forces. (A). Cascade Range The Cascade Mountains of Washington and Oregon are a series of towering volcanic peaks. The Appalachian, Rocky, and Himalayan Mountains, on the other hand, have a great vertical relief that involves complex folding on a very large scale. The crust was thickened in these places as compressional forces produced tight, almost vertical folds. Thus, folding is a major feature of these major mountain ranges, but faulting and igneous intrusions are invariably also present. Differential weathering of different rock types produced the parallel features of the Appalachian Mountains that are so prominent in satellite photographs. The folded sedimentary rocks of the Rockies are evident in the almost upright beds along the flanks of the front range. Question #17: The Teton Mountains of Wyoming and the Sierra Nevadas of California are classic examples of (A). fault block mountains. Compression and relaxation of compressional forces on a regional scale can produce large-scale faults, shifting large crustal blocks up or down relative to one another. Huge blocks of rocks can be thrust to mountainous heights, creating a series of fault block mountains. Fault block mountains rise sharply from the surrounding land along the steeply inclined fault plane. The mountains are not in the shape of blocks, however, as weathering has carved them into their familiar mountain-like shapes. The Teton Mountains of Wyoming and the Sierra Nevadas of California are classic examples of fault block mountains that rise abruptly from the surrounding land. Question #18: The volcanoes of the Cascade Mountain Range are (D). composite volcanoes. A composite volcano is built up of alternating layers of cinders, ash, and lava flows, forming what many people believe is the most imposing and majestic of earth's mountains. The steepness of the sides is somewhere between the steepness of the low shield volcanoes and the steep cinder cone volcanoes. The Cascade volcanoes are composite volcanoes, but the mixture of lava flows and cinders seems to vary from one volcano to the next.
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Shaping Earth's Surface

Questions	Answers with Rationalization
Question #1: Which of the following must occur before the others can take place? (A). weathering (B). erosion (C). transportation (D). exfoliation	Question #1: Which of the following must occur before the others can take place? (A). weathering Weathering prepares the way for erosion by breaking solid rock into rock fragments. The fragments are then eroded, physically picked up by an agent such as a stream or a glacier. After they are eroded the materials are removed by transportation. Thus weathering must occur before erosion or transportation can take place. Exfoliation is a specific type of mechanical weathering that occurs in granite. Other types of weathering do occur and exfoliation is not necessary before erosion and transportation can take place.
Question #2: The red soils of Georgia, Oklahoma, and other places are one result of (A). frost wedging. (B). exfoliation. (C). chemical weathering. (D). erosion.	Question #2: The red soils of Georgia and Oklahoma are one result of (C). chemical weathering. The ferromagnesian minerals contain iron, magnesium, and other metal ions in a silicate structure. Iron can react with oxygen to produce several different iron oxides, each with its own characteristic color. The most common iron oxide is hematite and has a deep red color. It is the presence of such iron oxides that color many sedimentary rocks and soils. The red soils of Georgia, Oklahoma, and other places are colored by the presence of iron oxides produced by chemical weathering.
Question #3: A cold, dry climate would probably result in mostly (A). chemical weathering. (B). mechanical weathering. (C). exfoliation. (D). erosion.	Question #3: A cold, dry climate would probably result in mostly (B). mechanical weathering. Chemical weathering is the alteration of minerals by chemical reactions with water, gases of the atmosphere, or solutions. Mechanical weathering, on the other hand, is the physical breaking up of rocks without any changes in their chemical composition. Chemical change is slowed by cold, dry conditions and mechanical weathering is not so temperature dependent. Thus there is more mechanical weathering than chemical in a cold, dry climate.
Question #4: The roots of a tree grow into cracks and wedge the rocks apart as they grow. This is an example of (A). chemical weathering. (B). hydration. (C). mechanical weathering. (D). exfoliation.	Question #4: The roots of a tree grow into cracks and wedge the rocks apart as they grow. This is an example of (C). mechanical weathering. Mechanical weathering, is the physical breaking up of rocks without any changes in their chemical composition. The roots of trees and shrubs mechanically wedge rocks apart as they grow into cracks, exerting pressure on the walls of the crack and making it larger and larger as the root grows.
Question #5: A cavern in a limestone formation was probably produced by (A). oxidation. (B). hydration. (C). carbonation. (D). frost wedging.	Question #5: A cavern in a limestone formation was probably produced by (C). carbonation. Carbonation is a reaction between carbonic acid and the minerals making up rocks. Rainwater is naturally somewhat acidic because it dissolves carbon dioxide from the air. This forms a weak acid known as carbonic acid, which is the same acid found in your carbonated soda pop. Carbonic acid rain falls on the land, seeping into cracks and crevices where it reacts with minerals. Limestone is easily weathered to a soluble form by carbonic acid and many limestones caves were produced by the chemical weathering of limestone by carbonation.

Question #6: An abundance of humus with balanced amounts of sand, silt, and clay would be found in (A). subsoil. (B). topsoil. (C). bedrock. (D). middle soil.	Question #6: An abundance of humus with balanced amounts of sand, silt, and clay would be found in (B). topsoil. A soil that has balanced amounts of sand, silt, and clay mixed with an abundance of humus is called loam. Loam is a great soil for gardening since it is fertile and well drained, yet holds enough moisture for sustained plant growth. Loam is usually found in the topmost layers of soil, so it is also referred to as topsoil. It is usually more fertile because it is closer to the source of humus, which is altered, decay-resistant organic matter.
Question #7: In general, deeper and richer soils should be found in which type of climate? (A). dry and warm (B). wet and cold (C). wet and warm (D). dry and cold	Question #7: In general, deeper and richer soils should be found in which type of climate? (C). wet and warm In general, soils formed in cold and dry climates are shallower with less humus than soils produced in wet and warm climates. This happens because chemical reactions occur at a faster pace in warmer, wetter soil than they do in dry, cooler soil. Wet and warm climate is also more conducive to plant growth, which would provide more organic matter for the formation of humus.
Question #8: Which of the following is most effective in modifying the earth's surface? (A). running water (B). landslides (C). wind (D). glaciers	Question #8: Which of the following is most effective in modifying the earth's surface? (A). running water Running water is the most important of all the erosional agents of gravity that remove rocks and rock materials to lower levels. Streams and major rivers are at work, for the most part, 24 hours a day every day of the year moving rock fragments and dissolved materials from elevated landmasses to the oceans.
Question #9: Rapids and waterfalls are characteristic features of a stream valley in a stage of (A). youth. (B). maturity. (C). old age. (D). rejuvenation	Question #9: Rapids and waterfalls are characteristic features of a stream valley in a stage of (A). youth. The development of a stream channel into a widening floodplain seems to follow a general, idealized aging pattern. When a stream is on a recently uplifted landmass, it has a steep gradient, a vigorous, energetic ability to erode the land, and characteristic features known as the stage of youth. Youth is characterized by a steep gradient, a V-shaped valley without a floodplain, and the presence of features that interrupt its smooth flow such as boulders in the stream bed, rapids, and waterfalls.
Question #10: Flooding is more common in a stream valley that is in a state of (A). maturity. (B). old age. (C). youth. (D). rejuvenation.	Question #10: Flooding is more common in a stream valley that is in a state of (B). old age. The development of a stream channel into a widening floodplain seems to follow a general, idealized aging pattern. Old age is marked by a very low gradient in extremely broad, gently sloping valleys. The stream now flows slowly in broad meanders over the wide floodplain. Floods are more common in old age since the stream is carrying a full load of sediments and flows sluggishly.
Question #11: You should expect more old age features where a stream (A). flows into the ocean; at its mouth.	Question #11: You should expect more old age features where a stream (A). flows into the ocean; at its mouth. Streams and rivers are dynamic systems that respond to local conditions, so it is possible to find an "old age feature" such as meanders in an otherwise youthful valley. In general, old age

(B). is created; at its source. (C). halfway between its mouth and source. (D). none of the above. Question #12: Today, glaciers cover about what percent of Alaska? (A). nearly 33% (B). about 25% (C). more than 10% (D).less than 3% Question #13: A ridge-like glacial deposit is the (A). moraine. (B). cirque. (C). delta. (D). sand dune. Question #14: A U-shaped mountain valley was most likely formed by a (an) (A). stream in its youth stage. (B). glacier. (C). moving sand dune. (D). earthquake. Question #15: A mountain stream with gray to blue-gray water from suspended silt is most likely a result of (A). mass wasting. (B). glacial erosion. (C). wind erosion. (D). sheet erosion. Question #16: The landscape is most likely to have sharp angular outlines in which type of climate? (A). wet, cool (B). wet, warm (C). dry, cool (D). alternating cool and warm Question #17: Which of the following is more highly acidic?	characteristics are observed near the mouth of a stream where it flows into an ocean, lake, or another stream. Youthful characteristics are observed at the source, where the water collects to first form the stream channel. As the stream slowly lowers the land, the old age characteristics will move slowly but surely toward the source. Question #12: Today, glaciers cover about what percent of Alaska? (D).less than 3% Today, the most extensive glaciers in the United States are those of Alaska, which covers less than 3 percent of the state's land area. Question #13: A ridge-like glacial deposit is the (A). moraine. A glacier does its erosional work by bulldozing, abrasion, and plucking. Bulldozing is the pushing along of rocks, soil, and sediments by the leading edge of an advancing glacier. Deposits of bulldozed rocks and other materials that remain after the ice melts are called moraines. Question #14: A U-shaped mountain valley was most likely formed by a (an) (B). glacier. As an alpine glacier moves downhill through a V-shaped valley the sides and bottom of the valley are eroded wider and deeper. When the glacier later melts, the V-shaped valley is now a U-shaped valley that has been straightened and deepened by glacial erosion. Question #15: A mountain stream with gray to blue-gray water from suspended silt is most likely a result of (B). glacial erosion. A glacier does part of its erosional work by abrasion, which occurs as rock fragments frozen into the moving glacial ice scratch, polish, and grind against surrounding rocks at the base and along the valley walls. One result of this abrasion is the pulverizing of rock into ever finer fragments, eventually producing a powdery, silt-sized sediment called rock flower. Suspended rock flower in meltwater from a glacier gives the water a distinctive gray to blue-gray color. Question #16: The landscape is most likely to have sharp angular outlines in which type of climate? (C). dry, cool Chemical weathering is more dominant in warm, moist climates and mechanical weathering is more dominant in dry climates. Thus, landforms in warm, moist climates tend to have softer, rounded outlines from the accumulation of clay minerals, sand, and other finely divided products of chemical weathering. The landforms in dry climates, on the other hand, tend to have sharp angular outlines from the mass movement of rock materials from vertical cliffs. Lacking as much chemical weathering, the landscapes in dry climate regions tend to have sharper outlines. Question #17: Which of the following is more highly acidic? (A). grapefruit Pure, unpolluted rain is naturally acidic. Carbon dioxide in the
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(A). grapefruit (B). acid rain (C). pure, unpolluted rain (D). oranges	atmosphere is absorbed by rainfall, forming carbonic acid with a pH in a range of 5.6 to 6.2, compared to the pH of 7 for pure water. The pH of what is generally recognized as acid rain is even lower, made more acidic by the addition of exhaust emissions of sulfur and nitrogen oxides. Acid rain generally has a pH range of 4.0 to 5.5. Oranges and grapefruit are more acidic than acid rain, with grapefruit being more highly acidic with a pH in the range of 3.0 to 3.2. Nonetheless, acid rain does make a detrimental environmental impact on lakes, forest, crops, materials, and human health. It does not take a very high acidity to be damaging and acid rain with the acidity of grapefruit juice would be even more damaging.
Question #18: Increased CO₂ would probably increase which type of landscape feature? (A). caves in limestone (B). sand dunes in the desert (C). moraines in the mountains (D). deltas along the coast	Question #18: Increased CO₂ would probably increase which type of landscape feature? (A). caves in limestone There might be weather changes brought about by increased CO₂ -- such as the greenhouse effect -- that might cause changes in the desert, mountains, or coastal regions, but CO₂ plays a more direct role in the formation of limestone caves. Rainwater is acidic because it dissolves carbon dioxide from the air, forming the weak acid known as carbonic acid. More CO₂ to dissolve in rainwater might mean more acid, which would increase the weathering of limestone by carbonation.

Geologic Time

Questions	Answers with Rationalization
Question #1: When the trunk of a tree has been completely replaced by minerals, the tree is (A). petrified. (B). carbonized. (C). molded. (D). a cast.	Question #1: When the trunk of a tree has been completely replaced by minerals, the tree is (A). petrified. There are two processes involved in the making of petrified fossils, and they are not restricted to just wood. The processes involve mineralization, which is the filling of pore spaces with deposits of calcium carbonate, silica, or pyrite, and/or replacement, which is the dissolving of the original material and depositing of new material an ion at a time. Petrified wood is formed by both processes over a long period of time. As it decayed, the original wood was replaced by mineral matter. Over time the "mix" of minerals being deposited changed and the various resulting colors appear to preserve the texture of the wood.
Question #2: Which of the following is not a fossil? (A). a frozen mammoth (B). bat droppings in a cave (C). pot shards of an ancient tribe (D). an insect entombed in amber	Question #2: Which of the following is not a fossil? (C). pot shards of an ancient tribe A fossil is any evidence of former life, so the term means more than fossilized remains. Evidence can include actual or altered remains of plants and animals such as an insect entombed in amber or a frozen mammoth. It could also be just simple evidence of former life such droppings from bats in a cave. The word "fossil" carries the meaning of any evidence of ancient organisms in the history of life. Pot shards from an ancient tribe are therefore not fossils.
Question #3: Most fossils are usually found in (A). new, recently formed sedimentary rock. (B). recently eroded sedimentary rock. (C). glaciers. (D). sand dunes.	Question #3: Most fossils are usually found in (B). recently eroded sedimentary rock. A fossil must be protected, formed, and then found and studied to reveal its part in the history of life. This means the rocks in which the fossil formed must now somehow make it back to the surface of the earth. This usually involves movement and uplift of the rock, and weathering and erosion of the surrounding rock to release or reveal the fossil. Most fossils are found in recently eroded sedimentary rocks, before weathering and erosion has had the time to destroy them.
Question #4: A logical principle used in interpreting the history of rock layers is (A). contact metamorphosis. (B). angular unconformity. (C). superposition. (D). crosscutting relationships.	Question #4: A logical principle used in interpreting the history of rock layers is (C). superposition. The principle of superposition is a logical and obvious principle that is applied to sedimentary rocks. Layers of sediments are usually deposited in succession in horizontal layers, which later are compacted and cemented into layers of sedimentary rock. An undisturbed sequence of horizontal layers is thus arranged in chronological order with the oldest layers at the bottom. Each consecutive layer will be younger than the one below it, with the understanding that the layers have not been turned over by deforming forces.
Question #5: An unconformity in a sedimentary rock sequence results when (A). a fault cuts across a layer of sedimentary rock. (B). rock layers of different thickness are formed. (C). erosion occurs for a long	Question #5: An unconformity in a sedimentary rock sequence results when (C). erosion occurs for a long period of time. As erosion wears away the rock layers at a site, the sediments produced are deposited some place else. Later, the sites of erosion and deposition may shift, and the sediments are deposited on top of the eroded area. When the new sediments later are formed into new sedimentary rocks, there will be a time lapse between the top of the eroded layer and the

period of time. (D). magma intrudes across some layers of sedimentary rock. Question #6: Index fossils serve to identify (A). how thick a given rock layer should be. (B). specific layers of rocks anywhere. (C). the composition of different rock layers. (D). the exact location of oceans that no longer exist. Question #7: Which of the following is most important in reading the history from layers of sedimentary rocks? (A). understanding that younger rock layers are at the bottom of a sequence (B). measuring the thickness of each rock layer in a sequence (C). determining the composition of the rock layers (D). finding rock layers that have not been disturbed Question #8: Fossils show that over time some life forms disappeared from the record and new forms appeared. This is described by the logical frame of reference known as the principle of (A). faunal succession. (B). unconformity. (C). superposition. (D). uniformity. Question #9: An index fossil is most useful if it is a fossil of an organism that existed a (A). brief time over a wide area. (B). brief time over a small area.	new layers. A time break in the rock record is called an unconformity. The unconformity is usually shown by a surface within a sedimentary sequence on which there was a lack of sediment deposition, or where active erosion may even have occurred for some period of time. When the rocks are later examined, that time span will not be represented in the record, and if the unconformity is erosional, some of the record once present will have been lost. Question #6: Index fossils serve to identify (B). specific layers of rocks anywhere. Distinctive fossils of plant or animal species that were distributed widely over the earth, but lived only a brief time are called index fossils. Index fossils, together with the other principles used in reading rocks, make it possible to compare the ages of rocks exposed in two different locations. This is called age correlation between rock units. Correlations of exposed rock units separated by a few kilometers are easier to do, but correlations have been done with exposed rock units that are separated by an ocean. Question #7: Which of the following is most important in reading the history from layers of sedimentary rocks? (C). determining the composition of the rock layers The understanding of geologic processes has been made possible through the development of various means of measuring ages and time spans in geologic systems. An understanding of geologic time leads to an understanding of geologic processes, which then leads to an understanding of the environmental conditions that must have existed in the past. Thus, the mineral composition, texture, and sedimentary structure of rocks are clues about past events, events that make up the history of earth. Question #8: Fossils show that over time some life forms disappeared from the record and new forms appeared. This is described by the logical frame of reference known as the principle of (A). faunal succession. The principle of faunal succession recognizes that life forms have changed through time. Old life forms disappear from the fossil record and new ones appear, but the same form is never exactly duplicated independently at two different times in history. This principle implies that the same type of fossil organisms that lived only a brief geologic time should occur only in rocks that are the same age. According to the principle of faunal succession, then, once the basic sequence of fossil forms in the rock record is determined, rocks can be placed in their correct relative chronological position on the basis of the fossils contained in them. The principle also means that if the same type of fossil organism is preserved in two different rocks, the rocks should be the same age. This is logical even if the two rocks have very different compositions and are from places far, far apart Question #9: An index fossil is most useful if it is a fossil of an organism that existed a (A). brief time over a wide area. Distinctive fossils of plant or animal species that were distributed widely over the earth, but lived only a brief time are called index fossils. Index
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(C). long time over a small area. (D). long time over a wide area. Question #10: The earth dating method that is believed to be most accurate is (A). the rate of sediment deposition. (B). radioactive decay. (C). cooling of the earth from its original molten state. (D). rate that salt is being added to the oceans. Question #11: Major blocks of time in the earth's geologic history are based on (A). geomagnetic reversals of the earth's magnetic field. (B). volcanoes and other events in the breakup of Pangaea. (C). periodic meteorite bombardments of the earth. (D). appearance and disappearance of particular fossils in the rock record. Question #12: The least number of fossils are found in rocks formed during which era? (A). Precambrian (B). Cenozoic (C). Mesozoic (D). Paleozoic Question #13: Most of the earth's geologic history took place during which era?	fossils, together with the other principles used in reading rocks, make it possible to compare the ages of rocks exposed in two different locations. This age comparison can be made with more confidence if the index fossil is from an organism that lived for only a brief time and more comparisons can be made if it lived over a wide area. Question #10: The earth dating method that is believed to be most accurate is (B). radioactive decay. The discovery of the radioactive decay process in the elements of minerals and rocks led to the development of a new, accurate geologic clock. This clock finds the radiometric age of rocks in years by measuring the radioactive decay of unstable elements within the crystals of certain minerals. Since radioactive decay occurs at a constant, known rate, the ratio of the remaining amount of an unstable element to the amount of decay products present can be used to calculate the time that the unstable element has been a part of that crystal. Potassium, uranium, and thorium are radioactive isotopes that are often included in the minerals of rocks, so they are often used as radioactive clocks. Question #11: Major blocks of time in the earth's geologic history are based on (D). appearance and disappearance of particular fossils in the rock record. The major blocks of time in Earth's geologic history are called eras, and each era is identified by the appearance and disappearance of particular fossils in the sedimentary rock record. There are four main eras, which are: (1) Cenozoic, which refers to the time of recent life. Recent life means that the fossils for this time period are similar to the life found on earth today. (2) Mesozoic, which refers to the time of middle life. Middle life means that some of the fossils for this time period are similar to the life found on earth today, but many are different from anything living today. (3) Paleozoic, which refers to the time of ancient life. Ancient life means that the fossils for this time period are very different from anything living on the earth today. (4) Precambrian, which refers to the time before the time of ancient life. Question #12: The least number of fossils are found in rocks formed during which era? (A). Precambrian The part of the earth's history with evidence of life makes up about 15 percent of the total record and about 85 percent of the record occurred during the Precambrian era. This does not mean that life appeared suddenly in the Cambrian period. The fossil record is incomplete since it is the hard parts of animals or plants that form fossils, usually after rapid burial. Thus, the soft-bodied life forms that existed during the Precambrian era would make Precambrian fossils exceedingly rare. The Precambrian fossils that have been found are chiefly those of deposits from algae, a few fungi, and the burrow holes of worms. Another problem in finding ancient fossils of soft-bodied life forms is that heat and pressure have altered many of the ancient rocks over time, destroying any fossil evidence that may have been present. Question #13: Most of the earth's geologic history took place during which era?
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(A). Paleozoic (B). Precambrian (C). Mesozoic (D). Cenozoic Question #14: Which of the following is a Mesozoic Period? (A). Tertiary (B). Permian (C). Jurassic (D). Silurian Question #15: Which era could be called The Age of the Reptiles? (A). Mesozoic (B). Cenozoic (C). Paleozoic (D). Triassic Question #16: According to the rock record, fossils of fish first appeared in which period? (A). Jurassic (B). Devonian (C). Permian (D). Ordovician Question #17: The dominant life form of the Cambrian Period was (A). sharks (B). trilobites (C). fish (D). reptiles Question #18: Which era is known as The Age of the Mammals? (A). Precambrian (B). Paleozoic (C). Mesozoic (D). Cenozoic	(B). Precambrian Geologic history is divided into four main eras. The Precambrian era was first, lasting the first 4 billion years, or about 85 percent of the total 4.6 billion years of geologic time. The Paleozoic lasted about 10 percent of geologic time, the Mesozoic about 4 percent, and the Cenozoic only about 1.5 percent of all geologic time. Question #14: Which of the following is a Mesozoic Period? (C). Jurassic The Mesozoic refers to the time of middle life. Middle life means that some of the fossils for this time period are similar to the life found on earth today, but many are different from anything living today. The Mesozoic began with the extinction of many types of plant and animal life of the Paleozoic era and closed with a great dying of land and marine life that resulted in the extinction of many species, including the dinosaurs. The Mesozoic Era is subdivided into the Cretaceous, Jurassic, and Triassic Periods. Question #15: Which era could be called The Age of the Reptiles? (A). Mesozoic The dinosaurs first appeared in the Triassic, outnumbering all the other reptiles until the close of the Mesozoic. Fossils of the first birds, the first mammals, the first flowering plants, and the first deciduous trees appeared in the rocks of this era. Like the close of the Paleozoic, the Mesozoic era ended with a great dying of land and marine life that resulted in the extinction of many species, including the dinosaurs. Throughout the Mesozoic era the reptiles dominated life on the earth, so the Mesozoic era could be called The Age of the Reptiles. Question #16: According to the rock record, fossils of fish first appeared in which period? (D). Ordovician Fossils of fish first appear in the Ordovician, then become abundant and diversified by the Devonian. Sharks were common, as was a primitive form of an air-breathing fish. Primitive evergreen and fern-like trees appeared on the land at this time, according to the Devonian fossils. Question #17: The dominant life form of the Cambrian Period was (B). trilobites The earliest abundant fossils are found in rocks from the Cambrian period, representing an abundance of oceanic life with over a thousand different species of animals. There is no fossil evidence of life of any kind living on the land during the Cambrian. The dominant life forms of the Cambrian ocean were trilobites and brachiopods. The trilobites, now extinct, made up more than half the life population during the Cambrian. Question #18: Which era is known as The Age of the Mammals? (D). Cenozoic As the Cenozoic era opened, the dinosaurs were extinct and the mammals became the dominant life form. The Cenozoic is thus called The Age of the Mammals.
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The Atmosphere of Earth

Questions	Answers with Rationalization
Question #1: The most abundant gas in the atmosphere is (A). argon. (B). oxygen. (C). nitrogen. (D). water vapor.	Question #1: The most abundant gas in the atmosphere is (C). nitrogen. A sample of pure, dry air is colorless, odorless, and composed mostly of the molecules of just three gases, nitrogen, oxygen, and argon. Nitrogen is the most abundant (about 78 percent of the total volume), followed by oxygen (about 21 percent), then argon (about 1 percent). The molecules of these three gases are well mixed, and this composition is nearly constant everywhere near the earth's surface.
Question #2: The atmospheric gas that exists in variable amounts is (A). oxygen. (B). water vapor. (C). nitrogen. (D). argon.	Question #2: The atmospheric gas that exists in variable amounts is (B). water vapor. The atmosphere contains variable amounts of water vapor. Water vapor is the invisible, molecular form of water in the gaseous state, which should not be confused with fog or clouds. Fog and clouds are tiny droplets of liquid water, not water in the single molecular form of water vapor. The amount of water vapor in the atmosphere can vary from a small fraction of a percent composition by volume in cold, dry air to about 4 percent in warm, humid air. This small, variable percentage of water vapor is essential in maintaining life on the earth.
Question #3: The height of a mercury column in a barometer depends on (A). the width of the glass tube. (B). the height of the glass tube. (C). atmospheric pressure alone. (D). the width and height of the glass tube and the density of mercury.	Question #3: The height of a mercury column in a barometer depends on (C). atmospheric pressure alone. The mercury barometer measures the atmospheric pressure from the balance between the pressure exerted by the weight of the mercury in a tube and the pressure exerted by the atmosphere. As the atmospheric pressure increases and decreases, the mercury rises and falls. The average height of the column at sea level is 760.0 mm (29.92 inches).
Question #4: About how much incoming solar radiation makes it through the atmosphere to reach the earth's surface? (A). about a tenth (B). about a quarter (C). about half (D). all of it	Question #4: About how much incoming solar radiation makes it through the atmosphere to reach the earth's surface? (C). about half Radiation from the sun must pass through the atmosphere and the atmosphere filters, absorbs, and reflects it before it can reach the surface. On the average, the earth as a whole reflects about 30 percent of the total radiation back into space, with two-thirds of the reflection occurring from clouds. About 20 percent of the radiation is absorbed by air and clouds, leaving about one-half of the incoming solar radiation that reaches the earth's surface.
Question #5: Incoming solar radiation is absorbed by the earth's surface, which emits the energy with (A). about the same wavelength. (B). wavelengths longer than visible light. (C). wavelengths shorter than visible light. (D). variable wavelengths ranging over a wide range of frequencies	Question #5: Incoming solar radiation is absorbed by the earth's surface, which emits the energy with (B). wavelengths longer than visible light. The incoming solar radiation that reaches the earth's surface is absorbed rocks, soil, water, and the ground. These materials become warmer as a result emitting the absorbed energy as infrared radiation, wavelengths longer than the visible part of the electromagnetic spectrum.

on both sides of visible light.

Question #6: The greenhouse effect occurs because outgoing radiation is partly absorbed by
(A). oxygen and water vapor.
(B). carbon dioxide and oxygen.
(C). carbon dioxide and water vapor.
(D). oxygen, nitrogen, and argon.

Question #7: Temperature increases with altitude in the
(A). stratosphere.
(B). troposphere.
(C). mesosphere.
(D). no, temperature never increases with altitude.

Question #8: A sea breeze is partly a result of
(A). density differences between soil and water.
(B). differences in specific heat of soil and water.
(C). storms out at sea.
(D). dust devils over adjacent land areas.

Question #9: One of the factors that influences the movement of warm and cool air masses is a wave in the
(A). prevailing westerlies.
(B). jet stream.
(C). northeast trade wind belt.
(D). intertropical convergence

Question #6: The greenhouse effect occurs because outgoing radiation is partly absorbed by
(C). carbon dioxide and water vapor.
The incoming solar radiation that reaches the earth's surface is absorbed rocks, soil, water, and the ground. These materials become warmer as a result emitting the absorbed energy as infrared radiation, wavelengths longer than the visible part of the electromagnetic spectrum. This longer-wavelength infrared radiation has a frequency that matches some of the natural frequencies of vibration of carbon dioxide and water molecules. This match means that carbon dioxide and water molecules readily absorb infrared radiation that is emitted from the surface of the earth. The absorbed infrared energy shows up as an increased kinetic energy, which is indicated by an increase in temperature. Carbon dioxide and water vapor molecules now emit infrared radiation of their own, this time in all directions. Some of this reemitted radiation is again absorbed by other molecules in the atmosphere, some is emitted to space, and significantly, some is absorbed by the surface to start the process all over again. The net result is that less of the energy from the sun escapes immediately to space after being absorbed and emitted as infrared. It is retained through the process of being redirected to the surface, increasing the surface temperature more than it would have otherwise been.

Question #7: Temperature increases with altitude in the
(A). stratosphere.
The temperature increases in the stratosphere as a result of interactions between high-energy ultraviolet radiation and ozone. Ozone is triatomic oxygen that is concentrated mainly in the upper portions of the stratosphere.

Question #8: A sea breeze is partly a result of
(B). differences in specific heat of soil and water.
Under identical conditions of incoming solar radiation, the temperature changes experienced by the water will be much less than the changes experienced by the adjacent land. There are three principal reasons for this difference: (1) The specific heat of water is much greater the specific heat of soil. This means that it takes more energy to increase the temperature of water than it does for soil. (2) Water is a transparent fluid that is easily mixed, so the incoming solar radiation warms a body of water throughout, spreading out the heating effect. Incoming solar radiation on land, on the other hand, warms a relatively thin layer on the top, concentrating the heating effect. (3) Water is cooled by evaporation, which helps keep it at a lower temperature than an adjacent landmass under identical conditions of incoming solar radiation.

Question #9: One of the factors that influences the movement of warm and cool air masses is a wave in the
(B). jet stream.
Jet streams are sinuous, meandering loops of winds that tend to extend all the way around the earth, moving generally from the west in both hemispheres at speeds of 160 km/hr (about 100 mi/hr) or more. A jet stream may occur as a single belt, or loop, of wind but sometimes it divides into two or more parts. The jet stream develops north and south

belt. Question #10: A news story reports a humidity of 60%. This is a report on the (A). water vapor capacity. (B). dew point. (C). absolute humidity. (D). relative humidity. Question #11: The temperature to which air must be cooled to result in a relative humidity of 100% is called the (A). dew point temperature. (B). condensation point temperature. (C). wind cooling factor. (D). absolute temperature. Question #12: If the temperature of an object is below 0° C (A). frost will fall from the air onto the object. (B). frost will form directly on the object. (C). the dew point cannot be reached. (D). dew will form, which freezes into frost. Question #13: Tiny droplets of water can form in air when the relative humidity reaches 100% only if the air (A). can hold more water vapor. (B). contains dust or smoke particles. (C). has not reached the dew point. (D). is not being cooled.	loops of waves much like the waves you might make on a very long rope. These waves vary in size, sometimes beginning as a small ripple but then growing slowly as the wave moves eastward. Waves that form on the jet stream bulge toward the poles (called a crest) or toward the equator (called a trough). Warm air masses move toward the poles ahead of a trough and cool air masses move toward the equator behind a trough as it moves eastward. The development of a wave in the jet stream is understood to be one of the factors that influences the movement of warm and cool air masses, a movement that results in weather changes on the surface. Question #10: A news story reports a humidity of 60%. This is a report on the (D). relative humidity. The relationship between the actual absolute humidity at a particular temperature and the maximum absolute humidity that can occur at that temperature is called the relative humidity. Relative humidity is a ratio between (1) the amount of water vapor in the air and (2) the amount of water vapor needed to saturate the air at that temperature. The relationship is (absolute humidity at the present temperature/maximum absolute humidity at present temperature) times 100% = relative humidity. Question #11: The temperature to which air must be cooled to result in a relative humidity of 100% is called the (A). dew point temperature. As air is cooled, its capacity to hold water vapor is reduced to lower and lower levels. Even without water vapor being added to the air, a temperature will eventually be reached at which saturation, 100 percent humidity, occurs. Further cooling below this temperature will result in condensation. The temperature at which condensation begins is called the dew point temperature. Question #12: If the temperature of an object is below 0° C (B). frost will form directly on the object. Dew and frost form on the tops, sides, and bottoms of objects. Dew and frost condense directly on objects and do not "fall out" of the air. Question #13: Tiny droplets of water can form in air when the relative humidity reaches 100% only if the air (B). contains dust or smoke particles. Water vapor molecules in the air are constantly colliding and banging into each other, but they do not just join together to form water droplets, even if the air is saturated. The water molecules need something to condense upon. Condensation of water vapor into fog or cloud droplets takes place on tiny particles present in the air. The particles are called condensation nuclei. There are hundreds of tiny dust, smoke, soot, and salt crystals suspended in each cubic centimeter of the air that serve as condensation nuclei.
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Question #14: Frost would be most likely to form
(A). under trees on a windy night.
(B). in the open on a calm, cloudy night.
(C). in the open on a calm, clear night.
(D). under trees on a calm, cloudy night.

Question #15: The air temperature is 35 degrees Fahrenheit and the wind is blowing 20 mph, giving a wind chill factor of 12 degrees Fahrenheit. This means
(A). this is how cold you will feel.
(B). water will freeze in a bowl on the ground.
(C). a car radiator will need antifreeze protection down to 12 degrees F.
(D). all of the above.

Question #16: Air moving from a higher altitude to a lower altitude on the side of a mountain
(A). is cooled since cool air is more dense.
(B). will have a higher absolute humidity.
(C). will have a higher relative humidity.
(D). is warmed by compression.

Question #17: The intertropical convergence zone has tropical rain forests because
(A). sinking air warms, increasing the relative humidity.
(B). rising air cools as it is pushed upward.
(C). it is a high pressure belt of descending air.
(D). dry rising air is moved horizontally by the jet stream.

Question #14: Frost would be most likely to form
(C). in the open on a calm, clear night.
Dew and frost form on "C" nights, nights that can be described by the three "C" words of clear, calm, and cool. Dew and frost also seem to form more in open areas rather than under trees or other shelters and on objects such as grass rather than on the flat, bare ground.

Question #15: The air temperature is 35 degrees Fahrenheit and the wind is blowing 20 mph, giving a wind chill factor of 12 degrees Fahrenheit. This means
(A). this is how cold you will feel.
The human body constantly produces heat to maintain a core temperature, and some of this heat is radiated to the surroundings. When the wind is not blowing your body heat is also able to warm some of the air next to your body. This warm blanket of air provides some insulation, protecting your skin from the colder air farther away. If the wind blows, however, it moves this air away from your body and you will feel cooler. How much cooler will depend on how fast the air is moving and upon the outside temperature -- which is what the wind chill factor tells you. Thus, wind chill is an attempt to measure the effect of combinations of low temperature and wind. A wind chill of 12 degrees is not saying that an object will be cooled from 35 degrees to 12 degrees. It is saying that is how cold you will feel because of the low temperature and the wind, which blew away your warm blanket of air.

Question #16: Air moving from a higher altitude to a lower altitude on the side of a mountain
(D). is warmed by compression.
Air moving from a higher altitude becomes compressed as it moves to lower elevations under increasing atmospheric pressure. Compression of air, or any gas for that matter, increases the temperature by increasing the kinetic energy of the molecules. This creates a wind called a Chinook, which is common to mountainous and adjacent regions. A Chinook is a wind of compressed air with sharp temperature increases that can melt away any existing snow cover in a single day. The Santa Ana is a well-known compressional wind that occurs in southern California.

Question #17: The intertropical convergence zone has tropical rain forests because
(B). rising air cools as it is pushed upward.
The earth receives more direct solar radiation in the equatorial region than it does at higher latitudes. As a result, the temperatures of the lower troposphere are generally higher in the equatorial region, decreasing with latitude toward both poles. The lower troposphere from 10 degrees north to 10 degrees south of the equator is heated, expands, and becomes less dense. Hot air rises in this belt around the equator, known as the intertropical convergence zone. The rising air cools because it expands as it rises, resulting in heavy average precipitation. The tropical rain forests of the earth occur in this zone of high temperatures and heavy rainfall.

Question #18: The major factor that sets the stage for wind to blow is

(A). the fact that warm air will hold more water vapor than cool air.

(B). the fact that air becomes cooler when water vapor is condensed.

(C). that rising air becomes warmer from compression.

(D). uneven heating of the earth's surface.

Question #18: The major factor that sets the stage for wind to blow is (D). uneven heating of the earth's surface.

Uneven heating of the earth's surface sets the stage for convection. As a local region of air becomes heated, the increased kinetic energy of the molecules expands the mass of air, reducing its density. This less dense air is buoyant and is pushed upward by nearby cooler, more dense air. This results in three general motions of air: (1) the upward movement of air over a region of greater heating, (2) the sinking of air over a cooler region, and (3) a horizontal air movement between the cooler and warmer regions. In general, a horizontal movement of air is called wind, and the direction of a wind is defined as the direction from which it blows.

Weather and Climate

Questions	Answers with Rationalization
Question #1: The principal source of water vapor in the earth's hydrologic cycle is (A). the ocean. (B). lakes, rivers, and soil. (C). plants and animals. (D). burning of wood, fossil fuels, and crop residue.	Question #1: The principal source of water vapor in the earth's hydrologic cycle is (A). the ocean The hydrologic cycle is considered to have four main events: (1) evaporation of water from the ocean, (2) transport of water vapor through the atmosphere, (3) condensation and precipitation of water on the lands and, (4) the return of water to the ocean by rivers and streams. This definition of the hydrologic cycle involves only the ocean and the lands, but water vapor also evaporates from the land and may condense and precipitate back to the land without ever returning to the ocean. This can be considered as a small subcycle within the overall hydrologic cycle. The ocean-land exchange is the major cycle, and there are actually many small subcycles that also exist.
Question #2: Which cloud shape usually occurs when a parcel of air is moved upward in an unstable atmosphere? (A). cirrus (B). stratus (C). cumulus (D). fogus	Question #2: Which cloud shape usually occurs when a parcel of air is moved upward in an unstable atmosphere? (C). cumulus An uplifted parcel of air will expand and cool according to the dry adiabatic lapse rate. In instability, it will be warmer and therefore less dense than the surrounding air at any altitude it is lifted to. Thus, it will continue in the direction moved after the uplifting force is removed. Cumulus clouds usually develop in an unstable atmosphere, and the rising parcels of air, called thermals, can result in a very bumpy airplane ride.
Question #3: A parcel of dry air is moved upward for a distance of 100 m, so it (A). warms 1 degree C. (B). cools 1 degree C. (C). cools 0.65 degree C. (D). warms 0.65 degree C.	Question #3: A parcel of dry air is moved upward for a distance of 100 m, so it (B). cools 1 degree C. Rising air cools adiabatically since the cooling results from work performed by the molecules of gases in the air. Descending air warms adiabatically since the warming results from work done on the molecules of gases in the air. The rate of cooling or warming in the absence of condensation is called the dry adiabatic lapse rate. This rate is about 10 degrees C for each increase or decrease of 1 km of altitude. Thus a parcel of air moved upward a distance of 100 m is one-tenth of 1,000 m so the cooling would be one-tenth of 10 degrees, which is 1 degree C.
Question #4: Dry air expands and cools as it is moved upward in an unstable atmosphere. When the temperature reaches the dew point the rate of movement will (A). decrease. (B). reverse directions. (C). increase. (D). go to zero.	Question #4: Dry air expands and cools as it is moved upward in an unstable atmosphere. When the temperature reaches the dew point the rate of movement will (C). increase. As air moves upward and cools from expansion, sooner or later the dew point is reached and the air becomes saturated. As some of the water vapor in the rising parcel condenses to droplets, the latent heat of vaporization is released. The release of latent heat warms the air in the parcel and decreases the density even more, accelerating the ascent. This leads to further condensation and the formation of towering cumulus clouds.
Question #5: An adiabatic air temperature change is associated with	Question #5: An adiabatic air temperature change is associated with (B). rising or descending air. The decrease in temperature of an expanding gas is called adiabatic cooling. Adiabatic cooling takes place without heat flowing out of the

(A). the loss or gain of radiant energy. (B). rising or descending air. (C). density differences. (D). mixing of warm and cool air.	gas, and the cooling results only from the energy lost by expansion. Similarly, when work is done on a parcel of gas by compressing it, the temperature increases from the compression. Adiabatic heating takes place without heat flowing into the gas, and the heating results only from the energy gained by compression. Thus in the atmosphere adiabatic air temperature changes are associated with rising (less pressure) or descending (more pressure) air.
Question #6: Showers or thunderstorms are usually associated with which type of cloud? (A). stratus (B). cirrus (C). cumulus (D). any of the above	Question #6: Showers or thunderstorms are usually associated with which type of cloud? (C). cumulus In general, the basic form of a cloud has meaning about the general type of precipitation that can occur as well as the coming weather. Cumulus clouds usually produce showers or thunderstorms that last only brief periods of time.
Question #7: Which kind of cloud does not produce any precipitation of any kind? (A). cirrus (B). cumulus (C). stratus (D). none of the above	Question #7: Which kind of cloud does not produce any precipitation of any kind? (A). cirrus In general, the basic form of a cloud has meaning about the general type of precipitation that can occur as well as the coming weather. Cirrus clouds do not produce precipitation of any kind, but they may have meaning about the coming weather, usually meaning an approaching warm front.
Question #8: An air mass is created when a large body of air (A). moves over land from the ocean. (B). remains over land or water for a long time. (C). moves over the ocean from land. (D). moves back and forth over land and water.	Question #8: An air mass is created when a large body of air (B). remains over land or water for a long time. An air mass is defined as a large, more or less uniform body of air with nearly the same temperature and moisture conditions. An air mass forms when a large body of air, perhaps covering millions of square kilometers, remains over a large area of land or water for an extended period of time. While it is stationary it acquires the temperature and moisture characteristics of the land or water through the heat transfer processes of conduction, convection, and radiation and through the moisture transfer processes of evaporation and condensation.
Question #9: Which if the following air masses would most likely be dry and warm? (A). maritime tropical (B). continental tropical (C). continental polar (D). maritime polar	Question #9: Which if the following air masses would most likely be dry and warm? (B). continental tropical Air masses are classified according to the temperature and moisture conditions where they originate. There are two temperature extreme possibilities, a polar air mass from a cold region and a tropical air mass from a warm region. There are also two moisture extreme possibilities, a moist maritime air mass from over the ocean and a generally dry continental air mass from over the land. Thus, there are four main types of air masses that can influence the weather where you live: continental polar (dry and cold), maritime polar (wet and cold), continental tropical (dry and warm), and maritime tropical (wet and warm).
Question #10: A cold front is the boundary between (A). the southward edge of a cold air mass. (B). the northward edge of a warm air mass.	Question #10: A cold front is the boundary between (D). a moving cold air mass and a warmer air mass. A cold front is formed when a cold air mass moves into warmer air, displacing it in the process. A cold front is generally steep, and when it runs into the warmer air it forces it to rise quickly. If the warm air is moist, it is quickly cooled adiabatically to the dew point temperature,

(C). a moving warm air mass and a cooler air mass. (D). a moving cold air mass and a warmer air mass.	resulting in large, towering cumulus clouds and thunderclouds along the front.
Question #11: An air mass that formed over central Canada in January would be classified as (A). continental polar. (B). maritime tropical. (C). continental tropical. (D). maritime polar.	Question #11: An air mass that formed over central Canada in January would be classified as (A). continental polar. A large body of air that remains over the cold, dry, snow-covered surface of Siberia for some time will become cold and dry. A large body of air that remains over a warm tropical ocean, on the other hand, will become warm and moist. Knowledge about the condition of air masses is important because they tend to retain the acquired temperature and moisture characteristics when they finally break away, sometimes moving long distances. An air mass that formed over central Canada is a continental polar air mass that could bring cold, dry air to your location.
Question #12: A wave in a slowly advancing cold or stationary front can result in an area of (A). high pressure. (B). low pressure. (C). clear weather. (D). anticyclones.	Question #12: A wave in a slowly advancing cold or stationary front can result in an area of (B). low pressure. A slowly advancing cold front and a stationary front often develop a bulge, or wave, in the boundary between cool and warm air moving in opposite directions. The wave grows as the moving air is deflected, forming a warm front moving northward on the right side and a cold front moving southward on the left side. Cold air is more dense than warm air, and the cold air moves faster than the slowly moving warm front. As the faster moving cold air catches up with the slower moving warm air, the cold air underrides the warm air, lifting it upward. This lifting action produces a low-pressure area at the point where the two fronts come together.
Question #13: Most cyclonic storms in the continental United states move (A). southward. (B). northward. (C). eastward. (D). westward.	Question #13: Most cyclonic storms in the continental United states move (C). eastward. A cyclonic storm is a fully developed low-pressure center. Since its formation, this low-pressure center has been moving, taking its associated stormy weather with it in a generally easterly direction. Such cyclonic storms usually follow principal tracks along a front. Since they are observed generally to follow these same tracks, it is possible to predict where the storm might move to next.
Question #14: An advancing warm front usually produces what kind of clouds at the leading edge? (A). stratus (B). cirrus (C). cumulus (D). no clouds are produced	Question #14: An advancing warm front usually produces what kind of clouds at the leading edge? (A). stratus A warm front forms when a warm air mass advances over a mass of cooler air. Since the advancing warm air is less dense than the cooler air it is displacing, it generally overrides the cooler air, forming a long, gently sloping front. Because of this, the overriding warm air may form clouds far in advance of the ground-level base of the front. This may produce high cirrus clouds a day or more in advance of the front, which are followed by thicker and lower stratus clouds as the front advances. Usually these clouds result in a broad band of drizzle, fog, and the continuous light rain usually associated with stratus clouds. This light rain (and snow in the winter) may last for days as the warm front passes.

Question #15: Winds tend to blow in what direction with respect to a low-pressure center in the Northern Hemisphere?
(A). counterclockwise
(B). clockwise
(C). perpendicular to the direction of movement
(D). parallel to the isobars

Question #16: Tornadoes are associated with what type of clouds?
(A). thunderstorms
(B). stratus
(C). cirrus
(D). none of the above

Question #17: Fairly accurate weather forecasts can be made up to
(A). 1 day.
(B). 3 days.
(C). 5 days.
(D). 7 days.

Question #18: The basic framework for describing the earth's climates is based on
(A). latitude.
(B). longitude.
(C). altitude.
(D). none of the above.

Question #19: Most of the southeastern part of the United States has a climate that is
(A). Mediterranean.
(B). humid continental.
(C). marine.
(D). humid subtropical.

Question #15: Winds tend to blow in what direction with respect to a low-pressure center in the Northern Hemisphere?
(A). counterclockwise
A cyclone is defined as a low-pressure center where the winds move into the low-pressure center and are forced upward. As air moves in toward the center, the Coriolis effect and friction with the ground cause the moving air to veer to the right of the direction of motion. In the Northern Hemisphere this rightward veering of moving air produces a counterclockwise circulation pattern of winds around the low-pressure center. The upward movement associated with the low-pressure center of a cyclone cools the air adiabatically, resulting in clouds, precipitation, and stormy conditions.

Question #16: Tornadoes are associated with what type of clouds?
(A). thunderstorms
A tornado is the smallest, most violent weather disturbance that occurs on the earth. Tornadoes occur with intense thunderstorms, resembling a long, narrow funnel or ropelike structure that drops down from a thundercloud and may or may not touch the ground.

Question #17: Fairly accurate weather forecasts can be made up to
(B). 3 days
The use of new computer technology has improved the accuracy of next-day forecasts tremendously, and the forecasts up to three days are fairly accurate. For forecasts of more than five days, however, the number of calculations and the effect of uncertainties increase greatly.

Question #18: The basic framework for describing the earth's climates is based on
(A). latitude.
The low, middle, and high latitudes provide a basic framework for describing the earth's climates, which are defined in terms of yearly temperature averages. It is necessary to define the basic climates in terms of temperature because land and water surfaces react differently to incoming solar radiation, creating a different temperature. In general, the low latitudes receive a high amount of incoming solar radiation that varies little during a year. Temperatures are high throughout the year, varying little from month to month. The middle latitudes receive a higher amount of incoming radiation during one part of the year and a lower amount during the other part. Overall temperatures are cooler than in the low latitudes and have a wide seasonal variation. The high latitudes receive a maximum amount of radiation during one part of the year and none during another part. Overall temperatures are low, with the highest range of annual temperatures.

Question #19: Most of the southeastern part of the United States has a climate that is
(D). humid subtropical.
The humid subtropical climate is one of short mild winters, humid summers, moderate precipitation.

Question #20: In the United States, the Mediterranean climate is found in (A). Florida. (B). Texas. (C). Arizona. (D). California.	Question #20: In the United States, the Mediterranean climate is found in (D). California. The Mediterranean climate has hot, dry summers with short, mild and wet winters and it occurs in California in the United States.
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Earth's Waters

Questions	Answers with Rationalization
Question #1: Of all the water that exists on the earth, the part available for human consumption and agriculture is (A). less than 1%. (B). about 3%. (C). about 80%. (D). nearly 98%.	Question #1: Of all the water that exists on the earth, the part available for human consumption and agriculture is (A). less than 1%. Over 97 percent of the earth's water is stored in the earth's oceans. This water contains a relatively high level of dissolved salts, which makes ocean water unfit for human consumption and unfit for most agricultural purposes. All other water, which is fit for human consumption and agriculture, is called freshwater. About two-thirds of the earth's freshwater supply is locked up in the ice caps of Greenland and the Antarctic and in glaciers. This leaves less than 1 percent of all the water found on the earth as available freshwater. There is a generally abundant supply, however, because the freshwater supply is continually replenished by the hydrologic cycle.
Question #2: How many oceans exist on the earth's surface? (A). 7 (B). 3 (C). 1 (D). 5	Question #2: How many oceans exist on the earth's surface? (C). 1 In general, the ocean is a single, continuous body of salt water on the surface of the earth. Although there is really only one big ocean on the earth, specific regions have been given names for convenience in describing locations. For this purpose, three principal regions are recognized, the (1) Atlantic Ocean, (2) Indian Ocean, and (3) Pacific Ocean. These regions are often subdivided further into North Atlantic Ocean, South Atlantic Ocean, and so on.
Question #3: The best source of extending today's supply of freshwater is (A). surface water. (B). recycling waste water. (C). groundwater. (D). the ocean	Question #3: The best source of extending today's supply of freshwater is (B). recycling waste water. Possible sources of extending today's supply of freshwater include the recycling of waste water and turning to the largest supply of water in the world, the ocean. About 90 percent of the water used by industries is presently dumped as a waste product. In some areas city waste water is already being recycled for use in power plants and for watering parks. A practically limitless supply of freshwater could be available by desalting ocean water, something which occurs naturally in the hydrologic cycle. The ocean, and the nature of seawater, is the topic of the following section. The treatment of seawater to obtain a new supply of freshwater is presently too expensive because of the cost of energy to accomplish the task. New technologies, perhaps ones that use solar energy, may make this more practical in the future. In the meantime, the best sources of extending the supply of freshwater appear to be the control of pollution, the recycling of waste water, and conservation of the existing supply.
Question #4: What has happened to the average salinity of seawater since the ocean formed about 3 billion years ago? (A). major increase (B). slight increase (C). little change (D). major decline	Question #4: What has happened to the average salinity of seawater since the ocean formed about 3 billion years ago? (C). little change You might wonder why the ocean basin has not become filled in by the continuous supply of sediments and dissolved materials that would accumulate over millions of years. The basin has not filled in because (1) accumulated sediments have been recycled to the earth's interior through plate tectonics and (2) dissolved materials are removed by natural processes just as fast as they are supplied by the rivers. Some of the dissolved materials, such as calcium and silicon, are removed by

Question #5: The chemical composition of seawater taken from oceans around the earth is found to (A). vary tremendously. (B). vary slightly. (C). uniformly not vary. (D). ...the answer is unknown. Question #6: A given amount of seawater will have the greatest density if it has (A). no sediments, decreased salinity, increased temperature. (B). sediments, increased salinity, decreased temperature. (C). sediments, increased salinity, increased temperature. (D). no sediments, decreased salinity, decreased temperature. Question #7: Waves always appear to approach the shore head-on because of (A). density differences. (B). reflection. (C). refraction. (D). rep currents. Question #8: Breakers form when (A). the top of a wave overruns the bottom part. (B). waves enter shallow, then deeper water. (C). approaching waves hit the rip current. (D). the bottom surface is rippled. Question #9: The largest waves are produced by winds that are	plants and animals to make solid shells, bones, and other hard parts. Other dissolved materials, such as iron, magnesium, and phosphorous, form solid deposits directly and also make sediments that settle to the ocean floor. Hard parts of plants and animals and solid deposits are cycled to the earth's interior along with suspended sediments that have settled out of the seawater. Studies of fossils and rocks indicate that the composition of seawater has changed little over the past 600 million years. Question #5: The chemical composition of seawater taken from oceans around the earth is found to (C). uniformly not vary. The same proportion, meaning the same chemical composition, is found in seawater of any salinity of any sample taken from any location anywhere in the world, from any depth of the ocean, or taken any time of the year. Seawater has a remarkably uniform composition that varies only in concentration. This means that the ocean is well mixed and thoroughly stirred around the entire earth. Question #6: A given amount of seawater will have the greatest density if it has (B). sediments, increased salinity, decreased temperature. The density of seawater is influenced by three factors: (1) the water temperature, (2) salinity, and (3) suspended sediments. Cold water is generally more dense than warm water, thus sinking and displacing warmer water. Seawater of a high salinity has a higher relative density than less salty water, so it sinks and displaces water of less salinity. Likewise, seawater with a larger amount of suspended sediments has a higher relative density than clear water, so it sinks and displaces clear water. Thus sediments, increased salinity, and decreased temperature will cause the greatest density of seawater. Question #7: Waves always appear to approach the shore head-on because of (C). refraction. Most waves move toward the shore at some angle. As the wave crest nearest the shore starts to slow, the part still over deep water continues on at the same velocity. The slowing at the shoreward side refracts, or bends, the wave so it is more parallel to the shore. Thus, waves always appear to approach the shore head-on, arriving at the same time on all parts of the shore. Question #8: Breakers form when (A). the top of a wave overruns the bottom part. As waves move from the deep ocean to the shore they pass over shallower and shallower water depths. After they reach water that is less than one-half the wavelength, friction progressively slows the bottom part of the wave. The wave front becomes steeper and steeper as the top overruns the bottom part of the wave. When the wave front becomes too steep the top part breaks forward and the wave is now called a breaker. In general, this occurs where the water depth is about one and one-third times the wave height. The zone where the breakers occur is called surf. Question #9: The largest waves are produced by winds that are
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- (A). strong, blow for a long time, with a long fetch.
- (B). medium, blow for a long time, with a short fetch.
- (C). strong, blow for a short time, with a short fetch.
- (D). medium, blow for a short time, with a long fetch.

Question #10: Ocean waves will cause a particle floating near the surface to move

- (A). with the waves.
- (B). in circles.
- (C). up and down.
- (D). back and forth.

Question #11: The average salt concentration of seawater is

- (A). 3.5%
- (B). 35%
- (C). 0.35%
- (D). none of the above.

Question #12: Artesian water is water that

- (A). comes in plastic bottles.
- (B). is particularly pure.
- (C). has a better flavor than most water.
- (D). has been under pressure.

Question #13: The continental divide of the United States separates

- (A). Northern and Southern states.
- (B). The Atlantic Ocean and Pacific Ocean watersheds.
- (C). North America and South America.
- (D). North America into any two equal parts.

- (A). strong, blow for a long time, with a long fetch.

The characteristics of an ocean wave formed by the wind depend on three factors: (1) the wind speed, (2) the length of time that the wind blows, and (3) the fetch, which is the distance the wind blows across the open ocean. Larger waves are produced by strong winds that blow for a longer time over a long fetch.

Question #10: Ocean waves will cause a particle floating near the surface to move

- (B). in circles.

A particle of water on the surface moves upward and forward with each wave crest, then backward and down on the following trough, tracing out a nearly circular path through this motion. The particle returns to its initial position, without any forward movement while tracing out the small circle. Note that the diameter of the circular path is equal to the wave height. Water particles farther below the surface also trace out circular paths as a wave passes. The diameters of these circular paths below the surface are progressively smaller with increasing depth. Below a depth equal to about half the wavelength there is no circular movement of the particles.

Question #11: The average salt concentration of seawater is

- (A). 3.5%

The amount of dissolved salts in seawater is measured as salinity. Salinity is defined as the mass of salts dissolved in 1.0 kg, or 1,000 g of seawater. The salt content is reported in parts per thousand and 1,000 g of seawater contains 35 g of dissolved salts (and 965 g of water), which is a salinity of 35 parts per thousand. This is the same concentration as a 3.5 percent salt solution.

Question #12: Artesian water is water that

- (D). has been under pressure.

An aquifer is a layer of sand, gravel, sandstone, or other highly permeable material beneath the surface that is capable of producing water. In some places an aquifer carries water from a higher elevation, resulting in a pressure on water trapped by impermeable layers at lower elevations. Groundwater that is under such a confining pressure is in an artesian aquifer. "Artesian" refers to the pressure, and groundwater from an artesian well rises above the top of the aquifer but not necessarily to the surface. Some artesian wells are under sufficient pressure to produce a fountainlike flow or spring.

Question #13: The continental divide of the United States separates

- (B). The Atlantic Ocean and Pacific Ocean watersheds.

Two adjacent watersheds are separated by a line called a divide. Rain that falls on one side of a divide flows into one watershed and rain that falls on the other side flows into the other watershed. A continental divide separates river systems that drain into opposite sides of a continent. The North American continental divide trends northwestward through the Rocky Mountains.

Question #14: On the average, how much precipitation becomes groundwater?

- (A). less than 1%.
- (B). about 15%.
- (C). about 70%.
- (D). almost 90%.

Question #15: Compared to the volume of all the surface water, the volume of all the ground water is about

- (A). 10 times more.
- (B). 20 times less.
- (C). 30 times greater.
- (D). the same.

Question #16: The ocean current flowing between the Mediterranean and the Atlantic Ocean is a result of

- (A). salinity differences.
- (B). temperature differences.
- (C). suspended sediments.
- (D). none of the above.

Question #17: The continental shelf is a part of the

- (A). seamount ridge.
- (B). ocean basin.
- (C). continental slope.
- (D). continent.

Question #18: Of the following features of the ocean the deepest part is the

- (A). submarine canyon.
- (B). continental shelf.
- (C). continental slope.
- (D). ocean basin.

Question #14: On the average, how much precipitation becomes groundwater?

- (A). less than 1%.

The amount of a rainfall that becomes runoff or groundwater depends on a number of factors, including (1) the type of soil on the surface, (2) how dry the soil is, (3) the amount and type of vegetation, (4) the steepness of the slope, and (5) if the rainfall is a long, gentle one or a cloudburst. Different combinations of these factors can result in from 5 percent to almost 100 percent of given rainfall running off, with the rest evaporating or soaking in the ground. On the average, however, about 70 percent of all precipitation evaporates back into the atmosphere, about 30 percent becomes runoff, and less than 1 percent soaks into the ground.

Question #15: Compared to the volume of all the surface water, the volume of all the ground water is about

- (C). 30 times greater.

Precipitation soaks into the ground, or percolates slowly downward until it reaches an area, or zone, where the open spaces between rock and soil particles are completely filled with water. Water from such a saturated zone is called groundwater. There is a tremendous amount of water stored as groundwater, which makes up a supply about thirty times larger than all the surface water on the earth.

Question #16: The ocean current flowing between the Mediterranean and the Atlantic Ocean is a result of

- (A). salinity differences.

The water in the Mediterranean has a high salinity because it is mostly surrounded by land in a warm, dry climate. The Mediterranean seawater, with its higher salinity, is more dense than the seawater in the open Atlantic Ocean. This density difference results in two separate currents that flow in opposite directions between the Mediterranean and the Atlantic. The greater density seawater flows from the bottom of the Mediterranean into the Atlantic, while the less dense Atlantic water flows into the Mediterranean near the surface. The dense Mediterranean seawater sinks to a depth of about 1,000 m (about 3,300 ft) in the Atlantic, where it spreads over a large part of the North Atlantic Ocean.

Question #17: The continental shelf is a part of the

- (D). continent.

The continental shelf is considered to be a part of the continent and not the ocean, even though it is covered with an average depth of about 130 m (about 425 ft) of seawater. The shelf slopes gently away from the shore for an average of 75 km (about 47 mi), but it is much wider on the edge of some parts of continents than other parts.

Question #18: Of the following features of the ocean the deepest part is the

- (D). ocean basin.

Beyond the continental slope is the bottom of the ocean floor, the ocean basin. Ocean basins are the deepest part of the ocean, covered by about 4 to 6 km (about 2 to 4 mi) of seawater.

The Universe

Questions	Answers with Rationalization
Question #1: The shape of Earth is that of a (A). perfectly round sphere. (B). pear-shaped form bulging at the poles. (C). irregular spheroid bulging at the equator. (D). flat table top.	Question #1: The shape of Earth is that of a (C). irregular spheroid bulging at the equator. The Earth is flattened at the poles and has an equatorial bulge, as do many other planets.
Question #2: Because of the shape of Earth's orbit around the Sun the overall Earth receives (A). 6 percent more solar energy in January. (B). 6 percent more solar energy in July. (C). the same amount of solar energy year around. (D). more solar energy in July and less in January.	Question #2: Because of the shape of Earth's orbit around the Sun the overall Earth receives (A). 6 percent more solar energy in January. Earth's orbit is slightly elliptical and the earth is about 2.5 million km closer to the Sun in January than in July. One result is a January Sun with an apparent diameter that is 3 percent larger than the Sun in July. The overall Earth receives about 6 percent more solar energy in January as a consequence. However, the length of day and angle of sunlight striking the earth's surface apparently has a greater effect on the climate than the increase in the amount of solar energy.
Question #3: As Earth moves in its orbit it has an orbital speed that is (A). slowest in January. (B). slowest at perihelion. (C). fastest in January. (D). fastest at aphelion.	Question #3: As Earth moves in its orbit it has an orbital speed that is (C). fastest in January. Earth's orbit is slightly elliptical, so the planet moves with a varying speed that is described by Kepler's laws of planetary motion. That is, the earth moves fastest when it is closer to the Sun in January, at perihelion, and moves slowest when it is farthest away from the Sun in early July, at aphelion.
Question #4: The relationship of Earth's axis to the plane of the ecliptic (A). is one direction half the time, then moving to the opposite direction. (B). is the same direction at all times. (C). reverses direction every month. (D). reverses direction every 3 months.	Question #4: The relationship of Earth's axis to the plane of the ecliptic (B). is the same direction at all times. The earth moves around the Sun in its orbit and all points in that orbit describe a flat surface called the plane of the ecliptic. The earth's axis is an imaginary line through the geographic poles, about which the earth turns during a daily rotation. The relationship between the earth's axis and the plane of the ecliptic is one of constant inclination. The earth's axis is inclined 23.5° from a line perpendicular to the plane, and this inclination remains the same throughout a given year, but with small changes over time.
Question #5: The orientation of Earth's axis to the background of stars (A). is one direction half the time, then moving to the opposite direction. (B). reverses direction every 3 months. (C). is the same direction at all times.	Question #5: The orientation of Earth's axis to the background of stars (C). is the same direction at all times. The earth's axis is an imaginary line through the geographic poles, about which the earth turns during a daily rotation. The relationship between the earth's axis and the background of stars is one of constant inclination. The orientation of Earth's axis to the stars is considered to be the same throughout the year as Earth moves around its orbit. Small changes do occur in the orientation over time, but in general the axis points in the same direction, remaining essentially parallel to its position during any day of the year.

(D). reverses direction every month.

Question #6: Days and nights vary in length throughout the year because of

- (A). daily rotation.
- (B). matching rotation and revolution.
- (C). a constant inclination and orientation of the axis.
- (D). yearly revolution.

Question #7: The Northern Hemisphere's summer solstice occurs when Earth moves in its orbit to point the North Pole

- (A). directly away from the Sun.
- (B). directly toward the Sun.
- (C). halfway between toward and away from the Sun.
- (D). none of the above.

Question #8: On the Northern Hemisphere's winter solstice the Sun at noon makes a shadow that is

- (A). shortest.
- (B). halfway between shortest and longest.
- (C). longest.
- (D). none of the above.

Question #9: Daylight and night are of equal length during

- (A). spring or autumnal equinox.
- (B). winter solstice.
- (C). summer solstice.
- (D). None of the above.

Question #10: Which of the following provides evidence that Earth rotates?

- (A). Coriolis effect.
- (B). solstices.
- (C). constant orientation of the axis.
- (D). all of the above.

Question #6: Days and nights vary in length throughout the year because of

(C). a constant inclination and orientation of the axis.
The constant orientation and inclination of the Earth's axis results in the axis pointing toward the Sun as Earth moves in one part of its orbit, then pointing away from the Sun six months later. The constant inclination and orientation of the axis, together with Earth's rotation and revolution, combine to produce three related effects: days and nights that vary in length, changing seasons, and climates that vary with latitude.

Question #7: The Northern Hemisphere's summer solstice occurs when Earth moves in its orbit to point the North Pole

(B). directly toward the Sun.
The North Pole points toward the Sun on June 21 or 22, and the North Pole receives sunlight for a full twenty-four hours. This is the summer solstice in the Northern Hemisphere, the beginning of the summer season with the longest daylight periods and the Sun at its maximum noon height in the sky.

Question #8: On the Northern Hemisphere's winter solstice the Sun at noon makes a shadow that is

(C). longest.
The Northern Hemisphere's winter solstice occurs on about December 22 and identifies the beginning of the winter season. At the winter solstice the Sun at noon has the lowest altitude, and the shadow from a vertical stick is longer than any other day of the year.

Question #9: Daylight and night are of equal length during

(A). spring or autumnal equinox.
As Earth moves in its orbit between point its North Pole toward the Sun on about June 22 and pointing it away on about December 22, there are two times when it is halfway between. At these times Earth's axis is perpendicular to a line between the center of the Sun and Earth, and daylight and nights are of equal length. These are called the equinoxes after the Latin meaning "equal nights." The spring equinox occurs on about March 21 and identifies the beginning of the spring season. The autumnal equinox occurs on about September 23 and identifies the beginning of the fall season.

Question #10: Which of the following provides evidence that Earth rotates?

(A). Coriolis effect.
The earth has a greater rotational velocity at the equator than at the poles. As an object leaves the surface and moves north or south, the surface has a different rotational velocity, so it rotates beneath the object as it proceeds in a straight line. This gives the moving object an apparent deflection to the right of the direction of movement in the Northern Hemisphere and to the left in the Southern Hemisphere. This apparent deflection caused by the earth's rotation is called the Coriolis effect.

Question #11: The slow wobble of the earth's axis is called
(A). the Coriolis effect.
(B). precession.
(C). Foucault effect.
(D). none of the above.

Question #12: The distance east or west from the prime meridian is called
(A). latitude.
(B). parallels.
(C). longitude.
(D). none of the above.

Question #13: An apparent solar day is
(A). the interval between two consecutive solar noons.
(B). slightly longer than a sidereal day.
(C). rarely exactly 24 hours long.
(D). all the above.

Question #14: If your favorite constellation rose at 8:00 PM one night, when would it rise two weeks later?
(A). around 7:00 PM
(B). at 8:00 PM
(C). around 9:00 PM
(D). it depends on the constellation

Question #15: North of the Tropic of Cancer, the sun appears directly overhead at noon
(A). daily.
(B). never.
(C). twice a year, at the equinoxes.
(D). twice a year, at the solstices.

Question #11: The slow wobble of the earth's axis is called
(B). precession.
The earth slowly wobbles on its axis, much like a spinning top that swings in a slow circle as it spins. It will take the earth about 26,000 years to complete one turn, or wobble, tracing out a big circle with it's axis. The wobbling of the axis is called precession.

Question #12: The distance east or west from the prime meridian is called
(C). longitude.
To identify a location you need two lines, an east- and west-running parallel and a line that runs pole to pole. The north-south running arcs that intersect at both poles are called meridians. There is no naturally occurring, identifiable meridian that can be used as a point of reference such as the equator serves for parallels, so one is identified as the referent by international agreement. The referent meridian is the one that passes through the Greenwich Observatory near London, England, and this meridian is called the prime meridian. The distance from the prime meridian east or west is called the longitude. The degrees of longitude of a point on a parallel are measured to the east or to the west from the prime meridian up to 180 degrees.

Question #13: An apparent solar day is
(D). all the above.
An apparent solar day is the interval between two consecutive crossings of the celestial meridian by the Sun, for example, from one local solar noon to the next solar noon. Since Earth is moving in orbit around the Sun, it must turn a little bit farther to compensate for its orbital movement, bringing the Sun back to local solar noon. As a consequence, the apparent solar day is about four minutes longer than the sidereal day.

Question #14: If your favorite constellation rose at 8:00 PM one night, when would it rise two weeks later?
(A). around 7:00 PM
Since Earth is moving in orbit around the Sun, it must turn a little bit farther to compensate for its orbital movement, bringing the Sun back to local solar noon. As a consequence, the apparent solar day is about four minutes longer than the sidereal day. This additional time accounts for the observation that the stars and constellations of the zodiac rise about four minutes earlier every night, appearing higher in the sky at the same clock time until they complete a yearly cycle. So, 4 minutes times 14 is 56 minutes, or about a hour earlier from 8:00 PM, which is 7:00 PM.

Question #15: North of the Tropic of Cancer, the sun appears directly overhead at noon
(B). never.
The parallels of 23.5 degrees north (Tropic of Cancer) and 23.5 degrees south (Tropic of Capricorn) of the equator identify the limits toward the poles where the Sun appears directly overhead during the course of a year. The Sun never appears directly overhead north of the Tropic of Cancer or south of the Tropic of Capricorn.

Question #16: South of the Antarctic Circle, the sun appears
(A). directly overhead only once a year
(B). directly overhead twice yearly.
(C). above the horizon all day at least once during December.
(D). above the horizon all day at least once during June.

Question #17: In the time 9:00 AM, the AM means
(A). before the meridian
(B). after the meridian
(C). astronomical motion
(D). after morning

Question #18: We always see the same side of the Moon because
(A). the Moon doesn't rotate.
(B). the period of rotation of both Earth and the Moon is the same.
(C). the Earth's rotation matches the Moon's revolution.
(D). the Moon's periods of rotation and revolution are the same.

Question #19: During which phase of the moon would we most likely see a solar eclipse?
(A). full
(B). first quarter
(C). last quarter
(D). new

Question #20: A lunar eclipse can only occur during which phase of the moon?
(A). 1st quarter
(B). full
(C). last quarter
(D). new

Question #21: If you were flying east on a jet across three times zones and your watch read 12:00 noon when you landed, you should
(A). set it to 9:00 AM.
(B). leave it at 12:00 for that is when the sun crosses the celestial meridian.
(C). set it 6:00 PM.
(D). set it to 3:00 PM.

Question #16: South of the Antarctic Circle, the sun appears
(C). above the horizon all day at least once during December.
The parallels of 66.5 degrees north (Arctic circle) and 66.5 degrees south (Antarctic circle) of the equator identify the limits toward the equator of where the Sun appears above the horizon all day during the summer. The Sun would appear above the horizon all day during the Southern Hemisphere summer (December), and moving northward this would decrease with fewer days of full light until reaching the limit of one day of 24-hour daylight at the Antarctic circle.

Question #17: In the time 9:00 AM, the AM means
(A). before the meridian
The apparent movement of the Sun across the celestial meridian identifies the even of time called noon. The morning hours before the Sun crosses the meridian are identified as ante meridiem (A.M) hours, which is Latin for "before meridian."

Question #18: We always see the same side of the Moon because
(D). the Moon's periods of rotation and revolution are the same.
The Moon rotates in the same period as the time of revolution, so the sidereal month is also the time required for one rotation. Because the rotation and revolution rates are the same, you always see the same side of the Moon from Earth.

Question #19: During which phase of the moon would we most likely see a solar eclipse?
(D). new
A solar eclipse occurs when the Moon is between the Sun and the Earth. The Moon is between the Sun and the Earth only during the new moon phase, so this phase is when you might expect a solar eclipse to occur.

Question #20: A lunar eclipse can only occur during which phase of the moon?
(B). full
A lunar eclipse occurs when the Sun, Earth, and Moon are lined up so Earth's shadow falls on the Moon. The Earth's shadow can fall on the Moon only when the Earth is between the Sun and the Moon, which also happens to be the alignment required for the full moon phase.

Question #21: If you were flying east on a jet across three times zones and your watch read 12:00 noon when you landed, you should
(D). set it to 3:00 PM.
When you cross a boundary between two time zones, the clock is set ahead one hour if you are traveling east and back one hour if you are traveling west. Since you were flying east and crossed three times zones you should set your watch ahead by three hours, from 12:00 noon to 3:00 PM.

The Solar System

Questions	Answers with Rationalization
Question #1: Kepler's first law of planetary motion describes planets moving in (A). ellipses. (B). perfect circles. (C). epicycles. (D). pulsing circles.	Question #1: Kepler's first law of planetary motion describes planets moving in (A). ellipses. Kepler's first law states that each planet moves in an orbit that has the shape of an ellipse, with the Sun located at one focus.
Question #2: Kepler's third law of planetary motion describes a relationship between the (A). shape of orbit and location of Sun. (B). orbital velocity and position in orbit. (C). distance from the Sun and length of year. (D). path of epicycle and position.	Question #2: Kepler's third law of planetary motion describes a relationship between the (C). distance from the Sun and length of year. Kepler's third law describes a relationship between the time required for a planet to move around the Sun and its average distance from the Sun. The relationship is that the time squared is proportional to the distance cubed. This means that the more distant a planet is from the Sun, the longer the time required to complete one orbit.
Question #3: According to the protoplanet nebular model, the first important event in the creation of the Solar System was (A). creation of the Earth. (B). massive stars ending their life cycle in supernova. (C). creation of the Sun. (D). accumulation of dust and solid matter in a huge, bulging disk.	Question #3: According to the protoplanet nebular model, the first important event in the creation of the Solar System was (B). massive stars ending their life cycle in supernova. Earth, the other planets, and all the members of the solar system are composed of elements that were manufactured by former stars. Hydrogen fusion in the core of large stars resulted in the formation of the elements up to iron. Elements heavier than iron were formed in supernova explosions of dying massive stars. Thus the first state of the formation of the solar system consisted of the formation of the elements heavier than hydrogen in many, many previously existing stars, including the supernovas of more massive stars. Over a long period of time these elements began to concentrate in one region of space as dust, gases, and chemical compounds, but hydrogen was still the most abundant element in the nebula that was destined to become the solar system.
Question #4: A terrestrial planet slightly larger than the Earth's Moon is the eighth largest planet known as (A). Uranus (B). Venus (C). Mercury. (D). Pluto	Question #4: A terrestrial planet slightly larger than the Earth's Moon is the eighth largest planet known as (C). Mercury. Mercury is the innermost planet, moving rapidly in a highly elliptical orbit that averages about 58 million km from the Sun. It is the eighth largest planet, with a diameter of less than 5,000 km, which means that it is slightly larger than Earth's Moon.
Question #5: The brilliant evening and morning "star" that appears near sunrise or sunset is the planet (A). Mercury. (B). Saturn (C). Jupiter (D). Venus.	Question #5: The brilliant evening and morning "star" that appears near sunrise or sunset is the planet (D). Venus. Venus is sometimes to the left of the Sun, appearing as the evening star, and sometimes to the right of the Sun, appearing as the morning star. It shines brightly because it is covered with clouds that reflect about 80 percent of the sunlight, making it the brightest object in the sky after the Sun and Moon.

Question #6: The planet with areas of dry riverbed features with dry tributaries suggesting a former abundance of liquid water is

- (A). Mercury.
- (B). Mars.
- (C). Venus.
- (D). Neptune.

Question #7: This planet has rings of dust-sized particles and more than 7 moons.

- (A). Saturn
- (B). Uranus
- (C). Neptune
- (D). any of the above.

Question #8: The planet with an exceptional tilt to its orbital plane that some do not consider to be a true planet.

- (A). Uranus
- (B). Neptune
- (C). Pluto
- (D). Saturn

Question #9: The disk-shaped region of small icy bodies some 30 to 100 A.U. from the Sun believed to be the source of some comets is called the

- (A). Kuiper Belt.
- (B). Oort Cloud.
- (C). Asteroid Belt.
- (D).None of the above.

Question #10: Most meteorites are

- (A). pieces from the asteroid belt.
- (B). dust from space.
- (C). fragments of comets.
- (D). rocks from deep space outside the Solar System.

Question #11: Which planet(s) has (have) no moons?

- (A). Mercury
- (B). Mercury and Venus
- (C). Venus and Pluto
- (D). All of the planets have at least one moon.

Question #12: Most of the mass of the Solar System is found in

Question #6: The planet with areas of dry riverbed features with dry tributaries suggesting a former abundance of liquid water is

(B). Mars.

Sinuuous, dry riverbed channels with dry tributaries are found in the flat regions of Mars that are pitted with impact craters. Liquid water may have been present on Mars in the past, but none is to be found today. Mares is much too cold and the air pressure is too low for liquid water to exist. The water-carved channels observed on Mars today were carved when Mars as much warmer with an active greenhouse effect and abundant water.

Question #7: This planet has rings of dust-sized particles and more than 7 moons.

(D). any of the above.

Saturn has 18 satellites and an unique, beautiful system of rings. Uranus has 15 satellites and a system of 10 narrow rings and several dusty bands. Neptune has 8 satellites and a system of rings.

Question #8: The planet with an exceptional tilt to its orbital plane that some do not consider to be a true planet.

(C). Pluto

The orbit of Pluto has the greatest tilt (17 degrees) and is the most eccentric of all the planets, placing it sometimes closer to the Sun than Neptune and sometimes placing it farther out than Neptune. The great tilt and crossing orbits lead some people believe the small object (seven moons in the solar system are larger) might be an escaped moon of Neptune rather than a true planet.

Question #9: The disk-shaped region of small icy bodies some 30 to 100 A.U. from the Sun believed to be the source of some comets is called the

(A). Kuiper Belt.

The small icy bodies in the Kuiper Belt are understood to be the source of short-period comets.

Question #10: Most meteorites are

(A). pieces from the asteroid belt.

Most meteorites generally come from particles that resulted from collisions between asteroids in the asteroid belt.

Question #11: Which planet(s) has (have) no moons?

(B). Mercury and Venus

Mercury and Venus, the two planets closest to the Sun, do not have moons.

Question #12: Most of the mass of the Solar System is found in

(B). the Sun.

The Sun has over 700 times the mass of the rest of the solar

(A). the terrestrial planets. (B). the Sun. (C). the comets and asteroids. (D). the gas giants. Question #13: The smallest planet in the solar system is (A). Mercury. (B). Mars. (C). Venus. (D). Pluto. Question #14: Which planet has its axis of rotation tilted so much that it appears to rotate on its side? (A). Uranus (B). Pluto (C). Venus (D). Saturn Question #15: The atmosphere of Venus is predominantly (A). nitrogen (B). carbon dioxide (C). methane (D). water vapor Question #16: Remnants of asteroids that survive the trip through the earth's atmosphere to strike the surface are called (A). meteors (B). meteorites (C). meteoroids (D). "shooting stars" Question #17: Which of the following is true about the surface of Mars? (A). There are numerous active volcanoes. (B). The polar ice caps are frozen water. (C). It is much too cold for liquid water to exist. (D). The greenish areas are due to low, scrubby vegetation. Question #18: Which of the following about Jupiter is not true? (A). Because Jupiter is mostly hydrogen, it is only slightly more massive than the earth.	system, including all the planets, moons, comets, asteroids, meteors, and meteorites. Question #13: The smallest planet in the solar system is (D). Pluto. The Earth's Moon, Io, Europa, Ganymede, Callisto, Titan, and Triton are all moon of the solar system, but all are larger than the smallest planet Pluto. Question #14: Which planet has its axis of rotation tilted so much that it appears to rotate on its side? (A). Uranus Uranus has an odd orientation of its axis to the plane of its orbit. Most planets have their axes of rotation tilted less than 30° from a vertical line to the plane of the orbit. Uranus's tilt is 82°, meaning that it is practically on its side. Moving in its orbit around the Sun on its side, one pole receives direct sunlight for 21 years, then the equator receives direct sunlight for the next 21 years, then the other pole receives the direct sunlight for the next 21 years. This would produce some interesting climatic pattern on the planet. Question #15: The atmosphere of Venus is predominantly (B). carbon dioxide The atmosphere of Venus consists mostly of carbon dioxide, a few percent of nitrogen, and traces of water vapor and other gases. Question #16: Remnants of asteroids that survive the trip through the earth's atmosphere to strike the surface are called (B). meteorites Meteoroids are fragments of comets and asteroids in space. A meteor is the visible trail of light and smoke left by a meteoroid as it heats up from air friction with the atmosphere. A meteoroid that survives its fiery trip through the atmosphere to strike the surface of the earth is called a meteorite. Question #17: Which of the following is true about the surface of Mars? (C). It is much too cold for liquid water to exist. Question #18: Which of the following about Jupiter is not true? (A). Because Jupiter is mostly hydrogen, it is only slightly more massive than the earth. Jupiter is, in fact, about 318 times more massive than Earth.
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(B). Much of the planet is composed of hydrogen compressed so greatly that it is liquid.

(C). The colored bands in the atmosphere are believed to result from atmospheric convection.

(D). Jupiter is mostly made of light elements like hydrogen and helium.

Question #19: Which of these statements about Kepler's Laws is true?

(A). A planet moves most rapidly when its orbit takes it closest to the sun.

(B). The time that is required for a planet to complete an orbit is proportional to the square of its radius.

(C). The retrograde motion of a planet is due to its motion on its epicycle as it also revolves around the sun.

(D). Planets travel in circular orbits.

Question #20: Which of the following statements about the length of a planet's day is true?

(A). The closer the planet is to the sun, the shorter is its day

(B). The more massive the planet, the longer its day.

(C). The four gas giants barely rotate at all.

(D). None of these are true.

Question #19: Which of these statements about Kepler's Laws is true?

(A). A planet moves most rapidly when its orbit takes it closest to the sun.

This follows from Kepler's second law of planetary motion, that a line between the Sun and a planet moves over equal areas of the ellipse during equal time intervals. This means that a planet must move faster when it is closest to the Sun and slowest when it is farthest from the Sun.

Question #20: Which of the following statements about the length of a planet's day is true?

(D). None of these are true.

Earth in Space

Questions	Answers with Rationalization
Question #1: The shape of Earth is that of a (A). perfectly round sphere. (B). pear-shaped form bulging at the poles. (C). irregular spheroid bulging at the equator. (D). flat table top.	Question #1: The shape of Earth is that of a (C). irregular spheroid bulging at the equator. The Earth is flattened at the poles and has an equatorial bulge, as do many other planets.
Question #2: Because of the shape of Earth's orbit around the Sun the overall Earth receives (A). 6 percent more solar energy in January. (B). 6 percent more solar energy in July. (C). the same amount of solar energy year around. (D). more solar energy in July and less in January.	Question #2: Because of the shape of Earth's orbit around the Sun the overall Earth receives (A). 6 percent more solar energy in January. Earth's orbit is slightly elliptical and the earth is about 2.5 million km closer to the Sun in January than in July. One result is a January Sun with an apparent diameter that is 3 percent larger than the Sun in July. The overall Earth receives about 6 percent more solar energy in January as a consequence. However, the length of day and angle of sunlight striking the earth's surface apparently has a greater effect on the climate than the increase in the amount of solar energy.
Question #3: As Earth moves in its orbit it has an orbital speed that is (A). slowest in January. (B). slowest at perihelion. (C). fastest in January. (D). fastest at aphelion.	Question #3: As Earth moves in its orbit it has an orbital speed that is (C). fastest in January. Earth's orbit is slightly elliptical, so the planet moves with a varying speed that is described by Kepler's laws of planetary motion. That is, the earth moves fastest when it is closer to the Sun in January, at perihelion, and moves slowest when it is farthest away from the Sun in early July, at aphelion.
Question #4: The relationship of Earth's axis to the plane of the ecliptic (A). is one direction half the time, then moving to the opposite direction. (B). is the same direction at all times. (C). reverses direction every month. (D). reverses direction every 3 months.	Question #4: The relationship of Earth's axis to the plane of the ecliptic (B). is the same direction at all times. The earth moves around the Sun in its orbit and all points in that orbit describe a flat surface called the plane of the ecliptic. The earth's axis is an imaginary line through the geographic poles, about which the earth turns during a daily rotation. The relationship between the earth's axis and the plane of the ecliptic is one of constant inclination. The earth's axis is inclined 23.5° from a line perpendicular to the plane, and this inclination remains the same throughout a given year, but with small changes over time.
Question #5: The orientation of Earth's axis to the background of stars (A). is one direction half the time, then moving to the opposite direction. (B). reverses direction every 3 months. (C). is the same direction at all times.	Question #5: The orientation of Earth's axis to the background of stars (C). is the same direction at all times. The earth's axis is an imaginary line through the geographic poles, about which the earth turns during a daily rotation. The relationship between the earth's axis and the background of stars is one of constant inclination. The orientation of Earth's axis to the stars is considered to be the same throughout the year as Earth moves around its orbit. Small changes do occur in the orientation over time, but in general the axis points in the same direction, remaining essentially parallel to its position during any day of the year.

(D). reverses direction every month.

Question #6: Days and nights vary in length throughout the year because of

- (A). daily rotation.
- (B). matching rotation and revolution.
- (C). a constant inclination and orientation of the axis.
- (D). yearly revolution.

Question #7: The Northern Hemisphere's summer solstice occurs when Earth moves in its orbit to point the North Pole

- (A). directly away from the Sun.
- (B). directly toward the Sun.
- (C). halfway between toward and away from the Sun.
- (D). none of the above.

Question #8: On the Northern Hemisphere's winter solstice the Sun at noon makes a shadow that is

- (A). shortest.
- (B). halfway between shortest and longest.
- (C). longest.
- (D). none of the above.

Question #9: Daylight and night are of equal length during

- (A). spring or autumnal equinox.
- (B). winter solstice.
- (C). summer solstice.
- (D). None of the above.

Question #10: Which of the following provides evidence that Earth rotates?

- (A). Coriolis effect.
- (B). solstices.
- (C). constant orientation of the axis.
- (D). all of the above.

Question #6: Days and nights vary in length throughout the year because of

(C). a constant inclination and orientation of the axis.
The constant orientation and inclination of the Earth's axis results in the axis pointing toward the Sun as Earth moves in one part of its orbit, then pointing away from the Sun six months later. The constant inclination and orientation of the axis, together with Earth's rotation and revolution, combine to produce three related effects: days and nights that vary in length, changing seasons, and climates that vary with latitude.

Question #7: The Northern Hemisphere's summer solstice occurs when Earth moves in its orbit to point the North Pole

(B). directly toward the Sun.
The North Pole points toward the Sun on June 21 or 22, and the North Pole receives sunlight for a full twenty-four hours. This is the summer solstice in the Northern Hemisphere, the beginning of the summer season with the longest daylight periods and the Sun at its maximum noon height in the sky.

Question #8: On the Northern Hemisphere's winter solstice the Sun at noon makes a shadow that is

(C). longest.
The Northern Hemisphere's winter solstice occurs on about December 22 and identifies the beginning of the winter season. At the winter solstice the Sun at noon has the lowest altitude, and the shadow from a vertical stick is longer than any other day of the year.

Question #9: Daylight and night are of equal length during

(A). spring or autumnal equinox.
As Earth moves in its orbit between point its North Pole toward the Sun on about June 22 and pointing it away on about December 22, there are two times when it is halfway between. At these times Earth's axis is perpendicular to a line between the center of the Sun and Earth, and daylight and nights are of equal length. These are called the equinoxes after the Latin meaning "equal nights." The spring equinox occurs on about March 21 and identifies the beginning of the spring season. The autumnal equinox occurs on about September 23 and identifies the beginning of the fall season.

Question #10: Which of the following provides evidence that Earth rotates?

(A). Coriolis effect.
The earth has a greater rotational velocity at the equator than at the poles. As an object leaves the surface and moves north or south, the surface has a different rotational velocity, so it rotates beneath the object as it proceeds in a straight line. This gives the moving object an apparent deflection to the right of the direction of movement in the Northern Hemisphere and to the left in the Southern Hemisphere. This apparent deflection caused by the earth's rotation is called the Coriolis effect.

Question #11: The slow wobble of the earth's axis is called

- (A). the Coriolis effect.
- (B). precession.
- (C). Foucault effect.
- (D). none of the above.

Question #12: The distance east or west from the prime meridian is called

- (A). latitude.
- (B). parallels.
- (C). longitude.
- (D). none of the above.

Question #13: An apparent solar day is

- (A). the interval between two consecutive solar noons.
- (B). slightly longer than a sidereal day.
- (C). rarely exactly 24 hours long.
- (D). all the above.

Question #14: If your favorite constellation rose at 8:00 PM one night, when would it rise two weeks later?

- (A). around 7:00 PM
- (B). at 8:00 PM
- (C). around 9:00 PM
- (D). it depends on the constellation

Question #15: North of the Tropic of Cancer, the sun appears directly overhead at noon

- (A). daily.
- (B). never.
- (C). twice a year, at the equinoxes.
- (D). twice a year, at the solstices.

Question #11: The slow wobble of the earth's axis is called (B). precession.

The earth slowly wobbles on its axis, much like a spinning top that swings in a slow circle as it spins. It will take the earth about 26,000 years to complete one turn, or wobble, tracing out a big circle with it's axis. The wobbling of the axis is called precession.

Question #12: The distance east or west from the prime meridian is called

(C). longitude.

To identify a location you need two lines, an east- and west-running parallel and a line that runs pole to pole. The north-south running arcs that intersect at both poles are called meridians. There is no naturally occurring, identifiable meridian that can be used as a point of reference such as the equator serves for parallels, so one is identified as the referent by international agreement. The referent meridian is the one that passes through the Greenwich Observatory near London, England, and this meridian is called the prime meridian. The distance from the prime meridian east or west is called the longitude. The degrees of longitude of a point on a parallel are measured to the east or to the west from the prime meridian up to 180 degrees.

Question #13: An apparent solar day is (D). all the above.

An apparent solar day is the interval between two consecutive crossings of the celestial meridian by the Sun, for example, from one local solar noon to the next solar noon. Since Earth is moving in orbit around the Sun, it must turn a little bit farther to compensate for its orbital movement, bringing the Sun back to local solar noon. As a consequence, the apparent solar day is about four minutes longer than the sidereal day.

Question #14: If your favorite constellation rose at 8:00 PM one night, when would it rise two weeks later?

(A). around 7:00 PM

Since Earth is moving in orbit around the Sun, it must turn a little bit farther to compensate for its orbital movement, bringing the Sun back to local solar noon. As a consequence, the apparent solar day is about four minutes longer than the sidereal day. This additional time accounts for the observation that the stars and constellations of the zodiac rise about four minutes earlier every night, appearing higher in the sky at the same clock time until they complete a yearly cycle. So, 4 minutes times 14 is 56 minutes, or about a hour earlier from 8:00 PM, which is 7:00 PM.

Question #15: North of the Tropic of Cancer, the sun appears directly overhead at noon

(B). never.

The parallels of 23.5 degrees north (Tropic of Cancer) and 23.5 degrees south (Tropic of Capricorn) of the equator identify the limits toward the poles where the Sun appears directly overhead during the course of a year. The Sun never appears directly overhead north of the Tropic of Cancer or south of the Tropic of Capricorn.

Question #16: South of the Antarctic Circle, the sun appears
(A). directly overhead only once a year
(B). directly overhead twice yearly.
(C). above the horizon all day at least once during December.
(D). above the horizon all day at least once during June.

Question #17: In the time 9:00 AM, the AM means
(A). before the meridian
(B). after the meridian
(C). astronomical motion
(D). after morning

Question #18: We always see the same side of the Moon because
(A). the Moon doesn't rotate.
(B). the period of rotation of both Earth and the Moon is the same.
(C). the Earth's rotation matches the Moon's revolution.
(D). the Moon's periods of rotation and revolution are the same.

Question #19: During which phase of the moon would we most likely see a solar eclipse?
(A). full
(B). first quarter
(C). last quarter
(D). new

Question #20: A lunar eclipse can only occur during which phase of the moon?
(A). 1st quarter
(B). full
(C). last quarter
(D). new

Question #21: If you were flying east on a jet across three time zones and your watch read 12:00 noon when you landed, you should
(A). set it to 9:00 AM.
(B). leave it at 12:00 for that is when the sun crosses the celestial meridian.
(C). set it 6:00 PM.
(D). set it to 3:00 PM.

Question #16: South of the Antarctic Circle, the sun appears
(C). above the horizon all day at least once during December.
The parallels of 66.5 degrees north (Arctic circle) and 66.5 degrees south (Antarctic circle) of the equator identify the limits toward the equator of where the Sun appears above the horizon all day during the summer. The Sun would appear above the horizon all day during the Southern Hemisphere summer (December), and moving northward this would decrease with fewer days of full light until reaching the limit of one day of 24-hour daylight at the Antarctic circle.

Question #17: In the time 9:00 AM, the AM means
(A). before the meridian
The apparent movement of the Sun across the celestial meridian identifies the even of time called noon. The morning hours before the Sun crosses the meridian are identified as ante meridiem (A.M) hours, which is Latin for "before meridian."

Question #18: We always see the same side of the Moon because
(D). the Moon's periods of rotation and revolution are the same.
The Moon rotates in the same period as the time of revolution, so the sidereal month is also the time required for one rotation. Because the rotation and revolution rates are the same, you always see the same side of the Moon from Earth.

Question #19: During which phase of the moon would we most likely see a solar eclipse?
(D). new
A solar eclipse occurs when the Moon is between the Sun and the Earth. The Moon is between the Sun and the Earth only during the new moon phase, so this phase is when you might expect a solar eclipse to occur.

Question #20: A lunar eclipse can only occur during which phase of the moon?
(B). full
A lunar eclipse occurs when the Sun, Earth, and Moon are lined up so Earth's shadow falls on the Moon. The Earth's shadow can fall on the Moon only when the Earth is between the Sun and the Moon, which also happens to be the alignment required for the full moon phase.

Question #21: If you were flying east on a jet across three time zones and your watch read 12:00 noon when you landed, you should
(D). set it to 3:00 PM.
When you cross a boundary between two time zones, the clock is set ahead one hour if you are traveling east and back one hour if you are traveling west. Since you were flying east and crossed three time zones you should set your watch ahead by three hours, from 12:00 noon to 3:00 PM.